



User Manual **SAM 940+**

Hand-Held Radioisotope Identifier



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CHAPTER 1. INTRODUCTION

The new, third generation, SAM 940+, meets the increasing demand for portability with a lightweight (1.2 Kg) unit that sacrifices neither speed nor quality of radionuclide identification. It is also designed to meet the requirements of ANSI 42.34 criteria for hand-held instruments.

Nucare's decades long experience in innovative RIID design drives a RIID that leverages the most advanced technologies available to deliver features such as: (a) source-less auto-stabilization; (b) real time reach-back; (c) detachable external probes. Features such as high resolution camera, multi-channel GPS, Gyro sensor, and a hot swappable battery pack complete a state-of-the-art unit.

The features of this new design assure the user of excellent performance that exceeds the demanding requirements of Homeland Security in the USA and beyond.



Figure 1.1.0. SAM940+

1.1. SAM 940+ EVOLUTION

The 3rd generation SAM 940+ is the most compact and lightweight hand-held RIID product ever produced. Volume and weight are a third less than previous models and yet advanced features such as multi-media support are included.

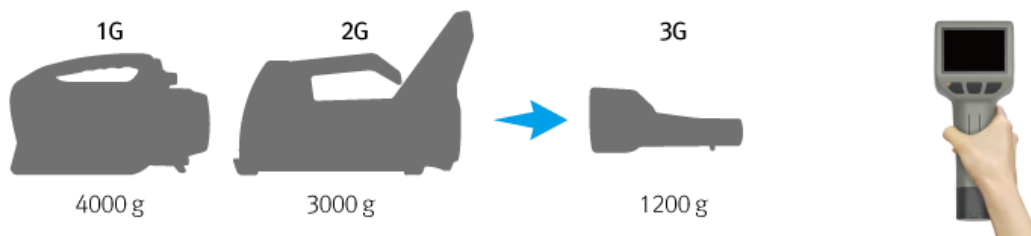


Figure 1.1.1. The 3rd generation hand-held RIID and its evolution

1.2. IMPORTANT SAFETY INFORMATION

The SAM 940+ has been meticulously engineered to provide safe, dependable, high performance for years. As with any kind of electrical equipment, following instructions and taking basic precautions will prevent harm to the operator and to the equipment.

Prior to using this equipment you should:

- ✓ Read and understand this manual thoroughly, especially the chapter on safety (**Chapter 2**).
- ✓ Read and understand thoroughly all warning labels on the equipment.
- ✓ Store all documents provided with the equipment in a safe place for future reference.

1.3. PRODUCT WARRANTY

Description of the Nucare, Inc. warranty for the SAM 940+ handheld RIID systems:

- ✓ The unit is warranted to be free of defects in material and workmanship, including: parts, labor and all unit elements.
- ✓ Unless otherwise indicated, the warranty is good for **12 months** after delivery to the original buyer only.

1.4. NOTIFICATION OF COPYRIGHT

The software in this device is protected by copyright laws and international treaty. You must treat the software like any other copyrighted material. Copyright laws prohibit making additional copies of the software for any reason other than specifically described in this manual. You may not copy the written materials accompanying the product without prior written consent from Nucare Inc.

CHAPTER 2. SAFETY

2.1. CAUTIONS AND NOTES

The following warnings and instructions are vital to the optimal operation of the SAM 940+ unit and should be made readily accessible for reference. Follow these warnings and instructions thoroughly to use the unit correctly and to protect the safety of the users.



Handling Instructions:

- ✓ The maintenance service of the unit should only be handled by qualified/trained personnel
- ✓ Cleaning the Camera and Flash lamp Windows: clean the windows with a soft, moist cloth to keep them clean and clear. Turn off the power before cleaning.



Safety Hazard Warnings:

- ✓ It can be very hazardous to open the unit on account of high voltage inside the unit (high voltage in the detector is up to 1000 Volts). Be sure to refer all servicing to qualified/trained personnel.



Disposal Instructions:

- ✓ If/when disposing the unit: observe local and national regulations for the disposal of units containing a lithium battery.
- ✓ Do not use household or municipal waste collection services for disposal of electrical and electronic equipment. EU countries require the use of separate recycling collection services.



WARNING:

- ✓ External equipment intended for connection to signal input, signal output or other connectors, shall comply with relevant EN Standard (e.g., EN 61326-1:2013, EN 301 489-1 V1.9.2 and EN 301 489-17 V2.2.1).
- ✓ If, in doubt, contact qualified technician or your local representative.

SAM 940+ CONTAINS FCC ID: UCSFC-2005-0033

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operations.

2.2. CALSSIFICATION

- ✓ EMC: Comply with EN 61326-1: 2013, EN 301 489-1 V1.9.2 and EN 301 489-17 V2.2.1)
- ✓ FCC Class: Class B Equipment (FCC CFR 47 part 15 subpart B, Section 15.101)
- ✓ Laser: Class II Equipment (Max. output: < 1mW, Wavelength: 650 – 660 nm)
- ✓ Degree of protection against electric shock: Type BF Applied Part
- ✓ Degree of protection against the ingress of water: IP65
- ✓ Equipment not suitable for use in the presence of a flammable anesthetic mixture with air or with oxygen or nitrous oxide.

2.3. LABELS

The label providing the manufacturer information and serial number of the system is located on the bottom of the unit (Figure 2.3.1.).

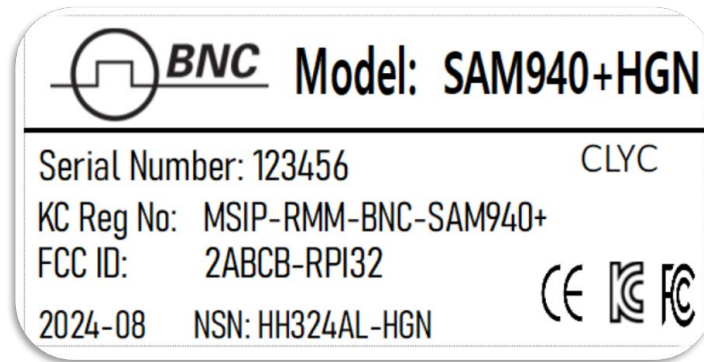


Figure 2.3.1. Label

CHAPTER 3. SYSTEM DESCRIPTION

3.1. SYSTEM COMPONENTS

The physical components of the SAM 940+ are listed in figure 3.1.1. The two software programs used with the SAM 940+ are described in **Chapter 4 and 7**. Also included is USB cables with universal adaptors for charging battery and connecting the SAM 940+ to a PC. A heavy duty carrying case and SAM 940+ components packaged in the custom-made sponge foam of the carrying case are shown in figure 3.1.2.



①	Gamma detector	②	LCD display	③	Neutron detector (optional)	④	Replaceable battery pack
⑤	AA battery holder	⑥	Universal adaptor	⑦	Belt holster	⑧	Camera

Figure 3.1.1. SAM 940+ Components



Figure 3.1.2. SAM 940+ Components packed in a carrying case

- ① RAD IQ™ SAM 940+ detector unit

- ② Universal power adaptor for RAD IQ™ SAM 940+
- ③ Replaceable battery pack (2 ea.) and emergency AA battery holder (1 ea.)
- ④ OTG communication USB cable
- ⑤ USB cable for battery charge
- ⑥ Belt holster, hand strap and battery charger
- ⑦ Heavy duty carrying case



WARNING: High current (3A) power adaptor and corresponding USB cables are provided for fast battery charging. Do not use any other power adaptor and USB cables. Using inappropriate adaptor and/or cable may slow down charging time and performance.

3.2. HIGH RESOLUTION CAMERA

Multi-media support is one of distinctive features of SAM 940+. Having visual information of sites and target objects greatly improves event for review.

SAM 940+ is equipped with a built-in camera (CMOS 8M pixel) module for recording media information that can be attached along with spectroscopic data.



Figure 3.2.1. Built in camera location and protective cover.



The camera lens is located as shown in figure 3.2.1. (red circle). The lens is water protected by a special optical cover. Keep the optical cover as clean as possible.

3.3. BELT HOLSTER AND HAND STRAP

The compact and light weight design enables easy transport of SAM 940+ in a belt holster. The belt holster is designed to monitor the display of SAM 940+ while wearing it.

The waist strap prevents the SAM 940+ from slipping out of your hand.



Figure 3.3.1. Belt holster and hand strap

3.4. POWER AND BATTERY CHARGE GUIDE

There are several ways to charge the SAM 940+ battery pack. Most effective ways are using a dedicated charger station provided or using an USB cable directly to the battery. Full charge may take up to 3-4 hours for each battery.

3.4.1. Remove/Insert a battery from/to the SAM 940+

SAM 940+ provides two rechargeable batteries. Following is instruction of removing/inserting the battery from/to the SAM 940+ unit.

3.4.1.1. Opening the end cap and removing the battery.

1. Twist the end cap clock wise and take out the end cap. (figure 3.4.1.)



Figure 3.4.1. Open the end cap

2. Tilt down the SAM 940+. (The battery will be sliding out by gravity).

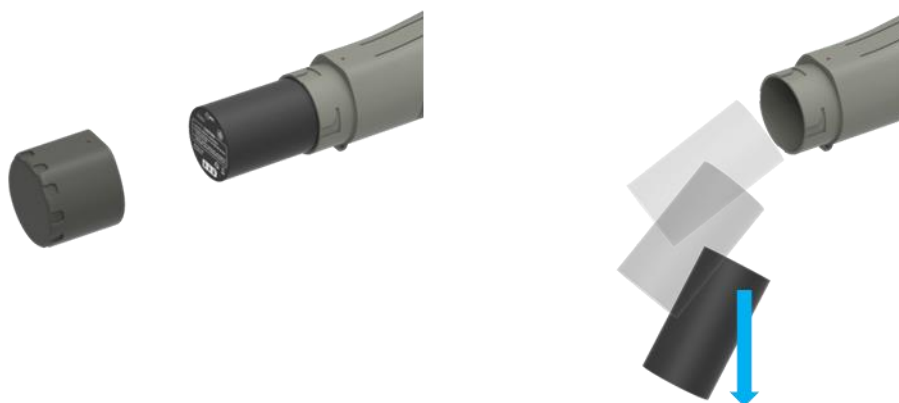


Figure 3.4.2. Battery out by tilting down.

3.4.1.2. Inserting the battery and closing the end cap.

1. Inserting the battery as shown in figure 3.4.3. (LED side is down).



Figure 3.4.3. Insert battery

2. Close the cap by twist counter-clock wise. The two red dots should be aligned (figure 3.4.4.)

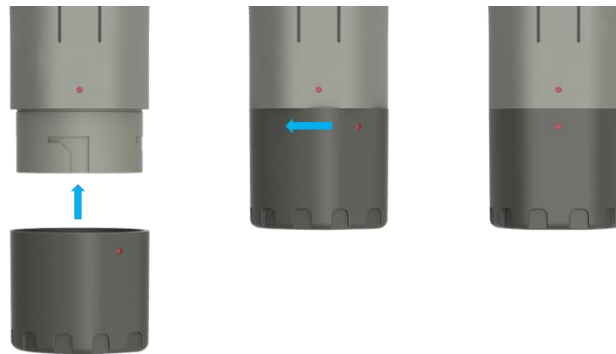


Figure 3.4.4. Close the end cap.

3.4.2. Directly charge to the battery

The swappable battery has a micro 5 pin USB port and battery status indicator LEDs. Using the port, the battery can be charged independently. The battery indicator LED shows the level of charging status when the LED power button is pressed.



Figure 3.4.5. Power switch and battery charger



The SAM 940+ battery has a built-in power protection circuit (PCM) which prevents the battery from both overcharging and over-discharging. The battery power indicator LEDs are turned on sequentially. All three LED is turned on when it is fully charged as shown in figure 3.4.5.

3.4.3. Charging battery using a dedicated charge station

The SAM 940+ comes with a dedicated battery charge station as shown in figure 3.4.6. The battery can be docked into the station in any azimuth angle. No alignment is needed.



The battery has smart contact design that enables docking the battery into the charge station without concerning about alignment.



Figure 3.4.6. Battery charge station

3.4.4. Charge though the SAM 940+

The battery can be charged without taking the battery out from the unit using the mini 5 pin USB port on the front face of the SAM 940+ as shown in figure 3.4.7.



Figure 3.4.7. USB port and charge the battery



The SAM 940+ comes with two different USB cables, i.e., micro 5 pin and mini 5 pin, as shown in figure 3.4.8. Be aware the difference and use an appropriate cable. Micro USB cable is for charging the battery and mini USB is for communication between the SAM 940+ and a PC.



Figure 3.4.8. Micro USB vs. mini USB cables.



The charging time might be longer than other methods if the unit is turn ON consuming power for operation or stand-by.

3.4.5. Power by Lithium Ion battery

Off-the-shelf **Lithium Ion battery** can be used for emergency cases when running out of power by the battery pack. Total of 4 AA size **Lithium Ion** batteries are needed to operate the SAM 940+ using the battery holder provided. The maximum stand-by operation time of the SAM 940+ with the 4 AA battery is up to two hours.



Figure 3.4.9. AA battery holder



Hazard: Internal high voltage (up to 1,000 volts)! Do not disassemble the unit. Servicing of the unit should be performed by NuCare Inc. qualified personnel.

3.5. SAM 940+ CONNECTIVITY

Another advanced features of SAM 940+ that is hardly found other products is its ability to connect to external devices. It supports connection to external detectors, network adaptor for 4G/LTE network and other devices via its USB port or a LIMO connector.

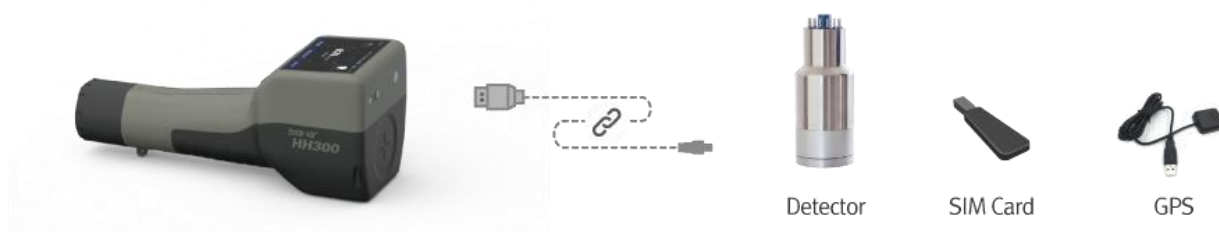


Figure 3.5.1. Example of detachable devices



Detachable device to the SAM 940+ has to be met with required communication protocol. Consult with Nucare engineer for details and feasibility.

3.6. SAM940+ SOFTWARE

The SAM 940+ system includes two dedicated software programs for conducting surveys and analyzing results in a computer.

3.6.1. PeakAbout IV for the detector unit

- PeakAbout IV is used for configuring and calibrating the system, and for real-time measurement and analysis of data acquired by the detector unit. It is based on the Android OS.
- Detailed instructions for operating PeakAbout IV are in **Chapter 4**.

3.6.2. PeakID IV for the PC

- PeakID supports management of Event Log files, data analysis including integration of data from multiple units, and data backup. It is compatible with PCs that run Windows 10 or higher.
- Detailed instructions for installing and operating PeakID IV are in **Chapter 7**.

3.7. PREREQUISITES FOR THE OPERATION OF THE SAM 940+

3.7.1. Update configuration parameters

There are few critical configuration parameters that is important in operating the SAM 940+. The SAM 940+ unit is delivered with configuration parameters set by factory default. However, the customized parameters that should be tailored to each user are recommended to edit before operating of the unit.

 PeakAbout IV SW itself allows user to edit some of configurable parameters. However, using PeakID IV SW installed in a PC with a mouse and a keyboard is more convenient in editing those. In addition, all the configurable parameters can be accessed/edited by PeakID IV SW.

 There are two ways to upload configuration parameters from a PC to the SAM 940+, i.e., using USB memory device or using WiFi network. The former is easier method while the later method requires pairing procedure between the SAM 940+ and a PC.

3.7.1.1. Using an USB

1. Edit configuration parameters using the PeakID IV application SW installed in a PC. **Refer to the section 7.9.3. User setup and 7.9.4. Administrator setup for instructions.**
2. Copy the configuration to an USB memory device (Refer to section 7.9.2.)
3. Upload the configuration into the SAM 940+. (Refer to the section 6.3.4.4.)

3.7.1.2. Using WiFi network

2. Edit parameters using the PeakID IV application SW installed in a PC. **Refer to the section 7.9.3. User setup and 7.9.4. Administrator setup.**
3. Pairing the SAM 940+ and a PC. (Refer to the section 6.3.4. Pairing.)
4. Upload edited configuration to the SAM 940+. Refer to the section **7.9.3. Upload/download configuration via WiFi network.**

3.7.1.2. Factory default parameters and its value

Factory default parameters are summarized in table 3.7.1.

Table 3.7.1. Factory default parameters

Privilege	Parameter	Factory default	Note
USER	User Name	BNC	
	User Location	BNC	
	Detector Info	TBD	Device dependent information
	SW version	TBD	Currently installed SW version
	Alarm Ringtone	Siren	
	Alarm Volume	100%	

ADMIN.	Isotope Library	ANSI NaI	
	Sleep mode time	300 sec.	
	Manual ID Time	60 sec.	
	Increment Time	10 sec.	
	Background Time	60 sec.	
	Calibration Counts	200,000 cnt.	
	Doserate Unit	Urem/h	
	Display Option	Linear	
	Alarm mode	FIX	
	Alarm Theshold	1000 cps.	
	Health Alarm	10 mSv/h	
	Neutron Alarm	0.8	
	Reachback Account	none	Reachback email information
	RadResponder Account	BNC	
	Device WiFi	none	
	Admin Password	1234	
	Time Zone	USA/LA	
	Sequential mode	Off	
	Seq. measurement	300	
	Seq. pause time	3	
	Seq. repeat	5	

CHAPTER 4. PEAKABOUT IV

4.1. INTRODUCTION

The main function of PeakAbout IV application software is to provide dose (and count) rate data for the gamma-ray. In addition it provides the operator with a convenient graphic status display and access to the analytical tools. It also manages the configuration and calibration of the SAM 940+ unit.

Intuitive, simple and hierarchical structure design of the new PeakAbout IV software make the operation of SAM 940+ easy and flawless. In addition, function buttons on handle provides extra convenience in handling the unit with wearing gloves.

Details of application software structure and its functions are described in the following sub sections.

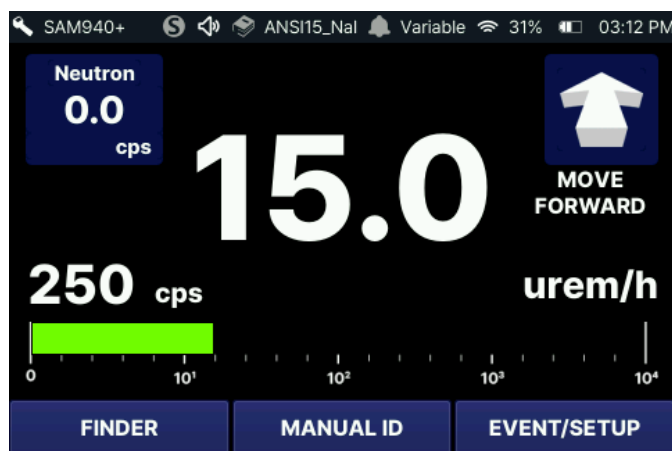


Figure 4.1.1. Main screen

4.2. FUNCTION BUTTONS

4.2.1. Function Buttons

The function buttons of SAM 940+ on its neck provides a natural grip with convenient control and operation of the unit. A user can operate the buttons with wearing a glove. Each function keys does actions corresponding to the menu in the bottom, for instance, **FINDER** ①, **MANUAL ID** ②, and **EVENT/SETUP** ③ action is executed when the function button left, middle and right is pressed, respectively.



Figure 4.2.1. Function buttons on the handle.

4.2.2. Double action function button.

In some menu screen, the function button does two actions. For example below (screen of typing in password for administrative access), each function button does two actions, i.e., LEFT/UP, OK/DONE and RIGHT/DOWN.

The second action, i.e., UP, DONE, and DOWN, can be effective by pressing the corresponding button and holding it for about 2-3 seconds. The action becomes active when the button is released.

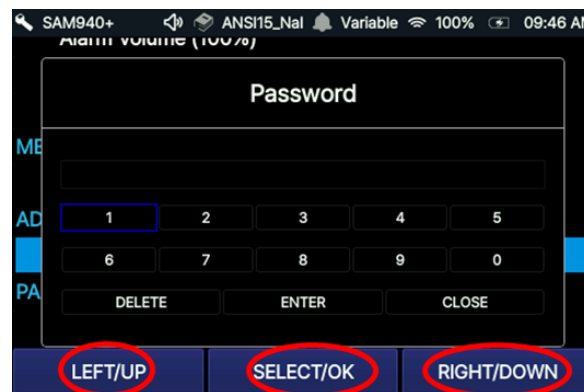


Figure 4.2.2. Double action example.

4.2.3. Keyboard Usage Guide.

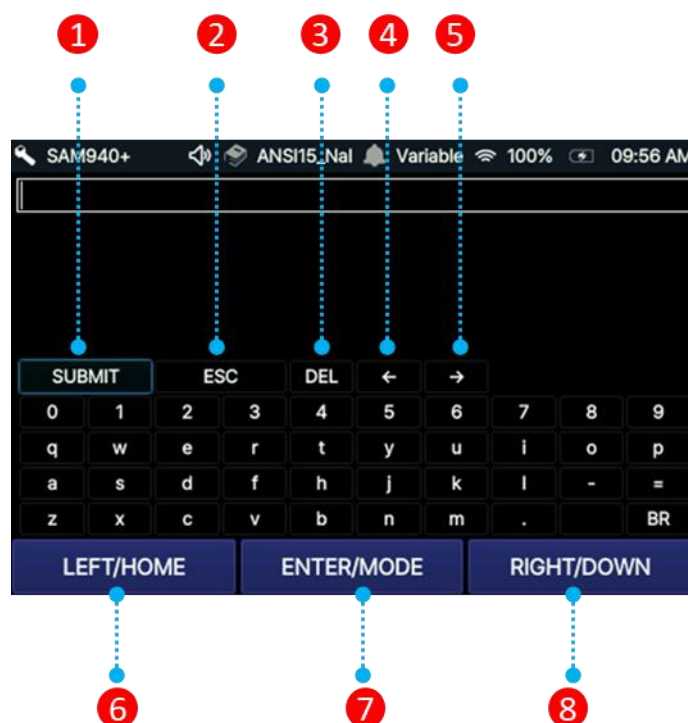


Figure 4.2.3: Keyboard screen

1	Submit button	Submit all the text you have entered
2	Esc	Close dialog
3	DEL	deletes the last character you entered or the character before the typing cursor
4	Left navigation	Move left one character
5	Right navigation	Move Right one character
6	LEFT/HOME	LEFT: Move to the left Hold HOME to quickly go to the 'Submit' button
7	ENTER/MODE	ENTER: select character you choose MODE: Switch among 3 modes: normal characters (shown figure 4.2.3), capital letters (shown figure 4.2.4), and special characters (shown figure 4.2.5)
8	RIGHT/DOWN	RIGHT: Move the right Hold DOWN button to move down

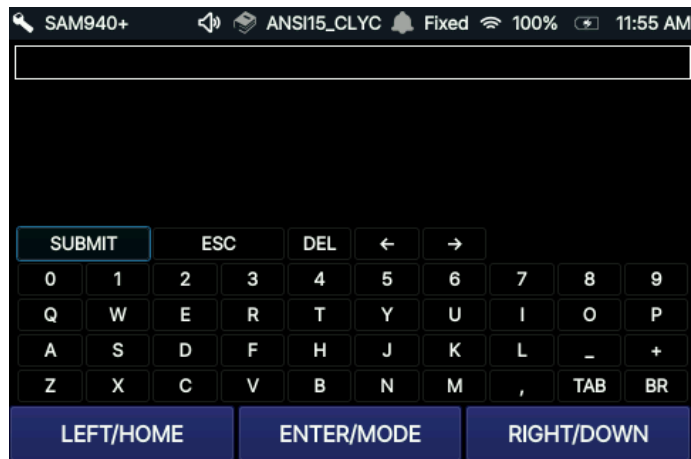


Figure 4.2.4: Capitalized characters mode

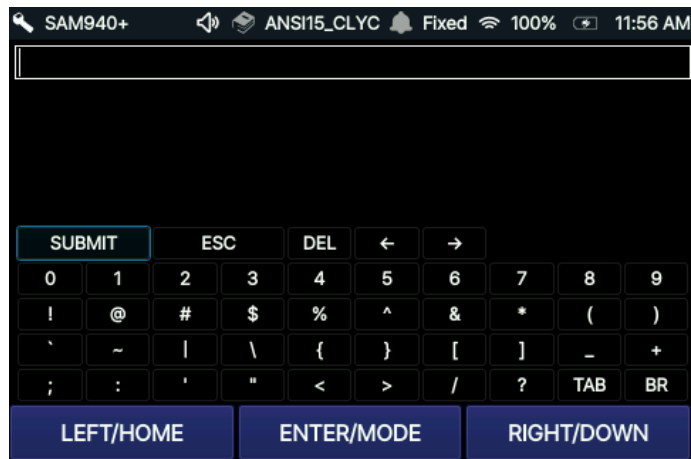


Figure 4.2.5: Special characters mode

4.3. MAIN SCREEN

The screen's main function is to display dose rate and count rate data for gamma-ray energy in real time, and neutron count rate data (if so equipped) as well. In addition to displaying dose rate and count rate data on a logarithmic scale main, the screen gives information on the system's operating status (Figure 4.3.1).

All real-time operations and menu access begin at the main screen.

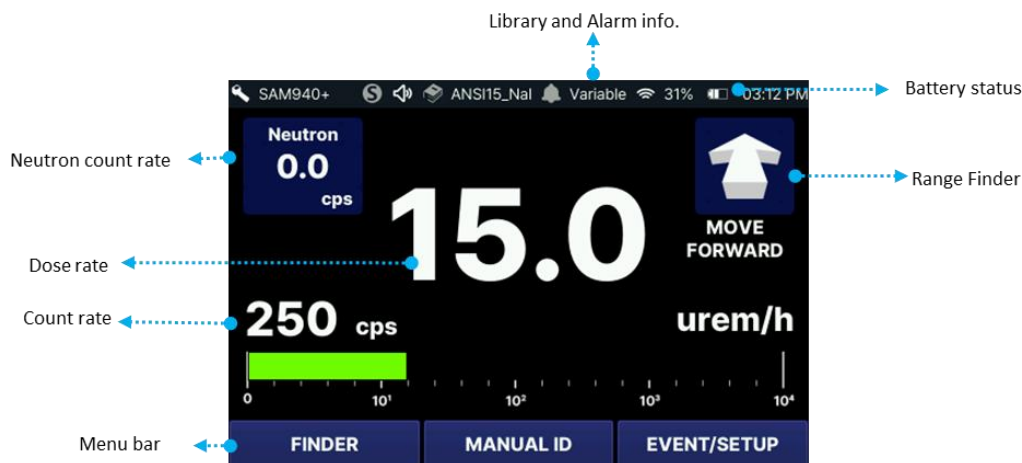


Figure 4.3.1. Main screen example.

4.4. MENU HIERARCHY

PeakAbout IV's menu hierarchy is summarized in figure 4.4.1. FINDER, MANUAL ID and EVENT LOG menu can be accessed from HOME screen. SETUP menu is hidden from the MAIN but it can be accessed by pressing EVENT LOG button for 2 seconds.

FINDER menu is divided into FINDER and READTIME. And FINDER screen is the primary display when FINDER menu is selected. FINDER and READTIME screen can be toggled each other.

MANUAL ID menu is divided into SPECTRUM and CLASSIFIER. And SPECTRUM screen is the primary. The two screens can be toggled each other. When the MANUAL ID is completed, EVENT REPORT screen is appeared automatically. REACHBACK and CAMERA menu can be accessed from the EVENT REPORT screen.

EVENT LOG menu is divided into EVENT ID and ENENT INFO. The two screens can be toggled each other. The primary screen is EVENT ID and REACHBACK screen can be accessible from it. CAMERA screen can be accessed from the EVENT INFO screen.

SETUP menu consists of USER setup, Measurement, Administrator setup and Paring.

Details of each menu is described in **chapter 5 and 6**.

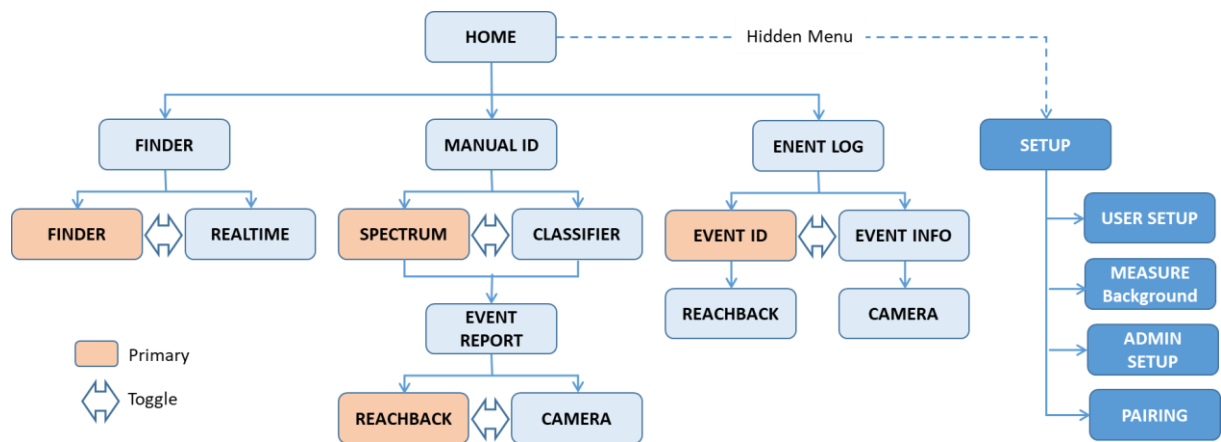


Figure 4.4.1. Menu structure

CHAPTER 5. BASIC OPERATING INSTRUCTIONS

5.1. PREPARE EQUIPMENT FOR OPERATION

5.1.1. Charge the battery

Charge the battery (refer to section 3.4. **Power and Battery Charger**). You may check the battery indicator for power status as shown in figure 5.1.1.



Figure 5.1.1. SAM940+ charge and power up example

5.1.2. Turn Power On.

Turn the power switch on. The power switch is located on the left side of the unit. Press the red button. (Figure 5.1.2).



Figure 5.1.2. Power switch location

5.2. AUTO CALIBRATION

The Auto Calibration and Auto Stabilization features adjust the calibration parameters of the SAM 940+ to correct for drift caused by environmental changes in ambient temperature. Auto Calibration and Auto Stabilization can recover peak shift up to $\pm 20\%$. If drift of more than $\pm 20\%$ between the stabilized peak and the original peak (Cs-137 calibration peak) occurs, then you must perform manual calibration using a Cs-137 source.

Auto Stabilization begins immediately after the completion or cancellation of Auto Calibration and executes again every three minutes, except when the system is in an alarm condition or when performing Manual Identification.

When Auto Calibration or Auto Stabilization is executed, the system automatically adjusts its gain using the 1461 keV energy peak emanated by the K-40 radionuclide from natural background and no other calibration source is needed. If necessary, such as in the rare case of a location where levels of potassium are unnaturally low (in a boat surrounded by water for instance), supplemental KCL (Potassium chloride) powder can be used to achieve stabilization more quickly.



Auto Calibration automatically executes when PeakAbout IV is initiated. We recommend waiting for Auto Calibration to complete (100%) before proceeding, Auto Calibration can be stopped by pressing Cancel if necessary. A pop-up message is appeared when the auto-calibration is completed. (figure 5.2.1).



Make sure that there are no radiation sources in the vicinity during Auto Calibration.

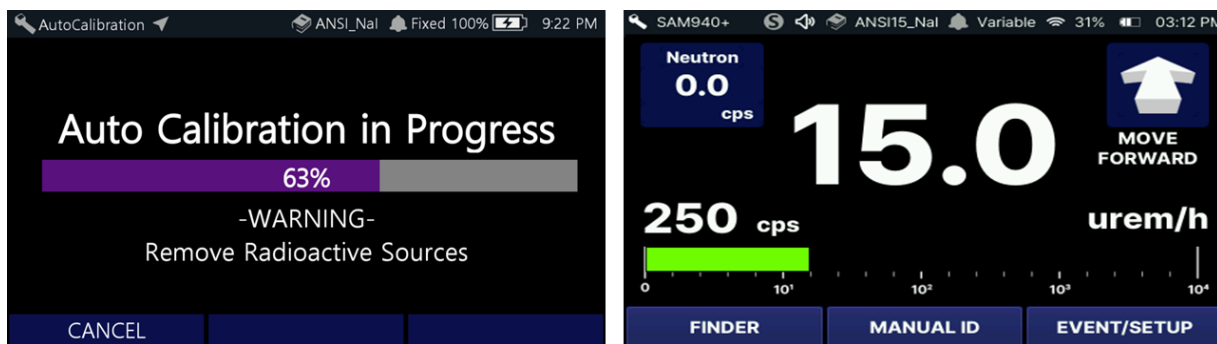


Figure 5.2.1. Auto Calibration example

5.3. PERFORM BACKGROUND MEASUREMENT

Reliable background information is critical to good nuclide identification and alarm performance. PeakAbout IV uses an advanced NORM rejection algorithm to minimize false alarms due to fluctuations of Naturally Occurring Radioactive Material in the background.

1. Make sure there are no radiation sources near the SAM 940+

2. Move to SETUP screen by selecting EVENT LOG and hold for 2 seconds. ①
3. Select /Measurement/Background ②

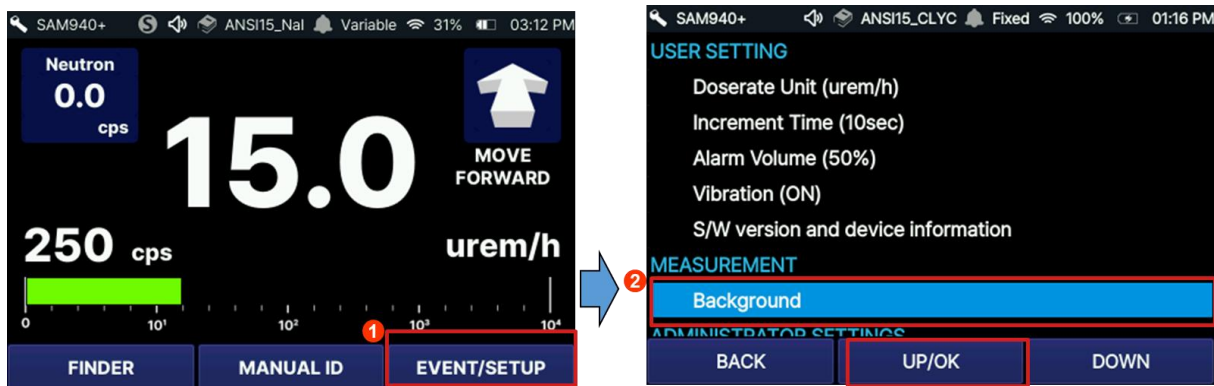


Figure 5.3.1. Background measurement

4. A spectrum will begin and will automatically update during the measurement (Figure 5.3.2.)
5. To cancel background measurement select BACK button.

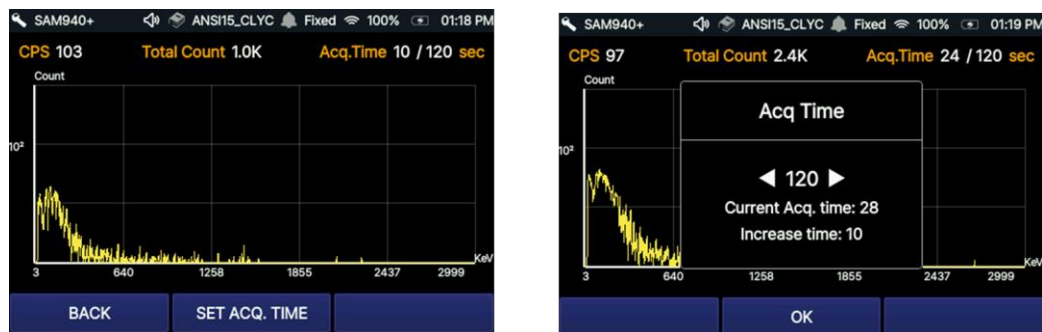


Figure 5.3.2. Background measurement and time setup



SETUP menu is hidden from the Main screen menu bar. To access, press and hold the EVENT LOG button for 2 seconds.



The background spectrum (displayed in yellow) is automatically updated during the measurement. Measurement time is defined in the Setup menu. The default time is 60 seconds. To change this setting, you must use the **PeakID IV**. Refer to **Section 7.9.4.6. Background Measurement Time**.



SET ACQ. TIME allows user to manually increase/decrease background measurement time as shown in figure 5.3.2.

5.4. BEGIN MONITORING RADIATION

The SAM 940+ automatically begins monitoring radiation when the Main screen opens.

1. Look at the alarm mode area on the status bar (Figure 5.4.1.). If Variable Alarm Mode is in effect, allow about 20 seconds for the unit to collect background and threshold data before surveying.

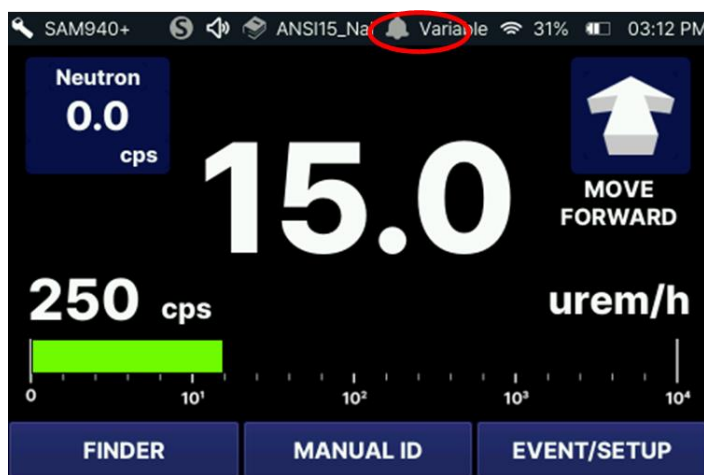


Figure 5.4.1. Status bar example with Fixed Alarm Mode in effect

2. The main displays both the dose rate (rem/h or Sv/h) and count rate (counts per second (cps)) of gamma activity. If so equipped, the count rate for neutron activity is also displayed on the Main screen. The gamma dose rate scale is logarithmic. PeakAbout IV automatically scales to $\mu\text{rem/h}$ or mrem/h ($\mu\text{Sv/h}$ or mSv/h) according to the detected activity level.
3. You may choose **FINDER** to view the **FINDER** or **REALTIME** screen during monitoring instead of the Main screen (Figure 5.6). The Realtime screen emphasizes count rate rather than dose rate information. Counts per second (cps) are displayed on a graph that moves to the left as it updates every second. The full graph represents the most recent 60 seconds of cps.
4. You might prefer to use the **REALTIME** ③ screen because the graph can alert you to a momentary increase in activity more effectively than the rapidly changing numerals on the Main screen. With the Finder screen, you also have a second chance to observe evidence of a source that you might not notice otherwise. The system will alarm regardless of which of these screens is enabled when the alarm threshold is exceeded.
5. **FINDER** ② provides directional information of detected isotopes. Combination of radiation event and internal gyro sensor response, PeakAbout IV estimates the direction of gamma source. The detail instruction of **FINDER** feature is described in 5.8. **Find Directional Information of Isotopes** section.



The dose rate unit (rem/h or Sv/h) is defined in the **PeakID IV Setup** menu. To change this setting, you must first log in as Administrator.



Figure 5.4.2. Example of Main screen and Finder screen

5.5. WHEN THE SAM940+ HAS ALARMED

When the system detects radioactivity, the readout on the main changes from white to red (Figure 5.5.1), and the audible alarm sounds. **There is no reason to panic when this happens**, as the alarm merely indicates that detected activity has only exceeded normal background dose rate or count rate. For concerns about personal safety, refer to **Section 7.9.4.8. Select Alarm Mode and Alarm Threshold**.

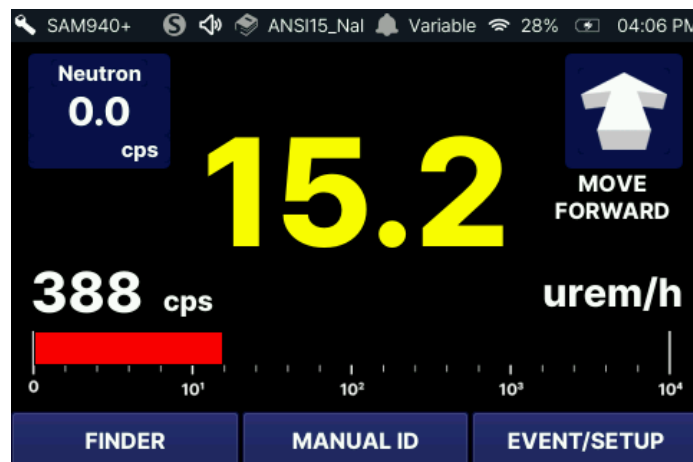


Figure 5.5.1. Alarm example

PeakAbout IV displays a helpful Range Finder (Table 5.5.1) on the Main screen. Use it to better locate the gamma source. To measure and record this kind of event, Manual ID must be performed. When the “IN RANGE” message displays, it is a good time to perform Manual ID. Refer to **Section 5.6. Perform Manual ID**.

Table 5.5.1. Range Finder messages

Range Finder Message	What It Means
	IN RANGE: Count rate/dose rate is within the optimal range for isotope identification. Manual ID should be performed in this range.
	MOVE FORWARD: Count rate/dose rate level is too low for isotope identification. Move closer to the radiation source.
	MOVE BACK: Count rate/dose rate level is higher than optimal for isotope identification. Move away from the radiation source.
	DANGER: Activity level is extremely high. Move back to a distance with a safer level or leave the area immediately.

5.6. PERFORM MANUAL ID

To measure and record an event, Manual Isotope Identification must be performed. Manual ID can be performed whether the SAM 940+ is in an alarm condition or not.

Figure 5.6.1. shows graphical illustration of ways to toggle between Main screen and Manual ID screen using function buttons.



Figure 5.6.1. Toggle between Main and Manual ID screen using the function buttons.

5.6.1. Initiate Manual ID

Initiate Manual ID by selecting MANUAL ID button on the Main screen. (figure 5.6.2)

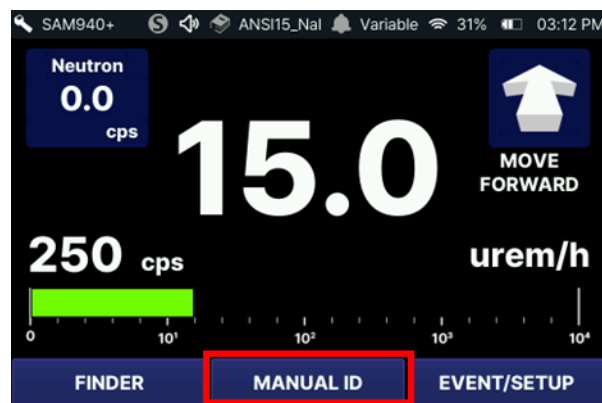


Figure 5.6.2. Initiate Manual ID.

When Manual ID is initiated, PeakAbout IV begins recording an event spectrum (Figure 5.6.3). The default measurement time is 60 seconds. While the Manual ID is running, you may decide that you will need to record a longer measurement. Without interrupting or restarting your Manual ID, you can extend or reduce the measurement time in increments of 10 seconds. Default measurement time can be set in PeakID. (Refer to **7.9.4. Administrator setup**).



Figure 5.6.3. Manual ID screen and setup acquisition time

5.6.2. Toggle between Spectrum screen and its Classification screen

Up to 10 isotopes can be identified simultaneously and presented on the Spectrum and Classification screens. The Spectrum screen displays the event spectrum. On the Classification screen, identified isotopes are displayed on a color-coded bar graph.

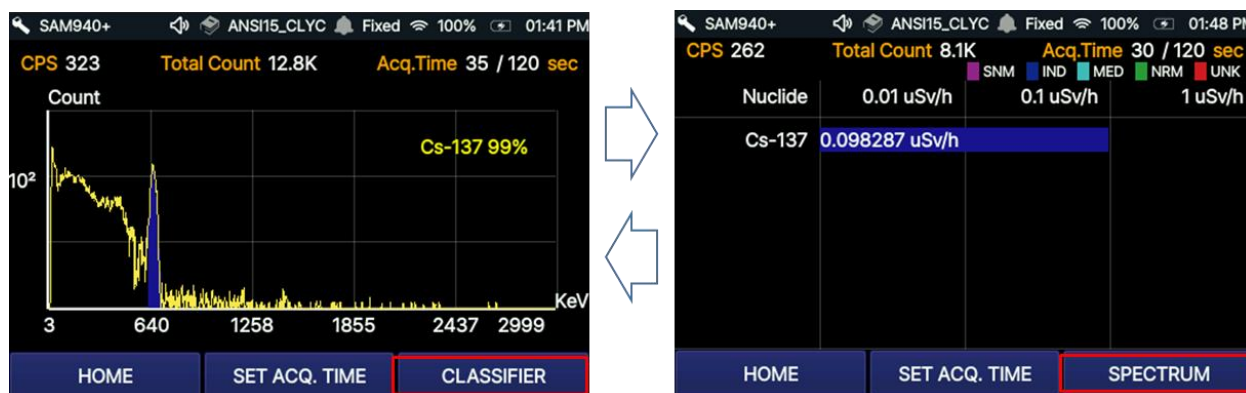


Figure 5.6.4. Manual ID screen and analysis screen

5.6.3. Spectrum screen

The Spectrum screen is shown in figure 5.6.5. and its contents are summarized in table below.



You can cancel an event before the measurement time has elapsed by touching the HOME key. The event data will not be saved, and you will return to the Main screen



Figure 5.6.5. Manual ID screen

1	Status bar (right)	Displays battery indicator and time
2	Acquisition time	Allows user to increase/decrease capture time and displays progress
3	Counts	Y axis displays counts on a self-adjusting, logarithmic scale
4	Channel/Energy	X axis displays energy on a linear or QCC scale
5	Nuclide name	Displays identity and confidence level of identified nuclide(s)
6	Photo peak	Color-coding corresponds to Classification screen (see Figure 5.6.6.)
7	Classifier	Toggle between Spectrum screen and Classification screen

5.6.4. Isotope classification screen.

The Isotope Classification screen is shown in figure 5.6.6. below. In this mode, the real-time Dose Rate measurement of identified nuclides is graphically displayed in a horizontally oriented bar graph. If multiple isotopes are identified, PeakAbout IV will place the most significant isotope (highest contributing dose rate) as the top-most bar, and the remaining isotopes in descending order to the least significant isotope (lowest contributing dose rate). Each bar will be named and displayed with the corresponding classification color as shown in figure 5.6.6. ④



Figure 5.6.6. Isotope identification screen

1	Status bar (right)	Displays battery indicator and time
2	Status bar (left)	Displays unit name, selected library, and alarm mode
3	Classification	Classifies identified isotopes as: SNM – Special Nuclear Material – magenta IND – Industrial Isotopes – blue MED – Medical Isotopes – cyan NRM – Naturally Occurring Radioactive Material – green UNK – Unknown Material – red
4	Event time and CPS	Time during event, acquisition time, and counts per second (CPS)
5	Dose rate	Displays dose rate of classified isotope on color-coded, logarithmic bar graph
6	Spectrum screen	Toggle between Classification screen and Spectrum screen

5.6.5 Identify Result

When measured time has elapsed, PeakAbout IV displays Identify result dialog showing Confident level, name and classifier identified of isotope. Click center button to turn off and on this dialog.

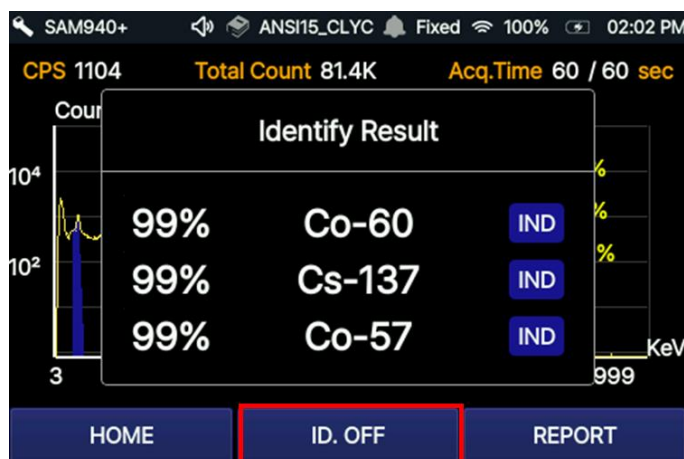


Figure 5.6.7. Identify Result example

Click Report button in left to navigate to Event Report screen .

5.6.5. Event report screen

When the measurement time has elapsed, PeakAbout IV displays the Event Report (Figure 5.6.7.). In this screen, you may send the event to reach-back or take photos.



Figure 5.6.8. Event Report example

1	Status bar (right)	Displays selected library, and alarm mode, battery indicator and time
2	Event Information	Total count, count rate and Acq. Time (ID progress)
3	Detected Nuclide	Displays identified isotope(s) with classification, dose rate, and confidence level
4	Media list	Allows review of attached photo(s)



To be identifiable, the isotopes must be in the selected isotope library (the name of the selected library appears on the status bar; see Figure 5.6.7.). The library setting is defined in the Administrator Settings menu in PeakID IV. To change this setting, you must first log in as Administrator. Refer to **Section 7.9.5. Library Setup**.



The default measurement time is defined in the Administrator Settings of PeakID IV. **Section 7.9.4. Administrator Setup**. The increment by which measurement time can be extended or reduced is defined under **Section 7.6 : Manual Identification Adjustment Time**. To access and change these settings, you must be logged in as Administrator.

5.7. TRANSMIT EVENT DATA TO REACH-BACK

When operating within range of Wi-Fi, the SAM 940+ allows you to transmit spectrum and event data directly to a prearranged location such as a U.S. DOE reach-back facility, group leader, command center, etc.

5.7.1. Transmit event to reach-back when manual ID is completed

Data from an Event that you select is emailed in N42-compatible XML format to the specified recipient.

1. Select **REACHBACK** button when manual ID is completed.
2. A dialog box will appear to select transfer method. Select ReachBack
3. Reachback progress is shown during the sending process
4. Transmit success message is appeared when transmission is completed.

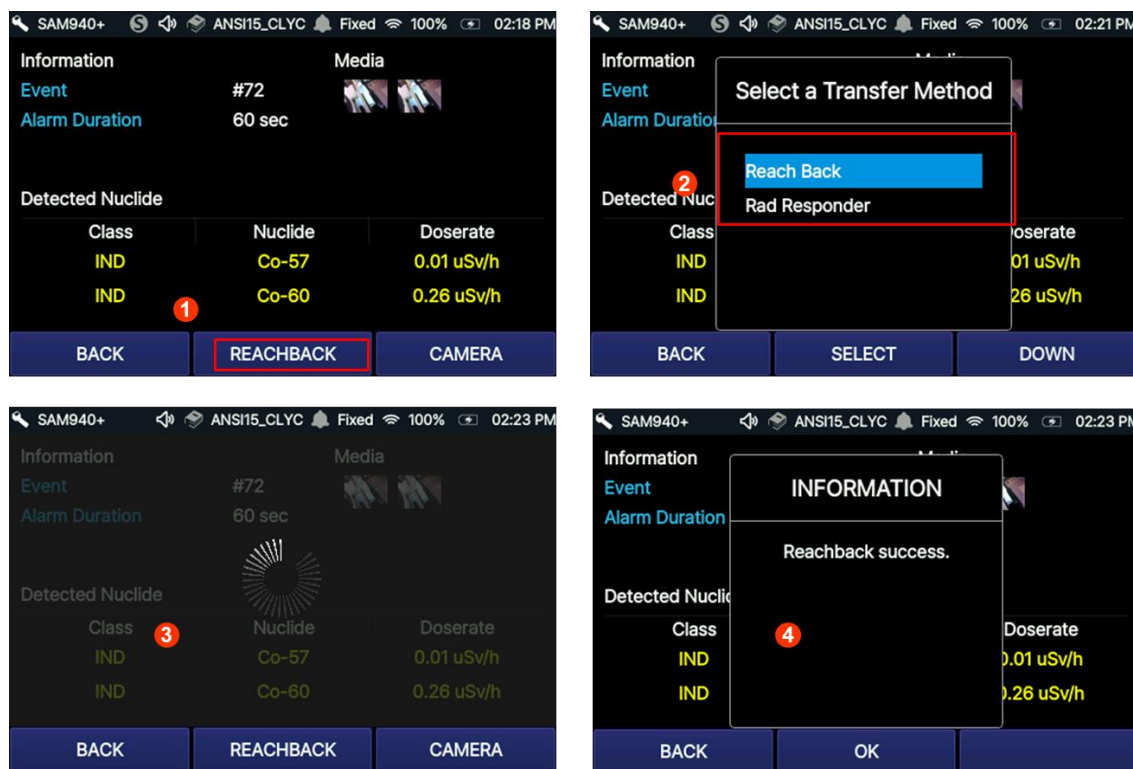


Figure 5.7.1. Transmit Event Report to Reach-back



Reach-back transmission can also be initiated from the Event Log. Refer to the instructions in **Section 6.2 : Use the Event Log to Transmit Event Data to Reach-back.**



Instructions for configuring Email Account settings are described in **Section 7.9.4.11. Reach Back Email Account.**

5.7.2. Transmit event to RadResponder when manual ID is completed

In the United States, we have partnered with FEMA to implement RadResponder, a free network for the rapid collection and management of radiological data during an emergency. When the SAM 940+ is operating within range of Wi-Fi, spectrum and event data can be uploaded to the RadResponder cloud in the appropriate format. RadResponder can be accessed on PDAs, smartphones, tablets, or any computer connected to the web, allowing it to be seamlessly employed at all levels of government during a radiological or nuclear emergency.

1. Select **REACHBACK** button when manual ID is completed.
2. A dialog box will appear to select transfer method. Select RadResponder
3. Reachback progress is shown during the sending process
4. Transmit success message is appeared when transmission is completed.

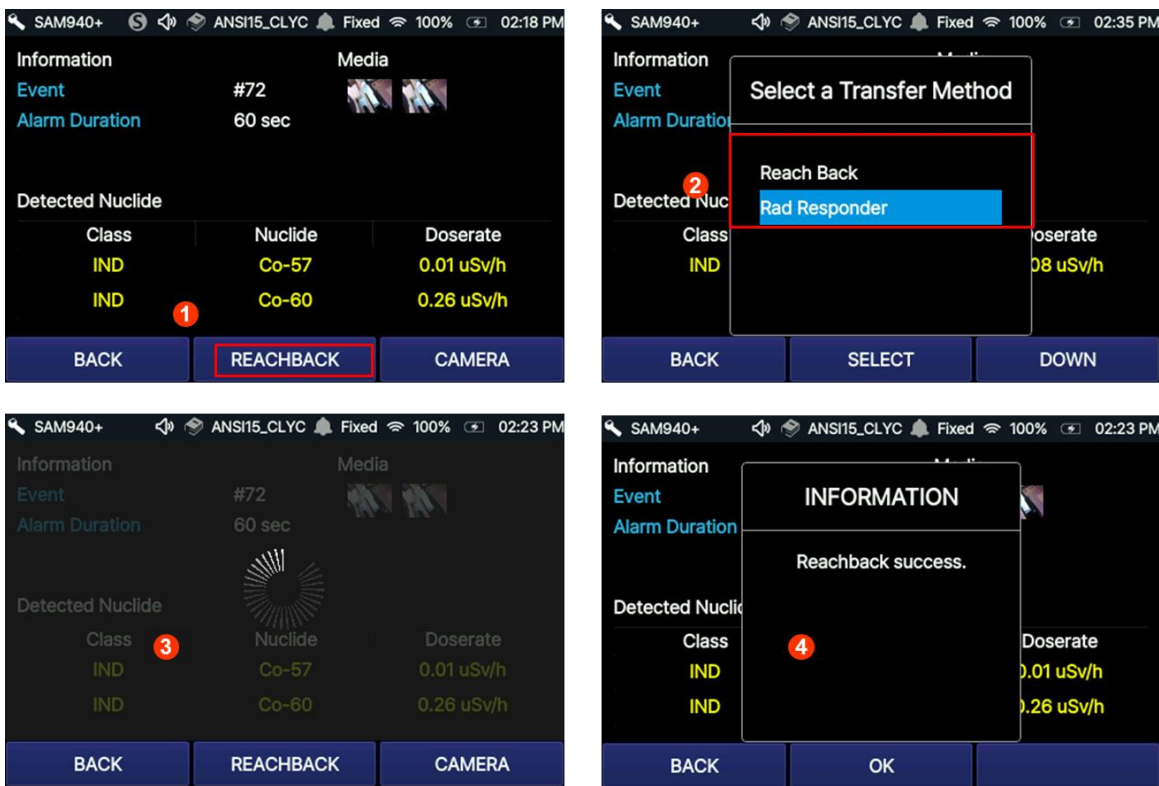


Figure 5.7.2. Transmit Event Report to RadResponder

 Before you can upload spectrum and event data to RadResponder, you must already have established a RadResponder account at www.RadResponder.net, and you must set up RadResponder Mode in PeakID IV.

5.8. ATTACH PHOTO IMMEDIATELY AFTER AN EVENT

Immediately upon completion of Manual Identification, the Event Report allows you to attach photos to the data that has been automatically saved in the Event file. (For instructions on adding media to a previously saved Event, refer to **Section 6.1. Use the Event Log to Access Event Data**).

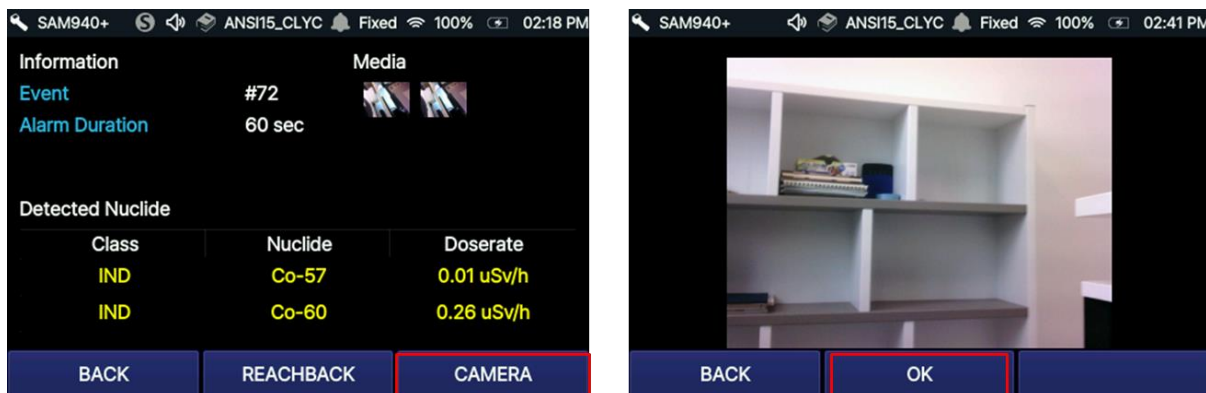


Figure 5.8.1. Attach photo to an event

5.9. FIND DIRECTIONAL INFORMATION OF ISOTOPES

The SAM 940+ provides directional information of isotope called **FINDER**. Combination of radial radiation activities and internal gyro sensor response, PeakAbout IV estimates the direction of gamma source. The **FINDER** screen can be accessed by selecting FINDER button from the main screens. The details of instruction of FINDER feature is described in following section.

5.9.1. Access to FINDERscreen

1. Access to **FINDER** screen by selecting **FINDER** button as shown in figure 5.8.1.

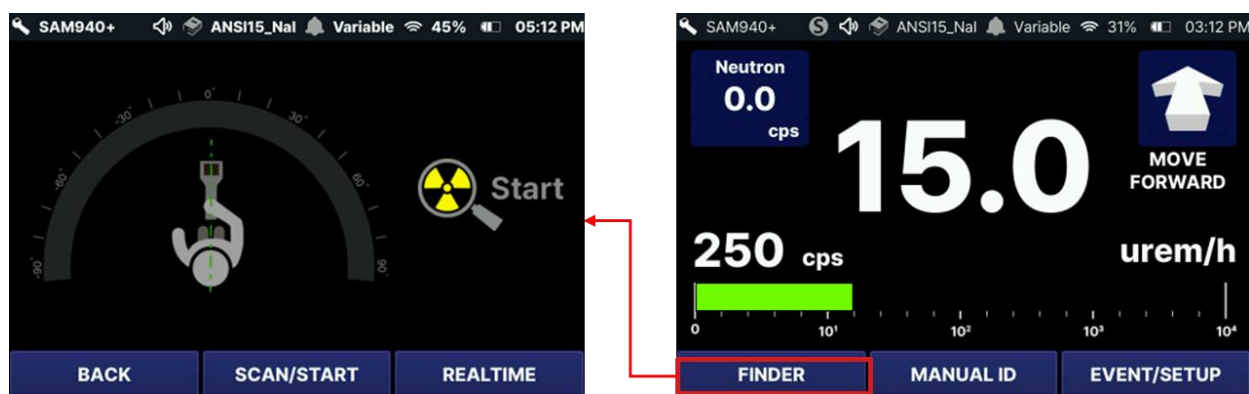


Figure 5.8.1. Access to **FINDER** screen

5.8.2. Start scan

1. Selecting SCAN START
2. Hold the SAM940+ firmly and position it in front of your body
3. Slowly sweep around the angle, with the maximum rotation from -90 degrees (counterclockwise) to 90 degrees (clockwise)



The current angle ①, count rate ② and rotation direction at the given direction will be displayed as shown in figure 5.8.2. To finish the scan, select SCAN STOP.

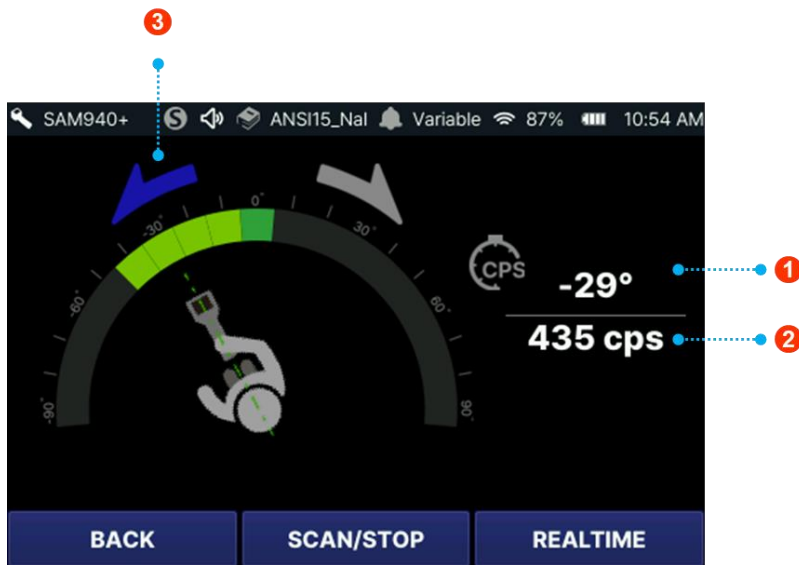


Figure 5.8.2. Scan progress status and count rate

5.8.3. FINDER report

Immediately upon completion of scan, the direction of isotope ① is displayed as an arrow as shown in figure 5.8.3. In addition to graphical indicator of the location, the range scan is shown in ②. The isotope location result is located at angle ③ and count rate ④.

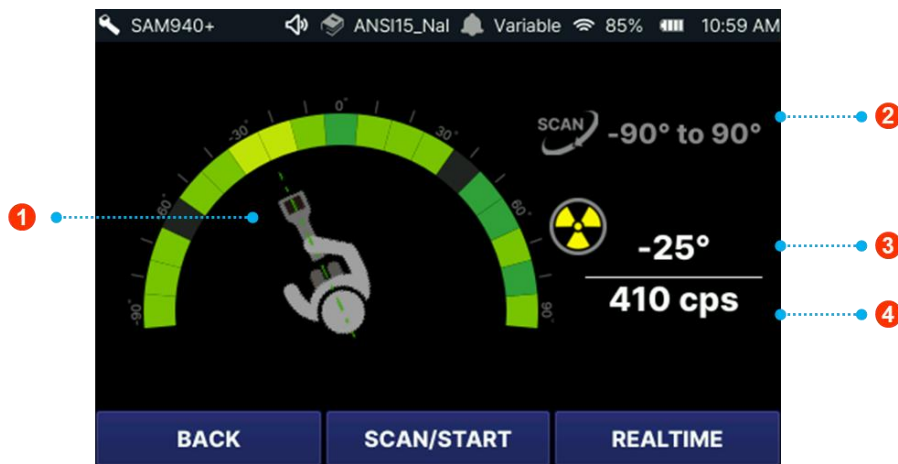


Figure 5.8.3. Scan result

5.8.4. Search fail cases

The SAM 940+ is unable to locate isotopes if the activity is too low, the scan is too fast, or the scan is conducted in normal background conditions. No source results are shown in example Figure 5.8.4



The activity of isotope should be greater or equal to double the background activity.

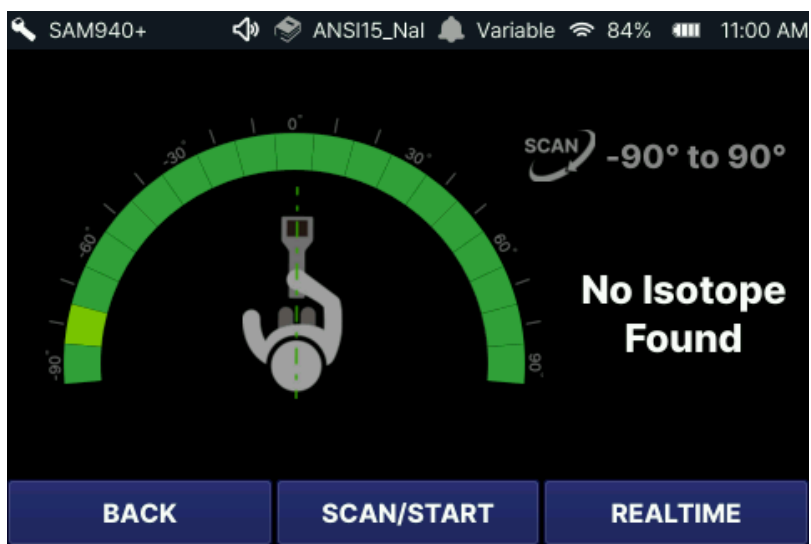


Figure 5.8.4. Search fail examples

5.10. FAVORITE EVENT

Favorite Event feature allows the user to tag an Event and find it later as a favorite.

5.10.1. Add Favorite Event

1. At the end of Manual ID, the ID report screen is showed up as shown in figure 5.10.1.
2. Select and hold the REACBACK button for 2 second.
3. The star icon is filled with blue color. And “add favorite” massage appear.

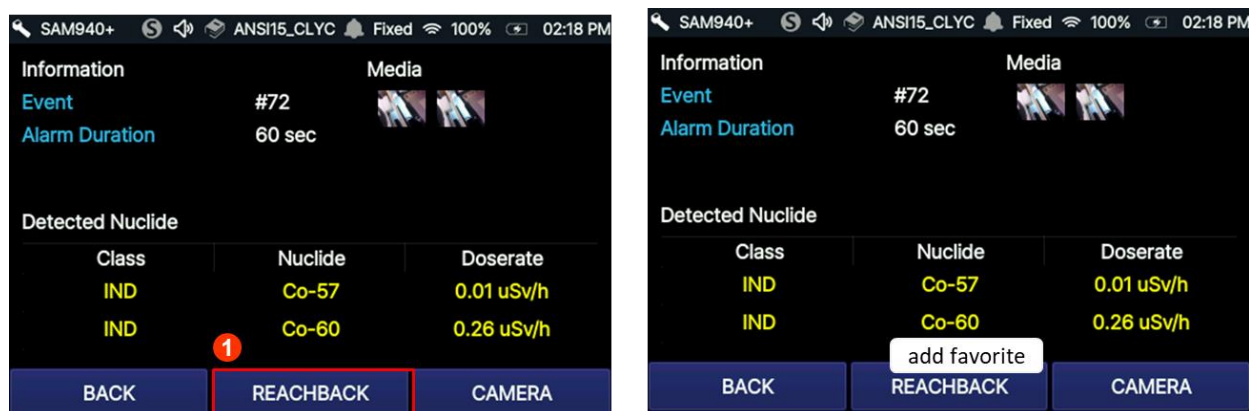


Figure 5.10.1. Add favorite example

5.10.1. Remove Favorite Event

1. Select and hold the REACBACK button for 2 second.
2. The star icon is turned to normal. And “remove favorite” massage appear.

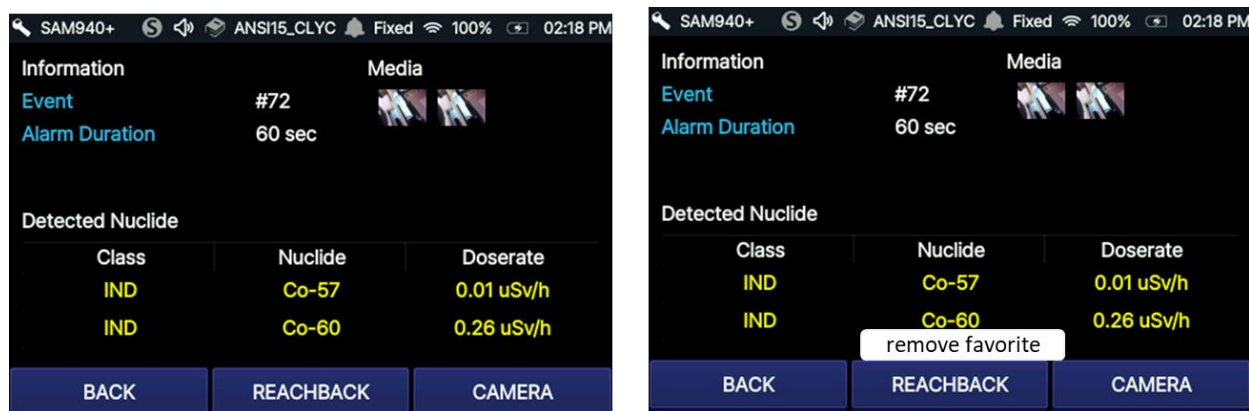


Figure 5.10.2. Remove favorite example



The favorite event is displayed in yellow color in the event log list as shown in figure 5.10.3.

No.	Date	Avg.Dose	Nuclide
171	24-09-02T15:28:06	0.13 uSv/h	Not Identify
170	24-08-29T17:23:19	1.25 uSv/h	Unknown
169	24-08-29T13:37:47	464.06 uSv/h	Not Identify
168	24-08-28T09:21:54	0.52 uSv/h	Cs-137
167	24-08-28T09:20:39	1.13 uSv/h	Cs-137
166	24-08-27T14:15:46	1.31 uSv/h	Cs-137
165	24-08-27T11:14:37	2.56 uSv/h	Unknown
164	24-08-27T10:57:14	0.93 uSv/h	Unknown
BACK		UP/OK	DOWN

Figure 5.10.3. Favorite event in yellow color in the log list

5.11. SLEEP MODE AND TURN OFF AFTER OPERATION

The SAM 940+ felt in “sleep mode” automatically when it is not in use for certain time (setup sleep mode time is defined in section 7.6). The sleep mode LED shown in figure 5.10.1. is turned on (yellow LED). It wakes up by pressing any functions keys.

It is recommended to turn off the SAM 940+ after operations are completed in order to reduce power drawn from the SAM 940+ battery.



Figure 5.11.1. Sleep mode LED and power switch

CHAPTER 6. OPERATING INSTRUCTIONS: EVENT LOG AND SETUP MENUS

6.1. USE THE EVENT LOG TO ACCESS EVENT DATA

When Manual ID is performed, PeakAbout IV automatically generates and saves an Event file that contains the results. Just as with a live Event Report, opening a saved Event allows you to analyze the saved spectrum; attach photos and transmit Event data.

Saved event file can be assessed by selecting **EVENT LOG** as shown in figure 6.1.1.



Event data can be downloaded by connecting the SAM 940+ to a PC that has PeakID IV installed on it (see Chapter 7.3.2. Connecting the SAM 940+ to a PC). Once the event data has been downloaded to the PeakID IV database, it is automatically deleted from the SAM 940+.

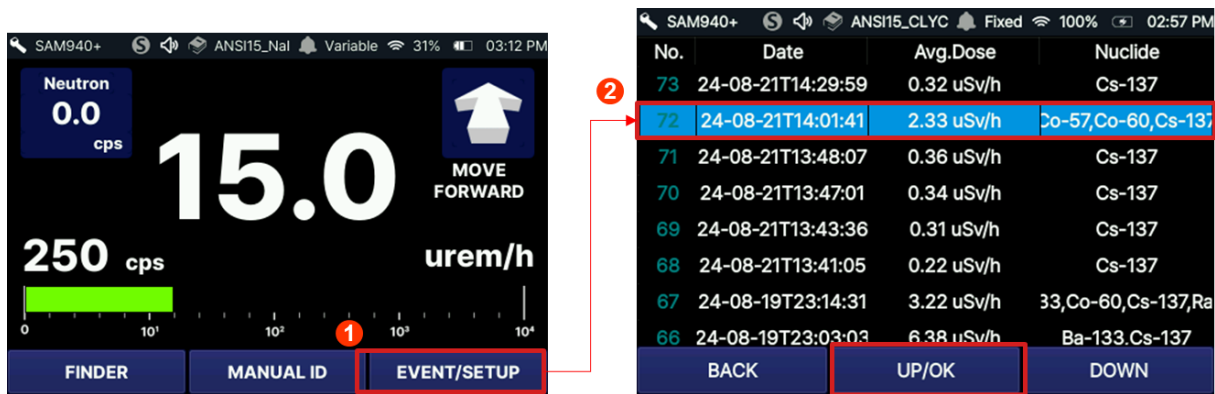


Figure 6.1.1. Access to event log

Once event is selected from the Event Log list and select OK button, the screen is changed to EVENT ID screen ② or toggled to EVENT INFO ③. The EVENT ID displays spectrum information along with detected nuclides, count rate. In this screen, reach back service is available. Another screen, EVENT INFO, ③ contains user information, location and average dose rate and maximum dose rate, and photo. Camera option is available from this screen.

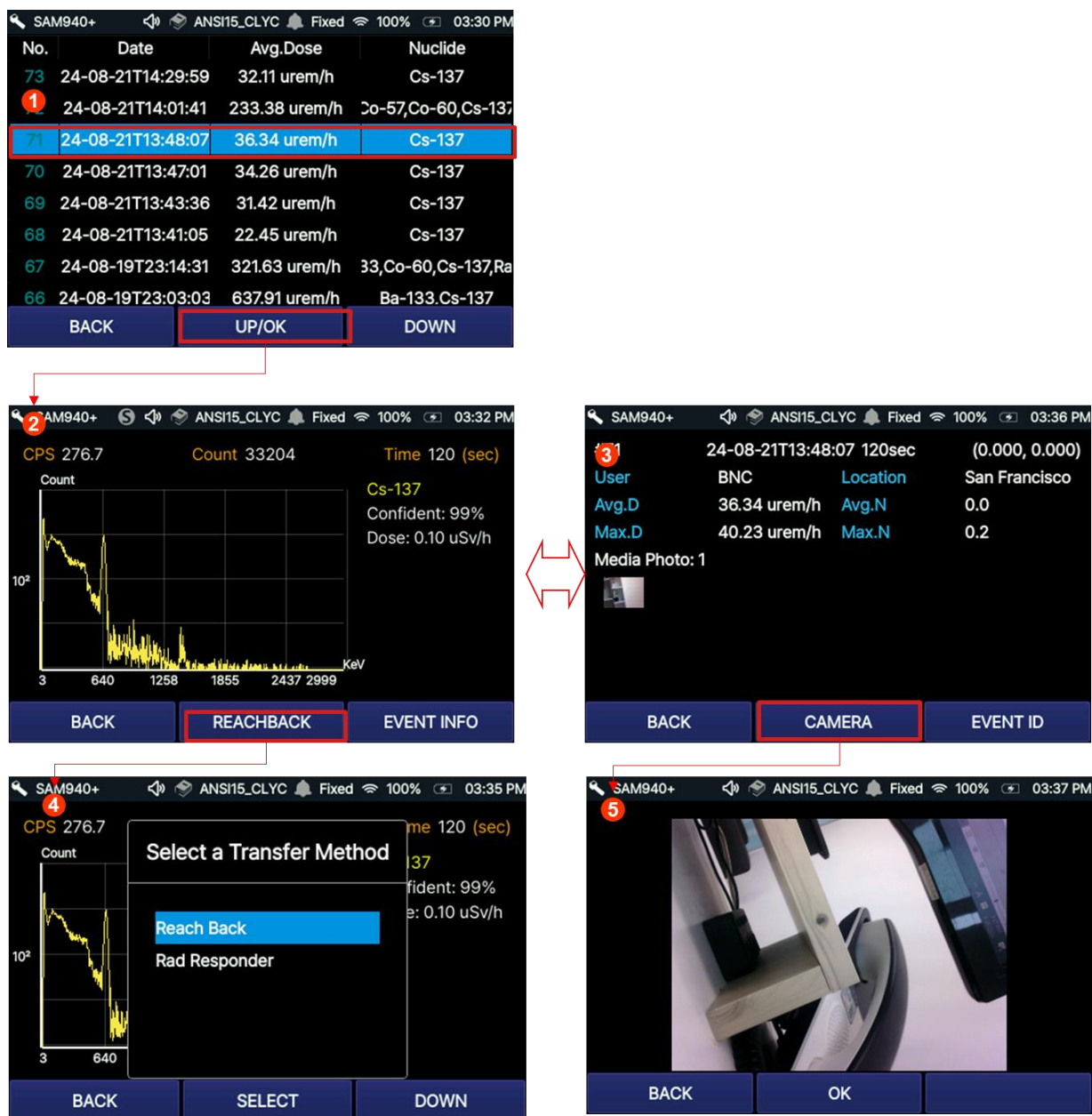


Figure 6.1.2. EVENT ID and EVENT INFO screen

6.2. USE THE EVENT LOG TO TRANSMIT EVENT DATA TO REACH BACK

Reach-back transmission can also be initiated from the Event Report. Instruction for configuring necessary settings are described in section 7.6.1.5.

6.2.1. Transmit event to reach-back

1. To open the Event Log, select **EVENT LOG** from the Main Screen (Figure 6.1.1.):
2. To locate an Event, scroll up/down using the function button and select desired event by press OK button (hold for 2 seconds). Refer to figure 6.1.2.
3. To transmit the Event, select REACHBACK button.
4. Select ReachBack

6.2.2. Transmit event to RadResponder

1. To open the Event Log, select **EVENT LOG** from the Main Screen (Figure 6.1.1.):
2. To locate an Event, scroll up/down using the function button and select desired event by press OK button (hold for 2 seconds). Refer to figure 6.1.2.
5. To transmit the Event, select REACHBACK button.
6. Select RadResponder.

6.3. USE THE SETUP MENU FOR USER SETTING

Setup Menus are intentionally hidden from the main tasks such as Radiation Monitoring and Manual ID to maximize ease-of-use during operation. This section discusses how to access and use these features.

For the convenience of user, PeakAbout IV allows the operator permission to access basic setup called User setting and some of advanced features which requires administrator login.



All the setup parameters not listed in PeakAbout IV is managed by PeakID IV application software. (Refer to **7.9. Configuration Menu GUI**).

SETUP screen of PeakAbout IV is shown in figure 6.3.1. and the features manageable by PeakAbout IV is summarized in table below.

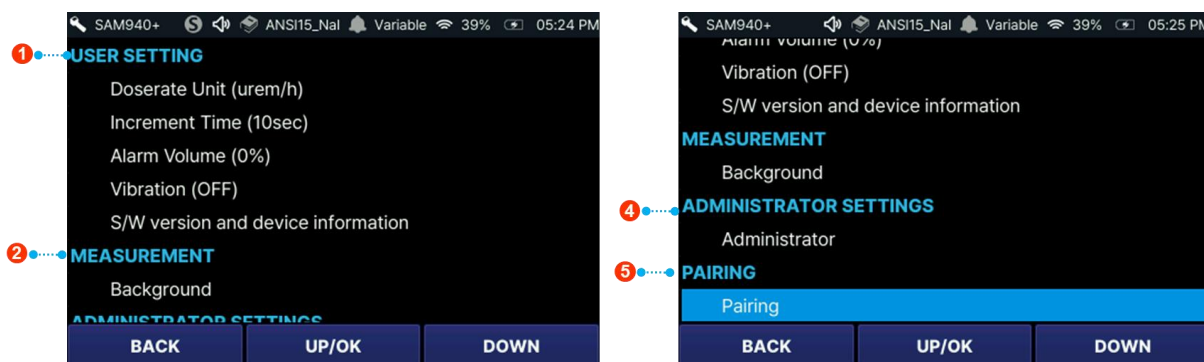


Figure6.3.1. SETUP screen

1	USER setting	Set increment time, alarm volume, S/W Update
---	--------------	--

2	Measurement	Perform Background measurement
3	Administrator	Set display option, password, calibration measurement and restore factory HW configuration.
4	Paring	WiFi connection to Windows program in a PC (PeakID SW)

6.3.1. User setting

6.3.1.1. Doserate Unit:

User can define doserate unit either urem/h or uSv/h as shown in figure 6.3.2.

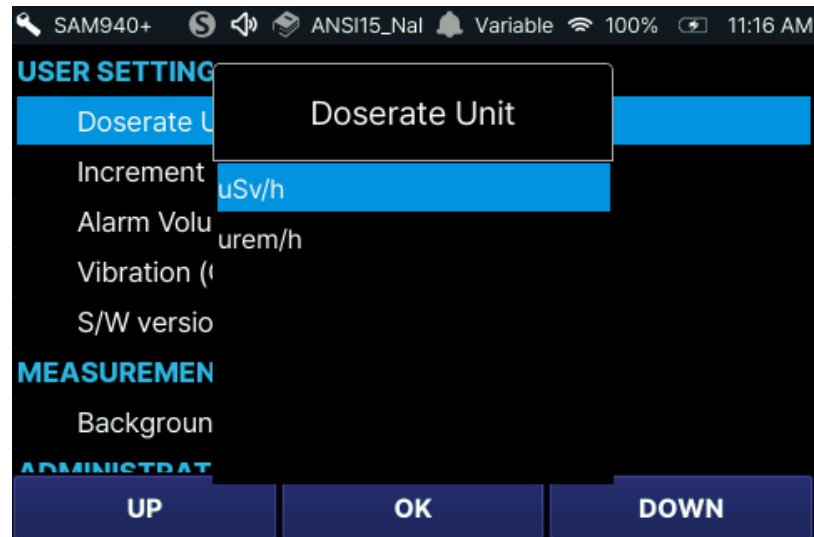


Figure 6.3.2. Increment time setup

6.3.1.2. Increment Time:

User can define increment time of measurement for Manual ID and Background measurement as shown in figure 6.3.3.

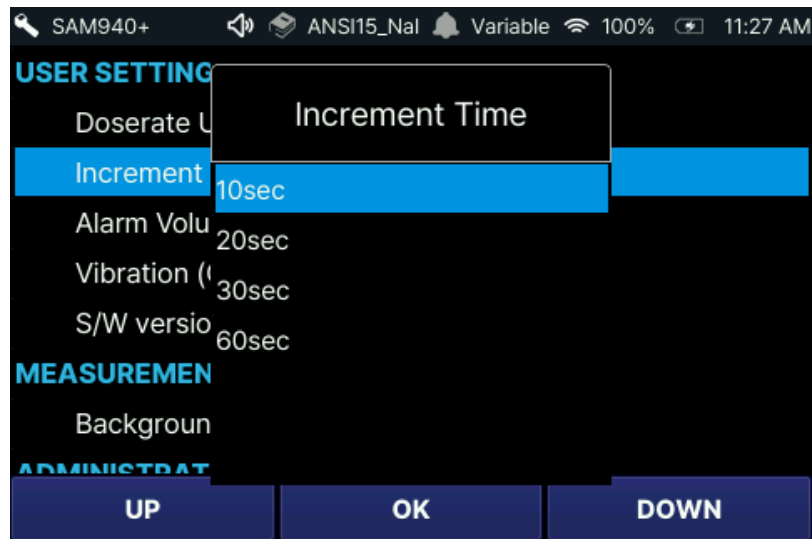


Figure 6.3.3. Increment time setup

6.3.1.3. Alarm volume

User can define alarm volume. 0 volume represents mute mode.



Alarm ringtone can be selected by PeakID IV application software.

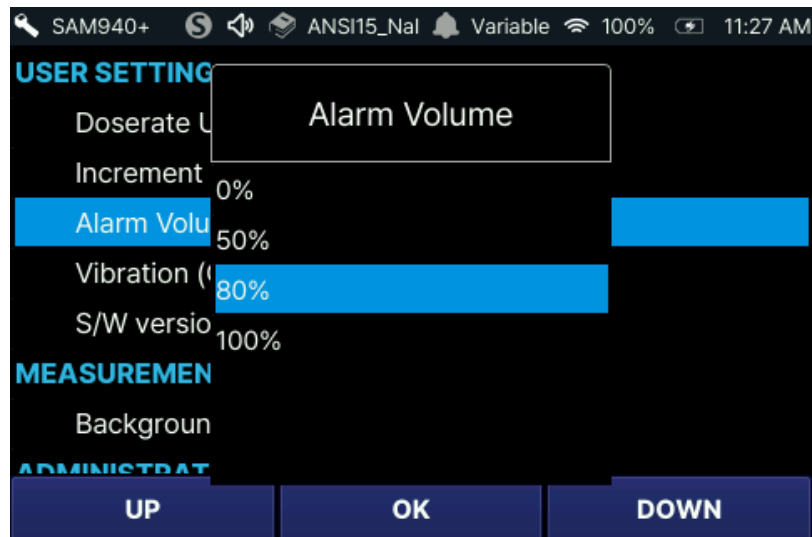


Figure 6.3.4. Alarm volume setup

6.3.1.4. Vibration

User can select vibration mode On/Off.



Figure 6.3.5. Vibration On/Off

6.3.1.5. SW version and device information:

The User Settings menu includes a Device Information sub-menu which contains detailed reference about your SAM 940+ and PeakAbout IV upgrade. Instructions for upgrading software are described in **Section 4.5: Upgrading PeakAbout IV**.

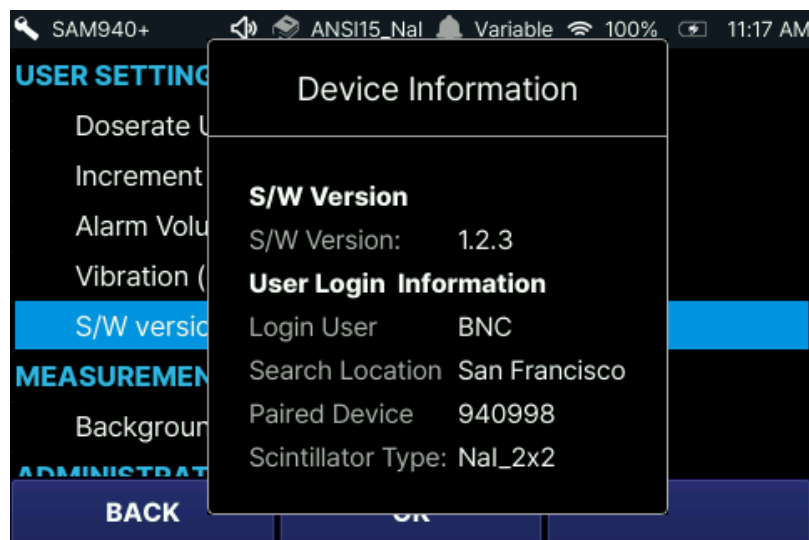


Figure 6.3.6. SW version and device information

6.3.2. Measurement

6.3.2.1. Background:

For instructions, refer to **Section 4.3.3: Perform Background Measurement** (Figure 6.3.6).



Figure 6.3.6. Background measurement

6.3.3. Access to Administrator Setup

You may access some of advanced features with Administrator privilege. Password is required to access this feature.



Some configurations are only accessed with administrator's privilege. The administrator's password set in factory is '1234'. Please change it more secured combination when you received the unit. The password may be changed in Administrator/New Password (refer to 7.9.2. **Administrator Setup**).

Type in password by using function button and select ENTER as shown in figure 6.3.7.

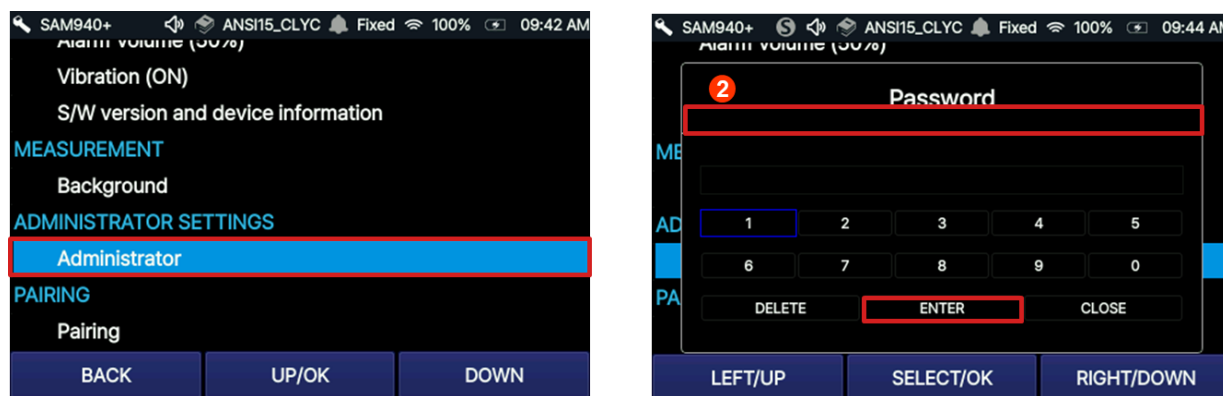


Figure 6.3.7. Administrative access

6.3.4. Administrator Setup Parameters

Administrator can manage some of configuration parameters listed in figure 6.3.8. They are categorized into spectrum, event and configuration management.

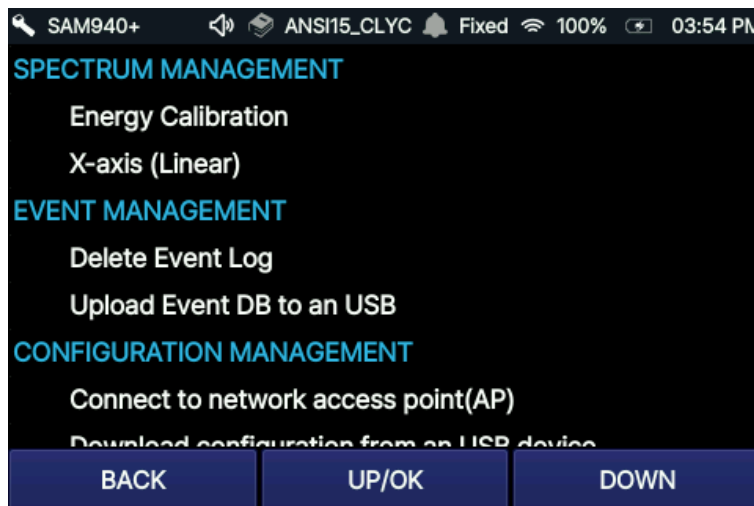


Figure 6.3.8. Administrative setup parameters

6.3.4.1. Spectrum management: Energy Calibration

While we recommend performing manual energy calibration to ensure optimal results, most normal operations can be conducted without this step. If a Cs-137 source is not available, skip this section.

1. Place a Cs-137 source in front of the detector and Initiate **Energy Calibration**
2. Remove the Cs-137 source and return to the Main screen.



Ideally, the Cs-137 source is active enough to generate approximately 2500 counts per second (cps). If the count rate is too high, increase the distance between the source and the detector unit. Decrease the distance if the count rate is low.



We recommend that total integrated counts in the manual calibration spectrum be no less than 100,000. The number of counts is defined in the Administrator Settings menu. Refer to Section 4.4.6.6 **Calibration Measurement Count**



If necessary, calibration count can be adjusted using SET COUNT function as shown in figure 6.3.9.

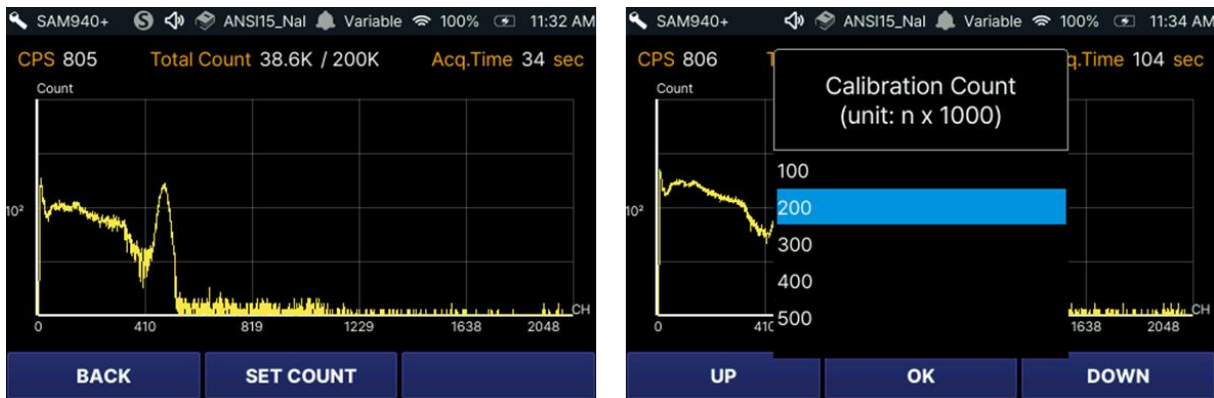


Figure 6.3.9. Adjust measurement counts

6.3.4.2. Spectrum Management: Spectrum display

SAM940+ provide a special display called QCC (Quadratic Compression Conversion) along with a typical linear scale display. QCC is one of adaptive display method that stretching/compressing vertical values such that compressing high energy while stretching low energy regions.

Linear and QCC display can be toggled by selecting X-axis display option as shown in figure 6.3.10.



Figure 6.3.10. Linear and QCC spectrum display

6.3.4.3. Event Management: Delete Event Log

All measured events in DB can be deleted by selecting this option.

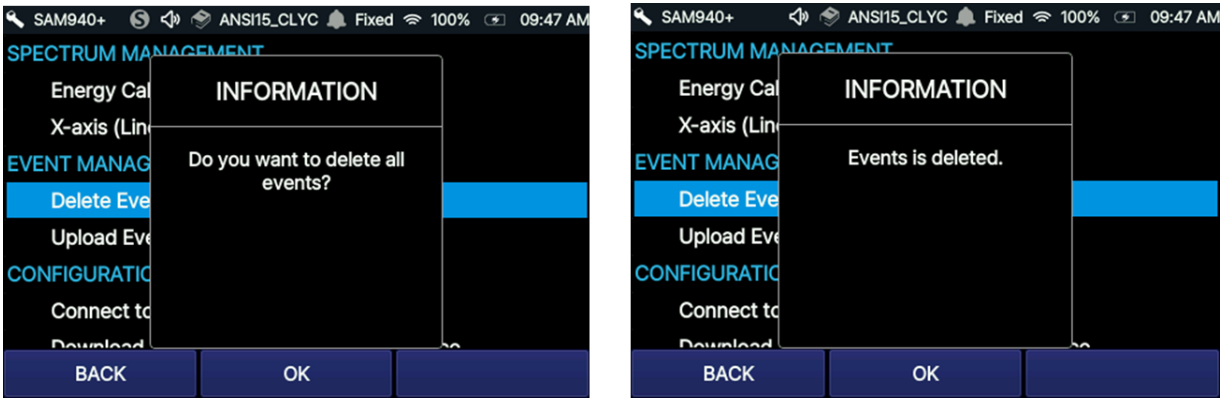


Figure 6.3.11. Delete Event Log

6.3.4.4. Event Management: Upload event DB to an USB device

User can upload Event DB in a SAM 940+ unit to an USB and upload it to a PC. Insert an USB and select 'Upload Event DB to an USB'. The Event DB can be analyzed by PeakID IV software.

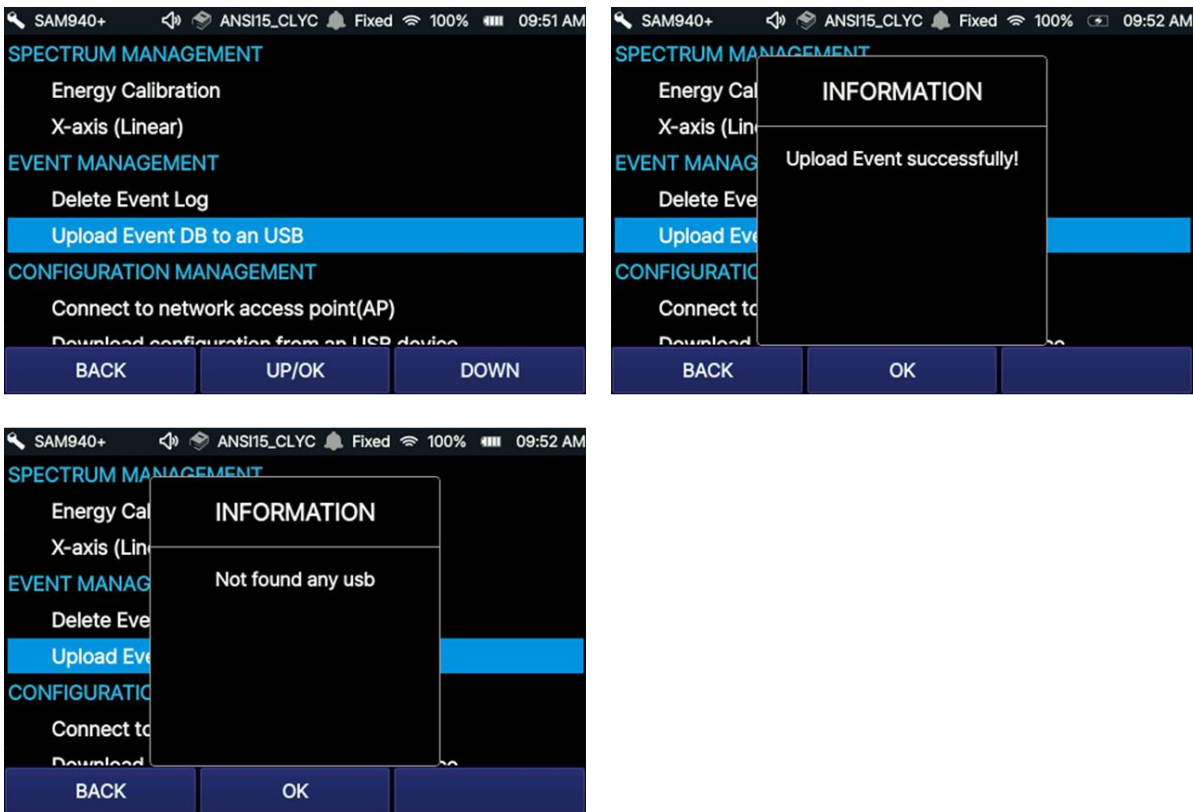


Figure 6.3.12. Upload Event DB to an USB device

In case USB is not found, please unplug the OTG cable and plug it back in.

6.3.4.5. Configuration Management: AP setup

In order to communicate to a PC and to enable Reach-back and RadResponder services, the SAM 940+ unit has to be connected appropriate network via WiFi connection. Following is procedure to setup network AP address.

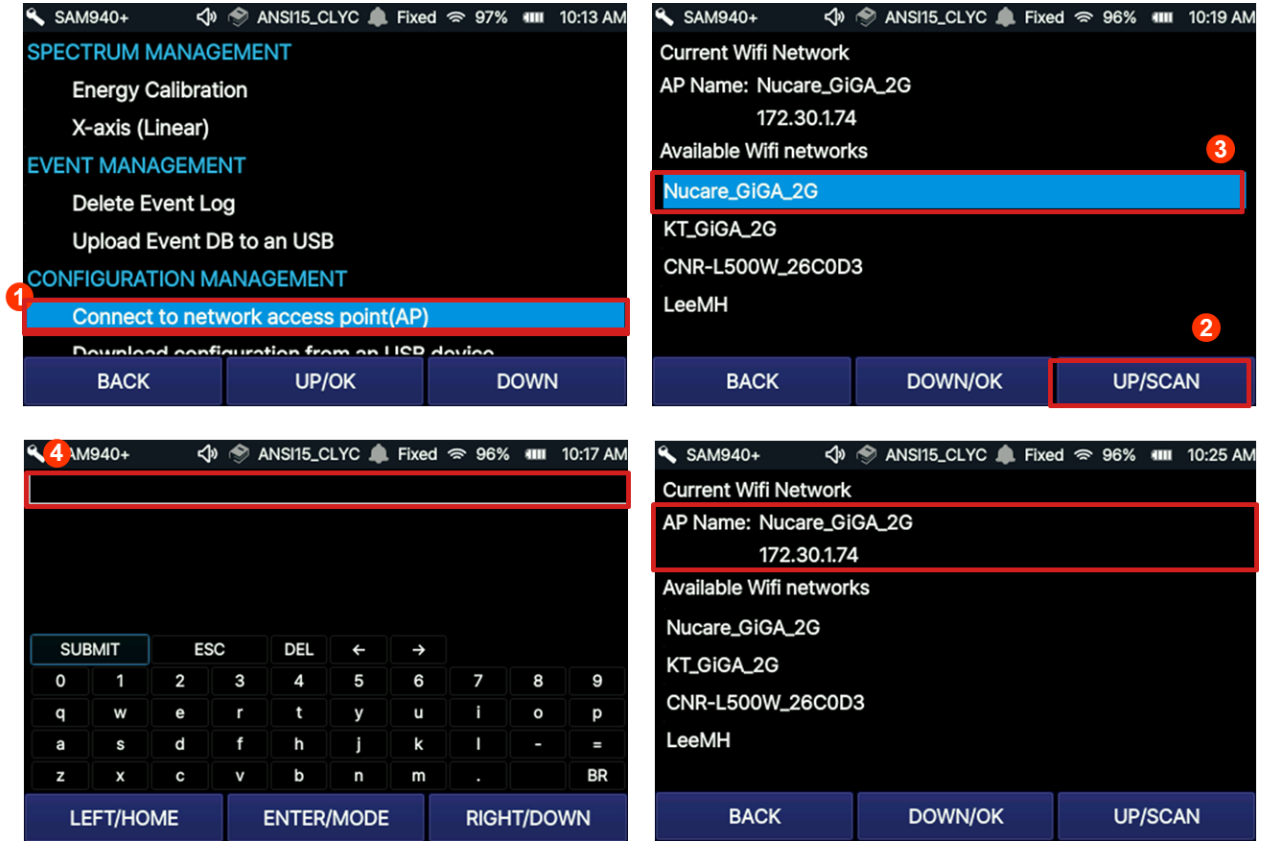


Figure 6.3.13. Setup AP network address



Unless switching to different WiFi network, the AP address setup is stored in memory of the unit and automatically connected to the defined WiFi network without this procedure.

6.3.4.6. Configuration Management: Download configuration from an USB device

The configuration parameters saved in PeakID IV in a PC can be downloaded from an USB. Insert an USB and select 'Download configuration from an USB device'.

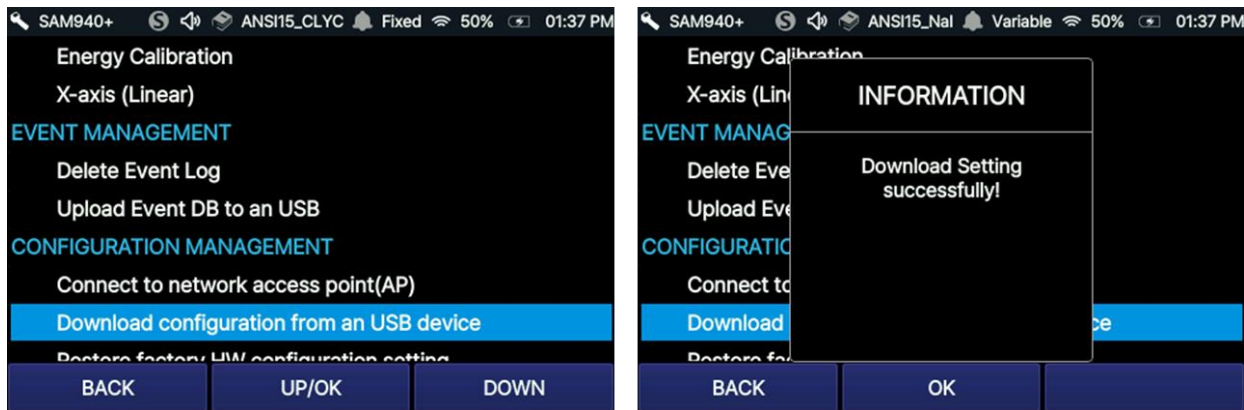


Figure 6.3.14. Download configuration from an USB device



In case USB is not found, please unplug the OTG cable and plug it back in.

6.3.4.7. Configuration Management: Restore factory HW configuration setting

The Restore Factory HW Configuration setting is used to reset parameters like calibration and high voltage bias for the installed detector among other settings. Situations in which this operation is useful are extremely rare, and the procedure should only be performed by a factory technician. Contact Nucare before performing this operation.

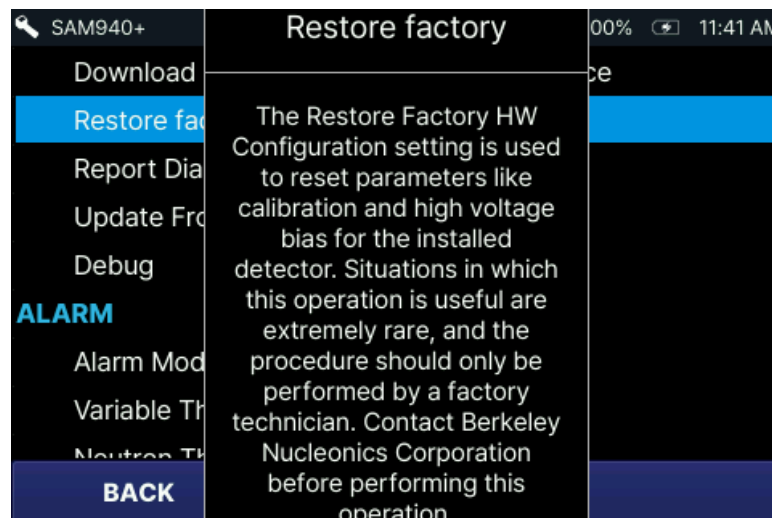


Figure 6.3.15. Restore factory HW configuration setting

6.3.4.8. Configuration Management: Report diagnostic DB

In case of scheduled check-up, malfunctioning unit and/or prevention of failures, a user can send diagnostic information to manufacturer. By selecting 'Report Diagnostic DB' necessary information is automatically transferred to a technician in service department of manufacturer.

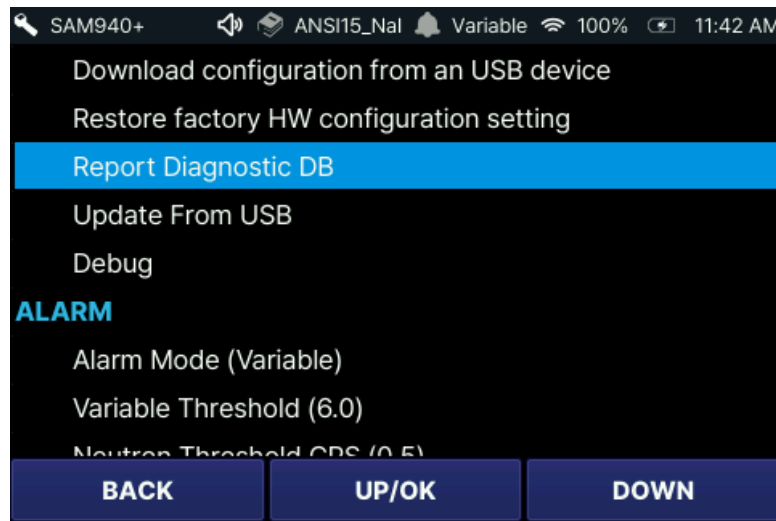


Figure 6.3.16. Report Diagnostic DB

6.3.4.9. Update From USB

Instructions for Updating the Device as Described Below:

1. Visit the website <https://bncsam.com/RESAM940PLUS.html> and download the PeakAbout IV software.
2. Copy the downloaded file to a USB memory stick.
3. Insert the USB memory stick into the device.
4. Access the Update from USB function as shown below.
5. The update process will begin automatically and will take approximately 5-10 minutes to complete.
6. After the update is finished, the device will automatically restart.

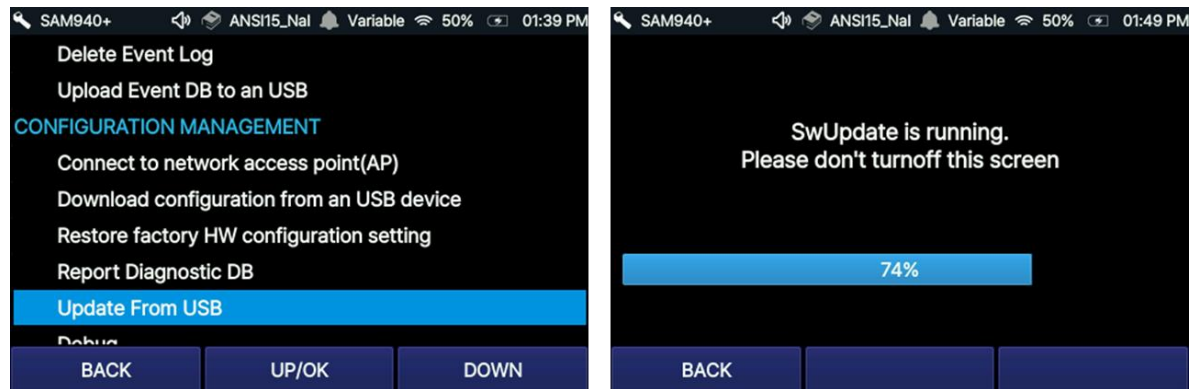


Figure 6.3.17. Update SW process

In case USB is not found, please unplug the OTG cable and plug it back in.

6.3.4.10. Fixed Mode

Fixed Alarm Mode is a basic method that assumes the background environment exhibits a very small amount of fluctuation. The alarm is triggered when the count rate exceeds the operator-determined threshold value. This method can cause false alarms if there is substantial background fluctuation due to Naturally Occurring Radioactive Material (NORM), as is often the case when the system is moving across a large area while actively monitoring.

The default threshold value for Fixed Alarm Mode is 1000 counts per second (cps), which is optimal for many applications. A typical background environment produces around 200 cps for a 3 x 3-inch NaI detector, or lower if a high-resolution detector option was chosen. A threshold value lower than 1000 is usually unnecessary and is more likely to generate false alarms. In situations where the background environment is known to be above normal, it may be necessary to select a higher value such 2000 or 3000 cps to minimize false alarms.

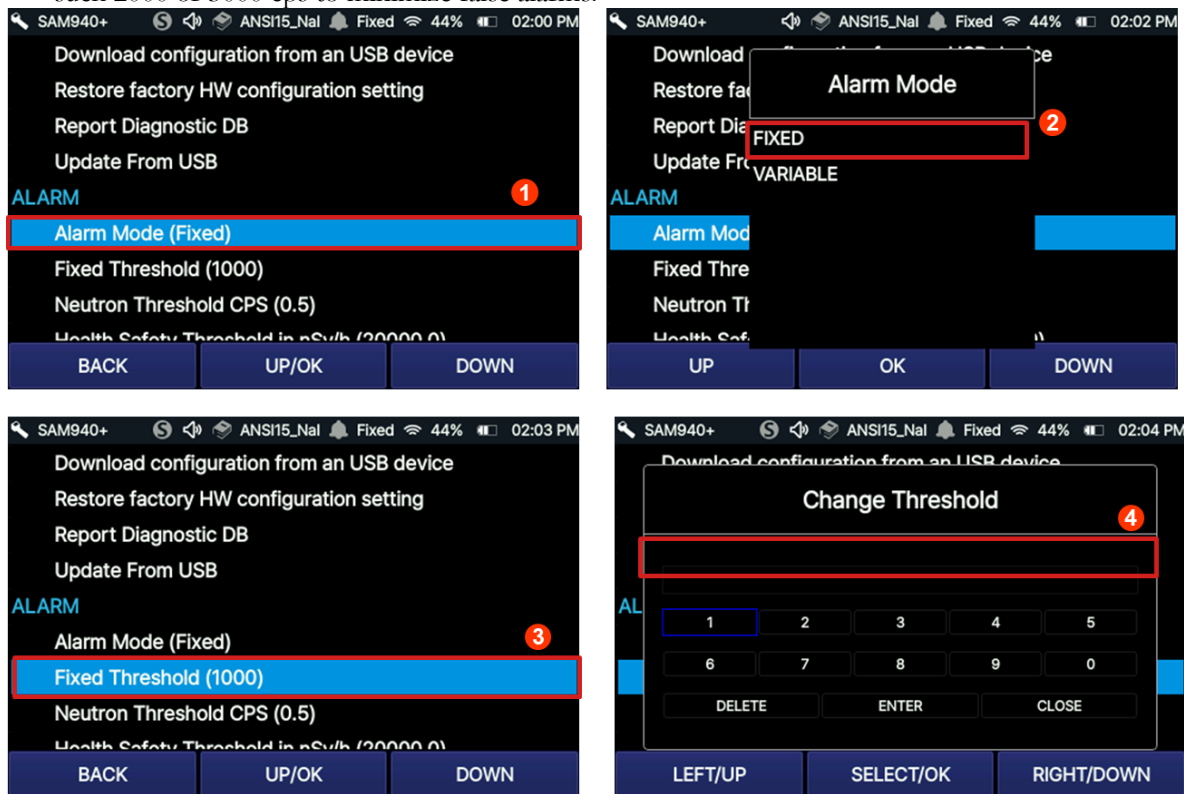


Figure 6.3.18. Fixed Alarm Mode

6.3.4.11. Variable Mode Threshold Sigma

Variable Alarm Mode is a more advanced method that continuously samples the previous 10 seconds' worth of background fluctuation and calculates the standard deviation above background (σ , or sigma) of the fluctuation. Using this method prevents alarms due to changing background (and eliminates the need to perform new background measurements). Instead, alarms are triggered when real gamma peaks exceed the standard deviation of the dynamic threshold value. We recommend using Variable Alarm Mode when moving across a large area that may be subject to changes in background.

The default threshold value for Variable Alarm Mode is 4 (Sigma) which is optimal for many applications.

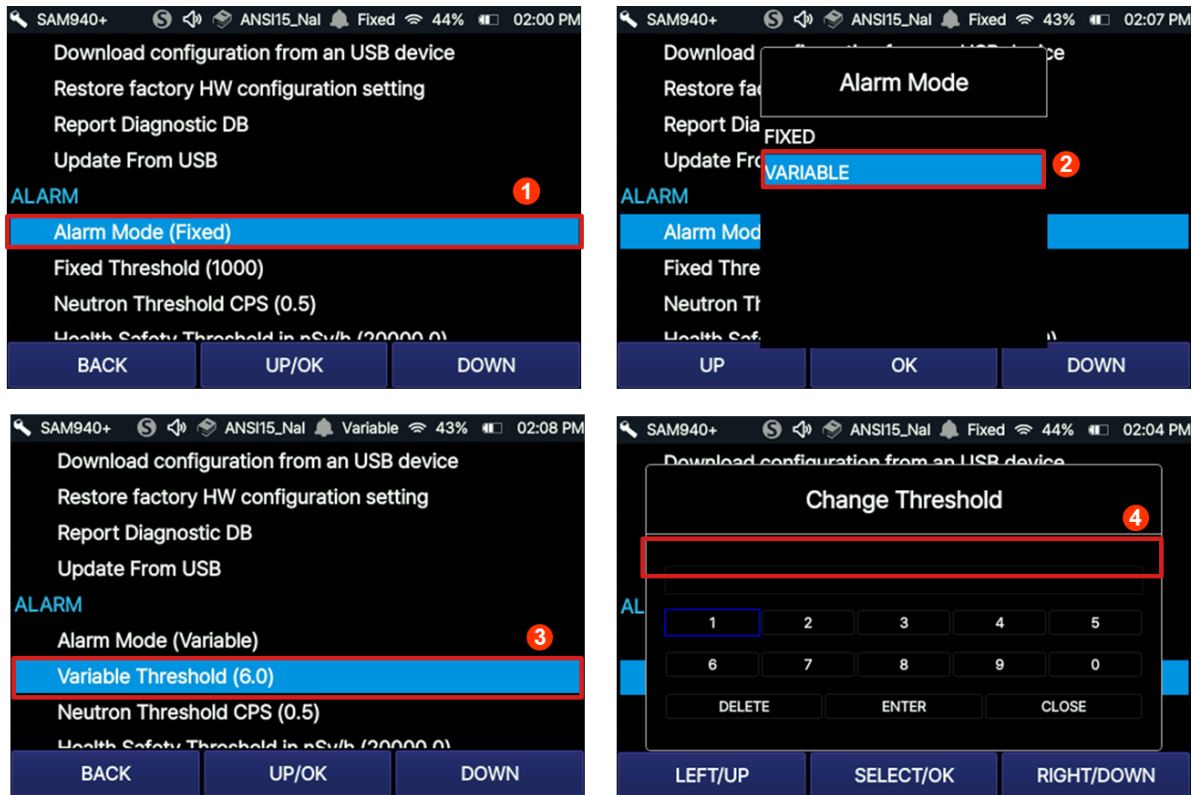


Figure 6.3.19. Variable Alarm Mode

6.3.4.12 Neutron Threshold CPS

If your SAM 940+ includes the neutron detection option, then it contains either solid-state sensor arrays on a ^6Li substrate (Domino) or a tube filled with ^3He gas, integrated with moderator packaging inside the detector unit. PeakAbout III will display neutron activity on the Main screen in counts per second (cps).

The default value for Neutron Threshold is 0.8 CPS which is optimal for most applications.

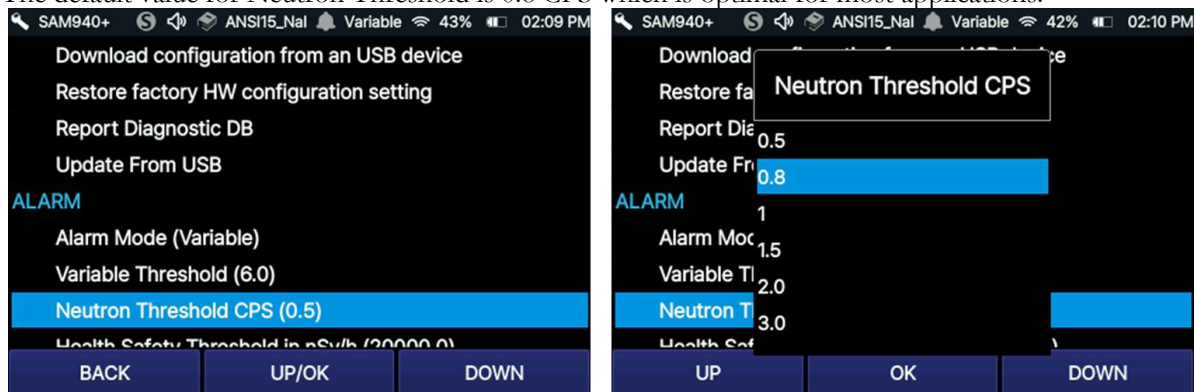


Figure 6.3.20. Neutron Threshold

6.3.4.13 Health Safety Threshold

For the user's safety, PeakAbout III features a separate Health Safety Alarm with a different ringtone and visual signal. When the safety threshold is exceeded, the distinctive Health Safety ringtone sounds, and the Operating Message displays "DANGER". The Health Safety Alarm takes precedence over all other alarm conditions.

The default value for Health Safety Threshold is 10,000 $\mu\text{rem/h}$ (10 mrem/h).

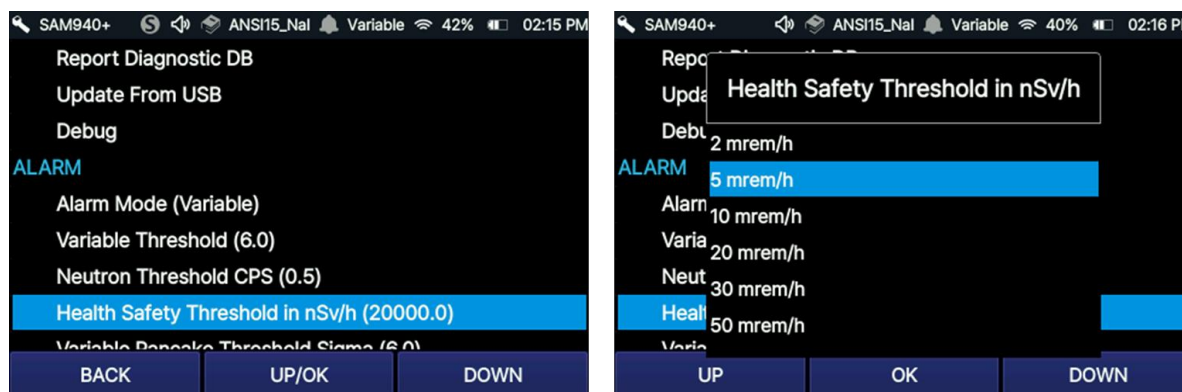


Figure 6.3.21. Health Safety Threshold

6.3.4.14 Variable Pancake Threshold Sigma

SAM 940+ also support connecting to another external sensor such as Pancake. The alarms for Pancake are triggered when real gamma peaks exceed the standard deviation of the dynamic threshold value.

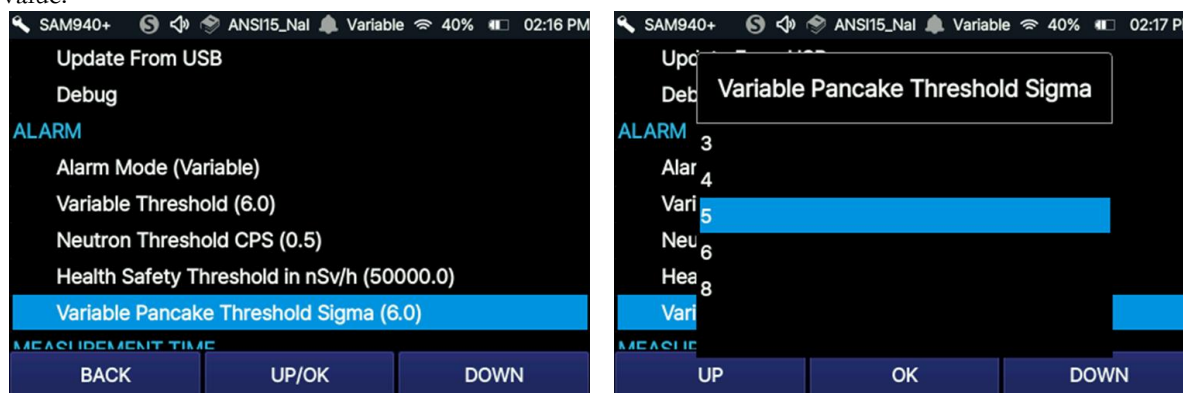


Figure 6.3.22. Variable Pancake Threshold Sigma

6.3.4.15 Measurement ID Time

To measure and record an event, Manual Isotope Identification must be performed. Manual ID can be performed whether the SAM 940+ is in an alarm condition or not.

The default Manual ID Measurement Time is 60 seconds.

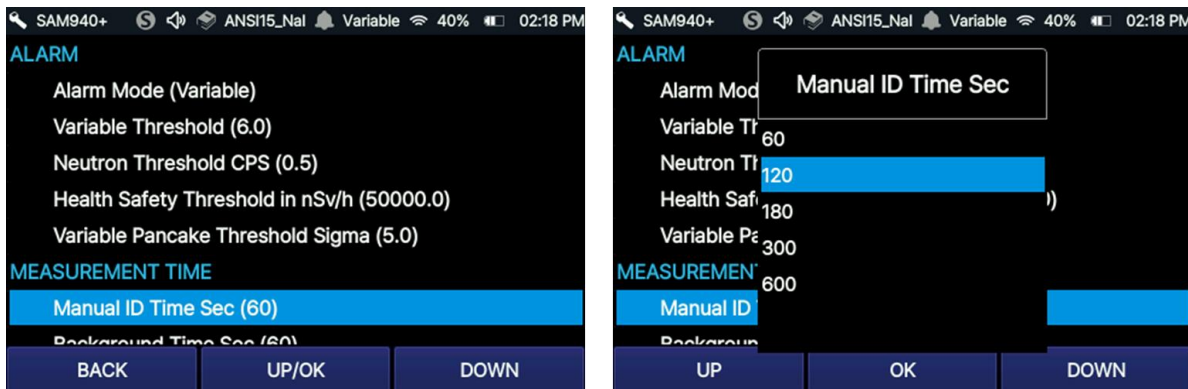


Figure 6.3.23. Manual ID Time Setting

6.3.4.16 Background Time

Reliable background information is critical to good nuclide identification and alarm performance. PeakAbout IV uses an advanced NORM rejection algorithm to minimize false alarms due to fluctuations of Naturally Occurring Radioactive Material in the background.

The default Background Measurement Time is 60 seconds.

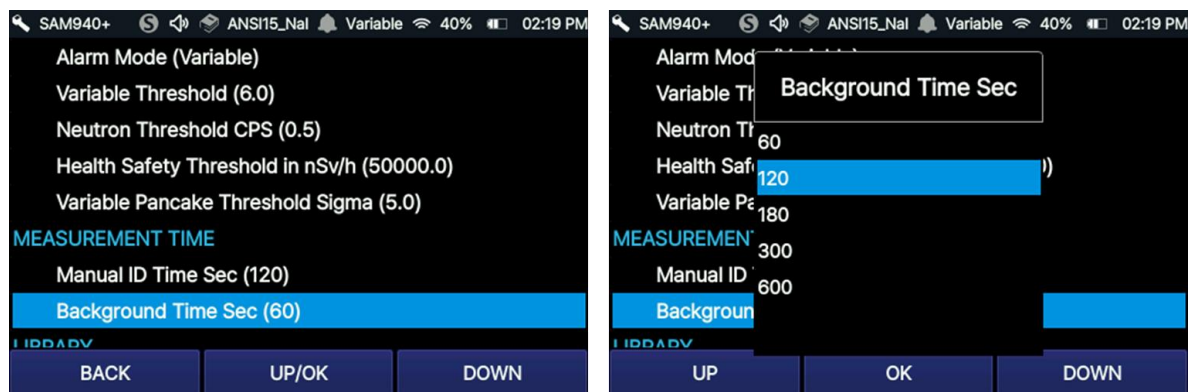


Figure 6.3.24. Background Time Setting

6.3.4.17 Select Library

The Library List in PeakAbout IV for SAM 940+ contains four libraries which are optimized for each detector: sodium iodide (NaI), cerium bromide (CeBr), CLLBC, CLYC detector.

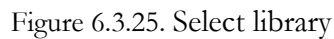
The default library is ANSI_NaI (ANSI_CeBr if your SAM940+ has a high-resolution detector).

- $$^{241}\text{Am}, ^{133}\text{Ba}, ^{57}\text{Co}, ^{60}\text{Co}, ^{137}\text{Cs}, ^{152}\text{Eu} \text{ (not ANSI required)}, ^{18}\text{F}, ^{67}\text{Ga}, ^{123}\text{I}, ^{125}\text{I}, ^{131}\text{I}, ^{111}\text{In}, ^{192}\text{Ir}, ^{40}\text{K}, ^{237}\text{Np},$$
- $$^{237}\text{Np} \text{ HE}, ^{239}\text{Pu}, ^{226}\text{Ra}, ^{99\text{m}}\text{Tc}, ^{232}\text{Th}, ^{201}\text{Tl}, ^{233}\text{U}, ^{233}\text{U} \text{ HE}, ^{235}\text{U}, ^{235}\text{U} \text{ HE}, ^{238}\text{U}$$

- $^{241}\text{Am}, ^{109}\text{Cd}, ^{57}\text{Co}, ^{60}\text{Co}, ^{134}\text{Cs}, ^{137}\text{Cs}, ^{152}\text{Eu}, ^{111}\text{In}, ^{192}\text{Ir}, ^{40}\text{K}, ^{85}\text{Kr}, ^{54}\text{Mn}, ^{22}\text{Na}, ^{204}\text{Tl}$

- $$^{77}\text{Br}, ^{58}\text{Co}, ^{60}\text{Co}, ^{67}\text{Co}, ^{51}\text{Cr}, ^{137}\text{Cs}, ^{18}\text{F}, ^{67}\text{Ga}, ^{124}\text{I}, ^{131}\text{I}, ^{192}\text{Ir}, ^{40}\text{K}, ^{99}\text{Mo}, ^{24}\text{Na}, ^{81}\text{Rb}, ^{82}\text{Rb}, ^{83}\text{Rb}, ^{124}\text{Sb}, ^{46}\text{Sc},$$
- $$^{75}\text{Se}, ^{153}\text{Sm}, ^{85}\text{Sr}, ^{89}\text{Sr}, ^{99\text{m}}\text{Tc}, ^{201}\text{Tl}, ^{133}\text{Xe}, ^{88}\text{Y}, ^{90}\text{Y}, ^{88}\text{Zr}, ^{89}\text{Zr}$$

- $$^{241}\text{Am}, ^{133}\text{Ba}, ^{57}\text{Co}, ^{60}\text{Co}, ^{137}\text{Cs}, ^{67}\text{Ga}, ^{131}\text{I}, ^{192}\text{Ir}, ^{40}\text{K}, ^{239}\text{Pu}, ^{226}\text{Ra}, ^{99\text{m}}\text{Tc}, ^{232}\text{Th}, ^{201}\text{Tl}, ^{235}\text{U}, ^{238}\text{U}, \text{DU}, \\ \text{HEU}, \text{RGPu}, \text{WGPu}$$



There are seven alarm ringtones to choose from: Danger, Siren, Morning call, Bell, Trumpet, Telephone call, and Beep.

Setup

ANSI15_NAI

Select Library (ANSI15_NAI)

ALARM SOUND

Alarm Ringtone (Danger)

SLEEP MODE

Sleep Mode Time Sec (300)

Sleep Mode (Normal)

GYROSENSOR CALIBRATION

Locator Offset Calibrate

CLOCK SETTING

Time Zone (America/Los_Angeles)

Setting Time (Wed Nov 01 19:29:56 PDT 2023)

Ringtone

Danger

Siren

Morning Call

Bell

Trumpet

Call

Beep

7:30 PM

Figure 6.3.26. Select Alarm Ringtone

6.3.4.19 Sleep Mode Time

To save power consumption and maximize operation time, the SAM 940+ falls in sleep mode when it is not used for a certain time.

The default sleep mode time is 300 seconds.

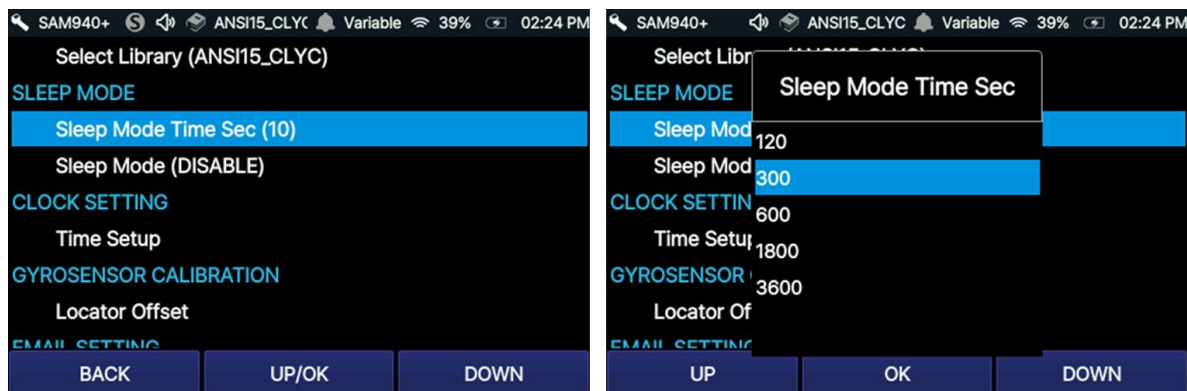


Figure 6.3.27. Setting Sleep Mode Time

6.3.4.20 Sleep Mode

PeakAbout provides two types of sleep mode:

- Enabled: Sleep mode will turn off the screen and disconnect from Wi-Fi
- Disabled: No Sleep mode is applied.

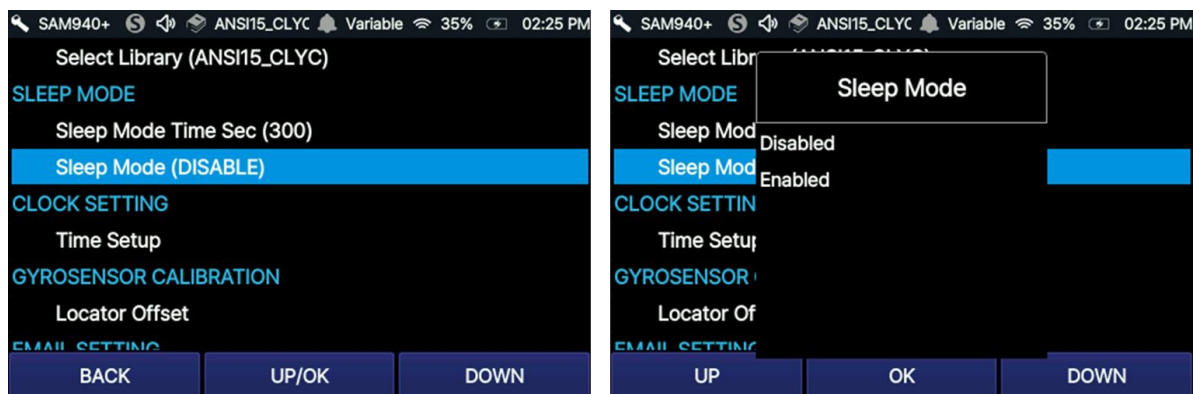


Figure 6.3.28. Sleep Mode Selection

6.3.4.21 Setting Time

User can setup time if time in system is not correct.

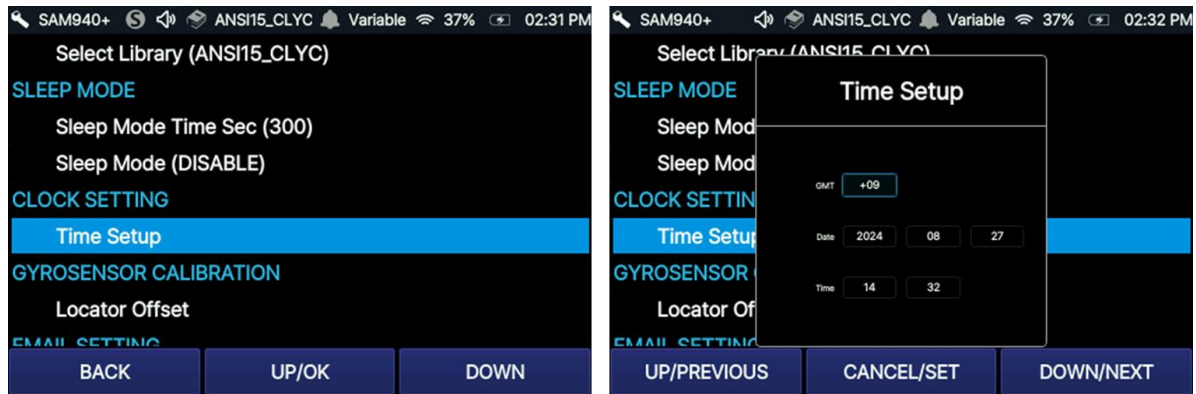


Figure 6.3.29. Setting time

6.3.4.22 FINDER Offset Calibrate

Setup Preference value for the Gyro Sensor (This procedure is to be performed when using the FINDER function for the first time, after a software update, or if the Gyro Sensor is not working correctly.)

Perform the setup by following these 5 steps:

- Step 1: Click START, then the user must rotate 1 full cycle.
- Step 2: After rotating 1 full cycle, click STOP. The offset value (x,y) is then calculated.
- Step 3: Repeat Step 1 & Step 2 at least two more times.
- Step 4: After repeating Steps 1 and 2 three times, the mean offset value is then calculated.
- Step 5: Click BACK to set the mean offset value as the default Preference for the Gyro Sensor.

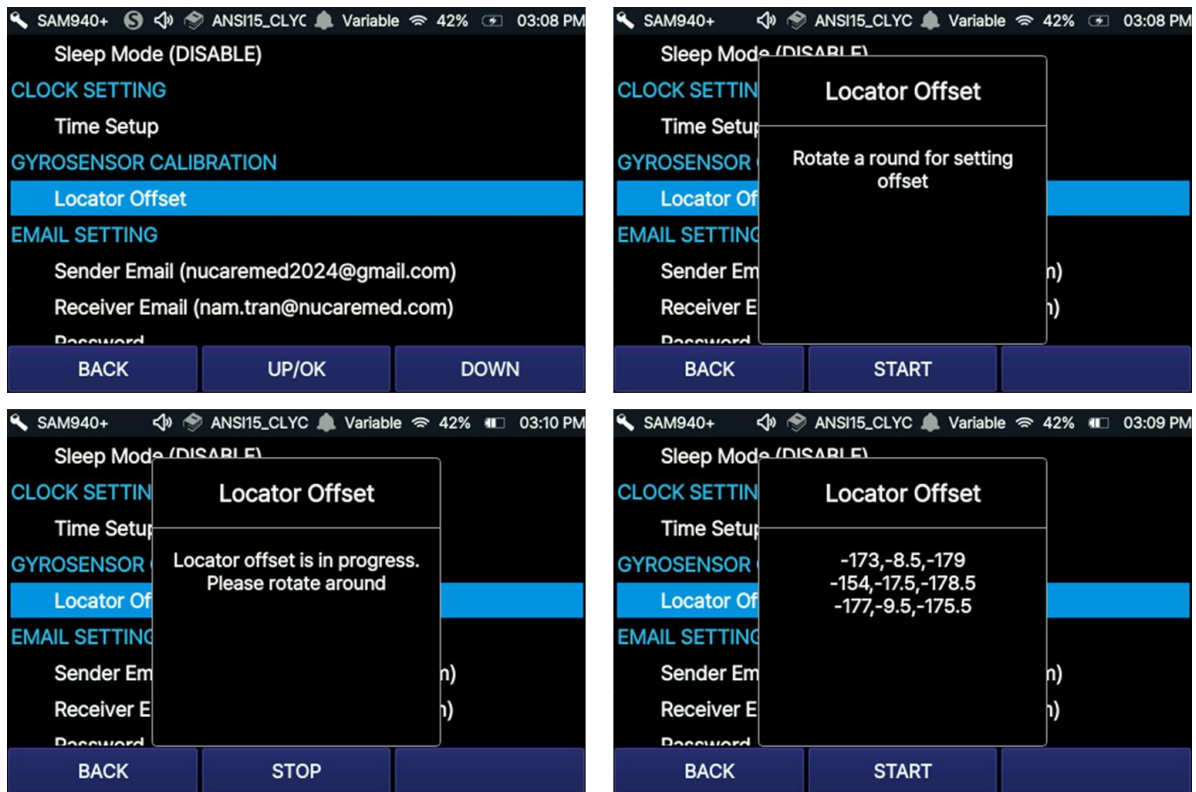


Figure 6.3.30. FINDER offset

6.3.4.23 Email Setting

Users can set the recipient's email and sender's information, including email address, password, server, port, and security settings.

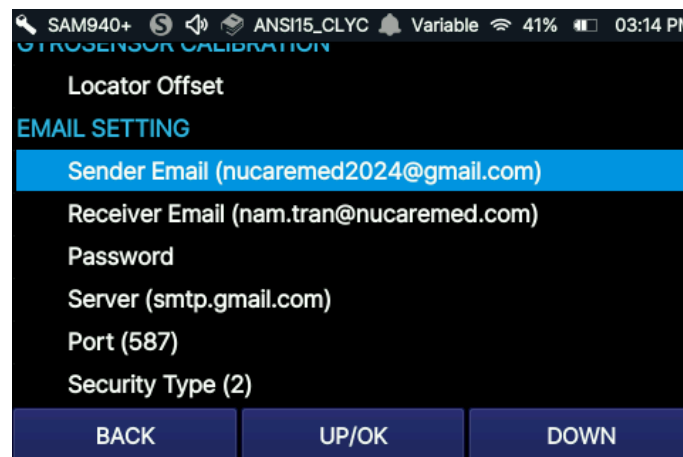


Figure 6.3.31. Email Setting screen

6.3.5. Paring

The SAM 940+ unit can be paired to a PC for remote control, DB management, configuration setup and etc.

1. Select **PARING** menu ①
2. Press the **Connect** button to open the socket port using TCP, allowing PeakID, as the client, to connect to the device ②. After pressing **Connect**, the pairing status will change from "Disconnected" to "Waiting for connection" Once PeakID has successfully connected, the pairing status will update to "Connected"
3. If you do not want to share data or wish to disconnect, you can press the **Disconnect** button.

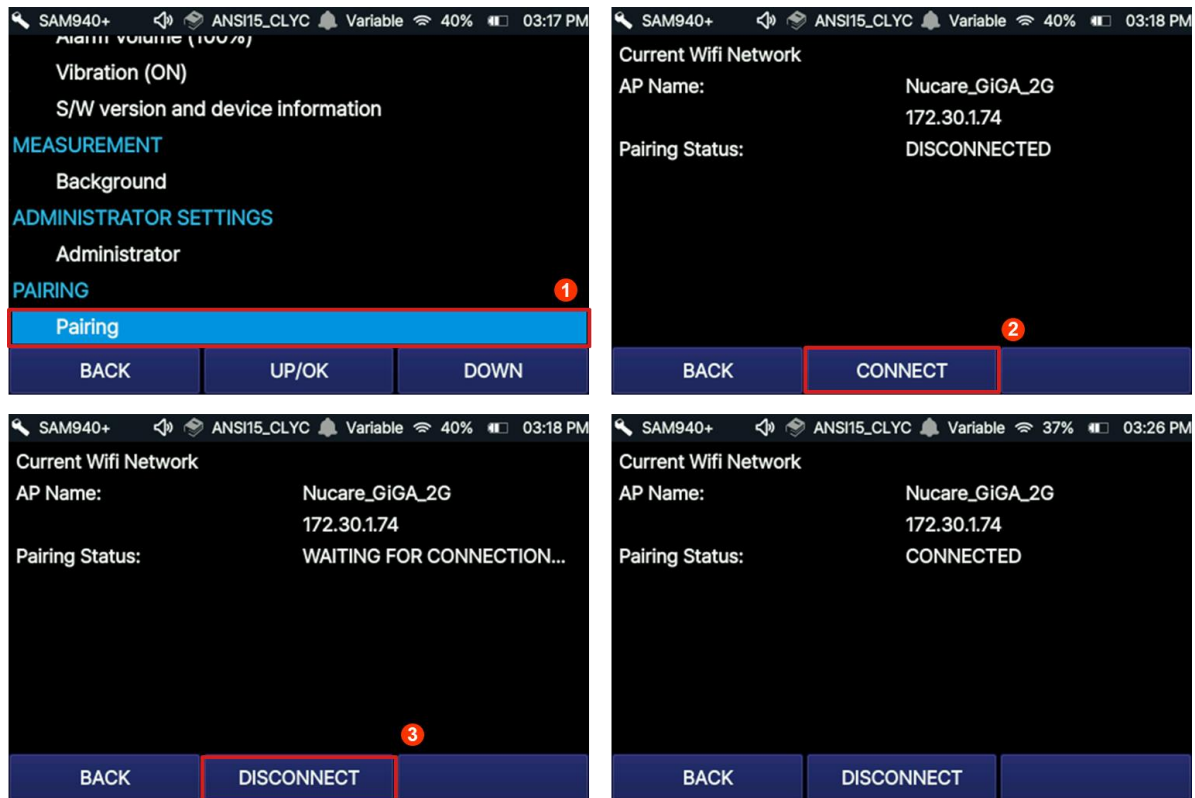


Figure 6.3.31. Paring to a PC



After the TCP connection is established, the device can both share data with PeakID and continue functioning normally, including features such as Alarm, Manual ID, Settings, and Event Log view.



If the connection is unstable or difficult to establish, close the TCP socket and reopen it by pressing the **Disconnect** button and then **Connect** again

CHAPTER 7. PEAKID IV APPLICATION

PeakID IV application SW remotely controls SAM 940+ as similar fashion as the PeakAbout IV does. In addition, PeakID IV supports management of library, Event files, data analysis including integration of data from multiple units, and data backup. It is compatible with PCs that run Windows 10 or higher. PeakID IV Installer can also be downloaded onto a PC from the Internet.

7.1. DOWNLOADING PEAKID IV

1. Using an Internet browser (Internet Explorer, Firefox, Chrome, etc.), type the following URL (Figure 7.1) address to open the BNCSAM web page:
<http://www.bnccsam.com/RESAM940PLUS.html>

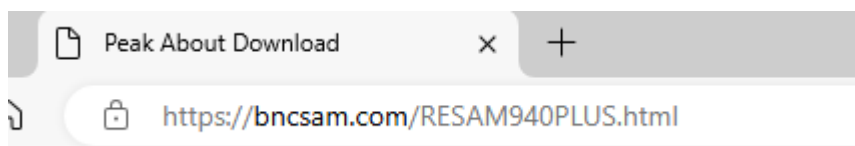


Figure 7.1.1. PeakID URL

2. Enter the serial number from the label on the detector unit, then click **PeakID IV** (Figure 7.1.2.).

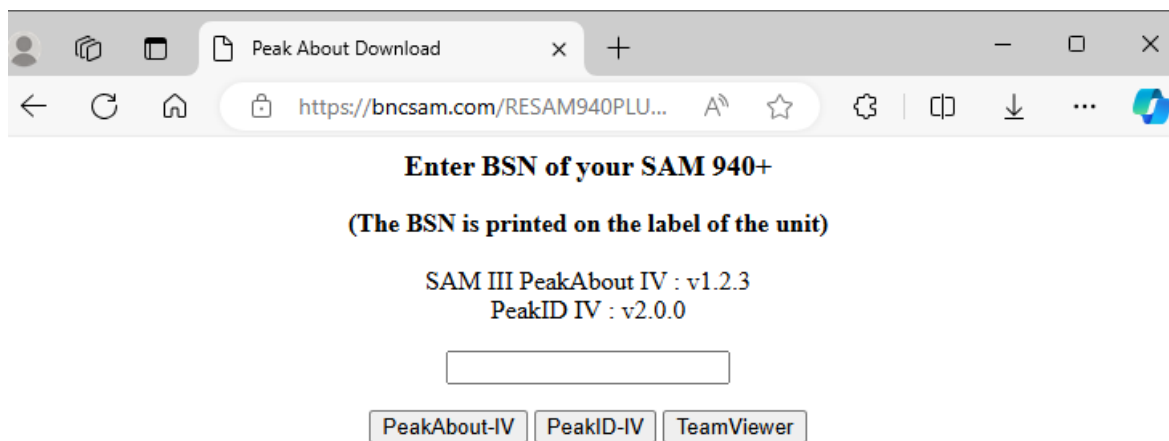


Figure 7.1.2. PeakID download page

3. Save the PeakID IV Installer on the PC (Figure 7.1.3).

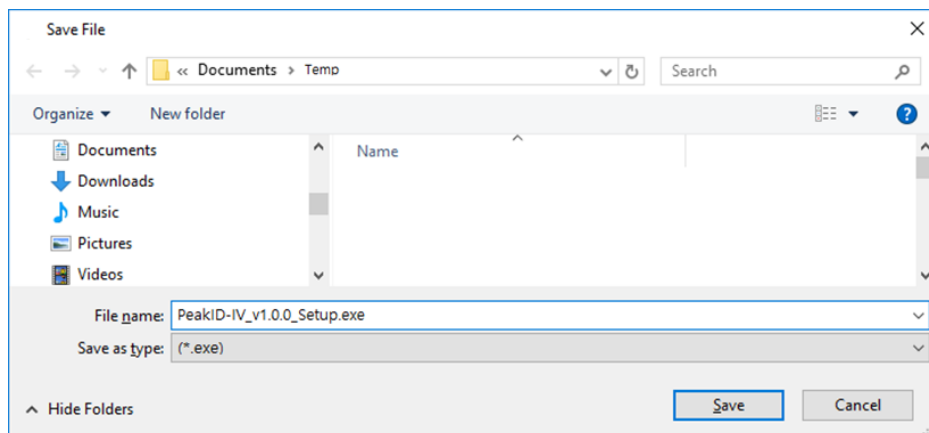


Figure 7.1.3. Save PeakID Installer

7.2. INSTALLING PEAKID IV

7.2.1. Installing PeakID IV

Installing PeakID IV on the PC is a simple process that makes use of an automated installer.

1. Go to the location where the PeakID IV Installer is saved. Click on PeakID IV.exe.
2. Follow the prompts in the InstallShield Wizard (Figure 7.2.1).

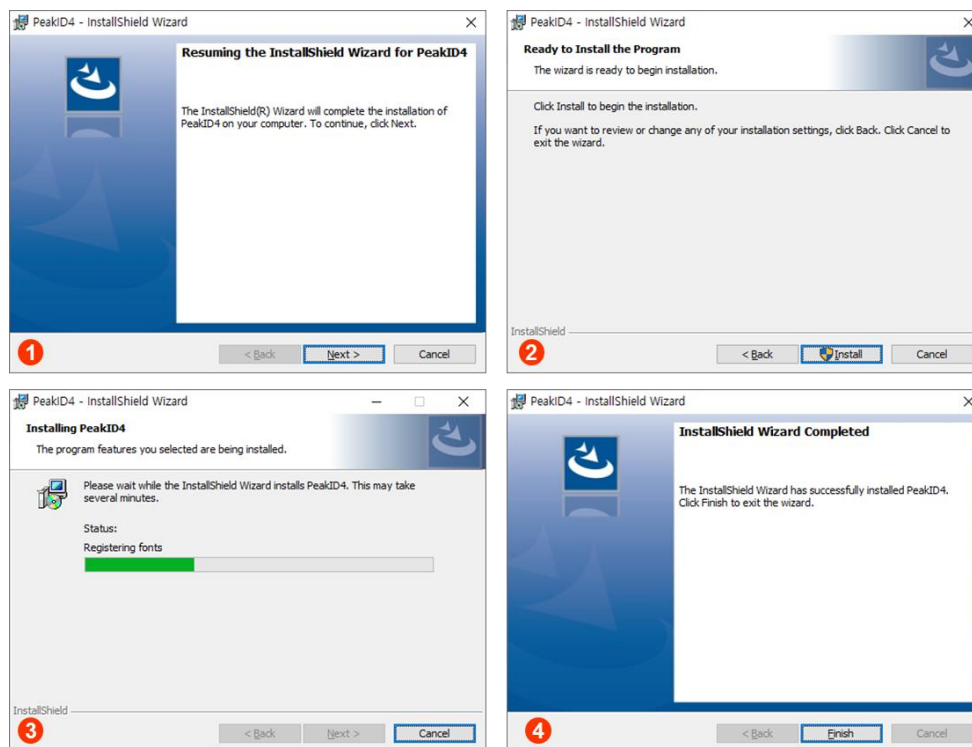


Figure 7.2.1. InstallShield Wizard for PeakID (initial screen)

7.3. CONNECTING SAM940+ AND PC

The SAM 940+ can be paired to a PC via WiFi network connection and USB

7.3.1. Pairing the SAM 940+ to a PC via WiFi Network

Once the SAM 940+ is paired to a PC, manual ID, Event analysis and Configuration setting can be done from a PC. Follow the instruction below to pair the SAM 940+ to a PC.

1. Click **Pairing Status** bar ① located at the bottom of PeakID IV main GUI.

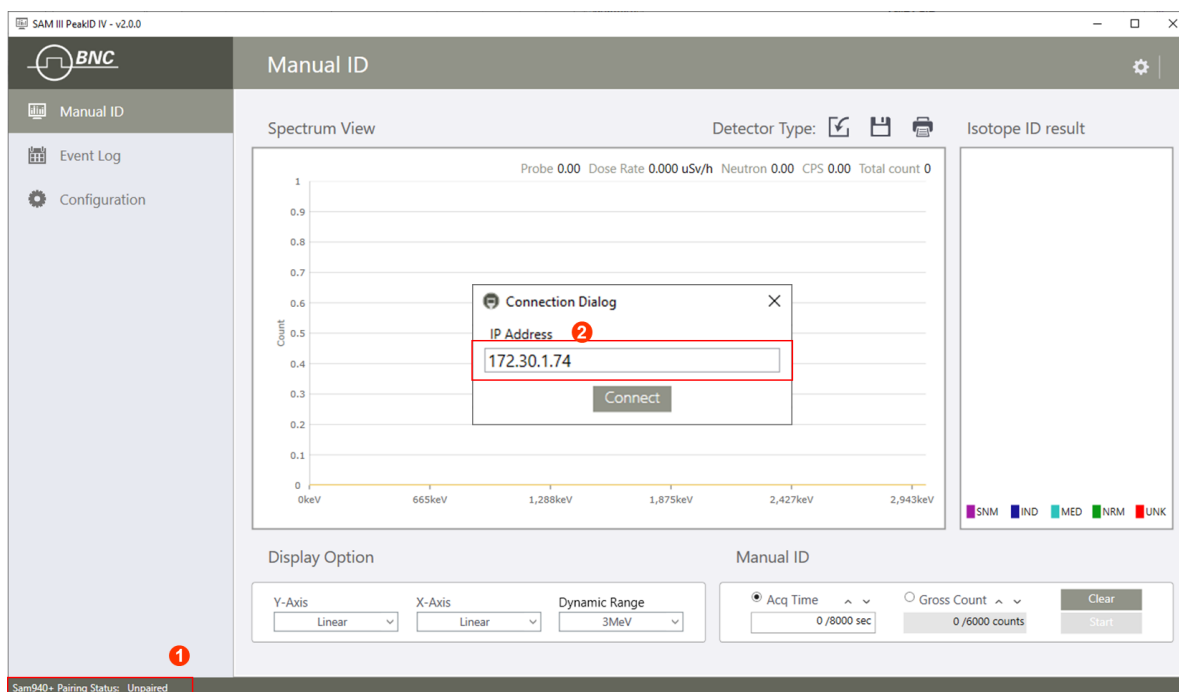


Figure 7.3.1. Connection bar

2. Input IP address of Device. Device should be opened Connect in Pairing mode. Press Connect button to pair with SAM 940+.
3. The connection process is illustrated in the image below.

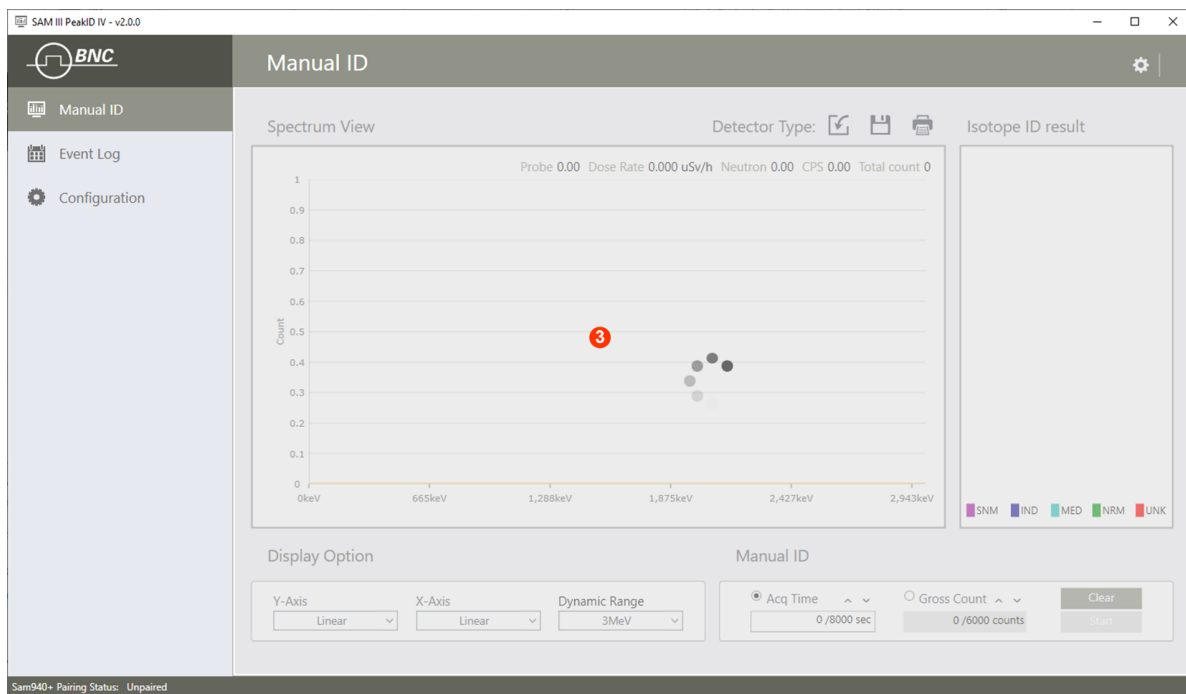


Figure 7.3.2. Connection progress

4. After the connection is complete, the status bar will change to: Paired.

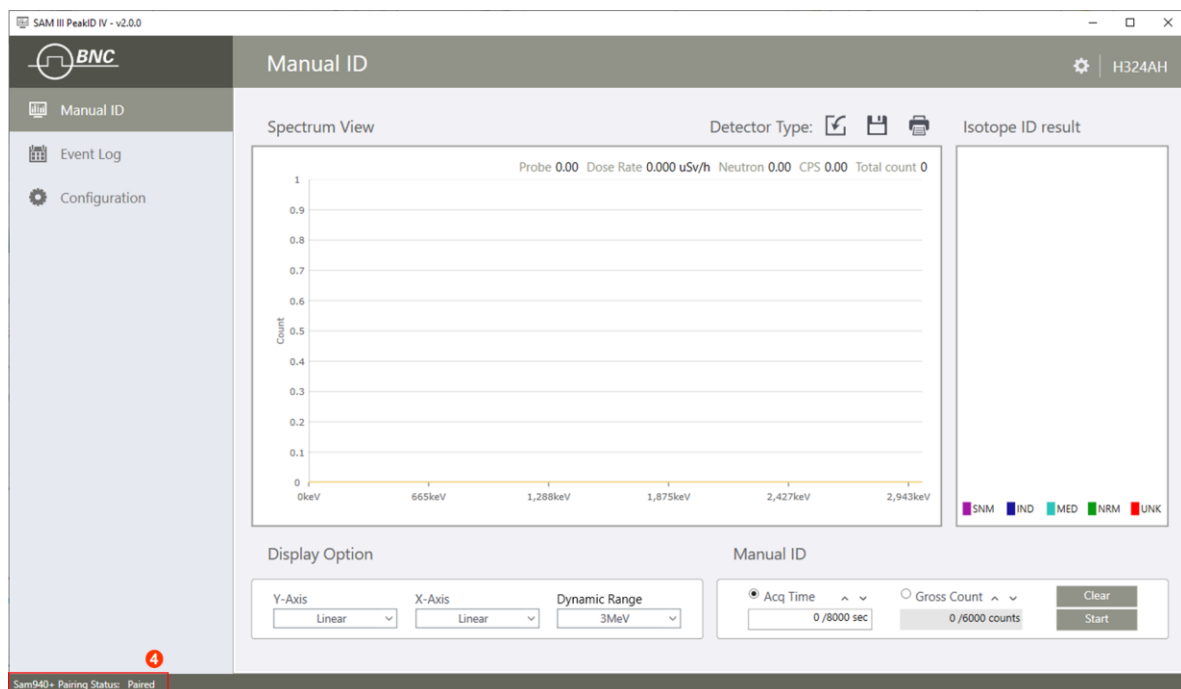


Figure 7.3.3. Connection success



The PeakAbout IV is disabled temporarily while the SAM 940+ is connected to a PC. To resume to normal operation, disconnect the SAM 940+ from the PC by selecting **DISCONNECT** button.

7.3.2. Connect the SAM 940+ to a PC via USB Cable

When the SAM 940+ is connected to a computer via USB cable, the device is recognized as a USB drive named **RAQ IQ USB** shown in Figure 7.3.4, and events can be loaded and analyzed, and configuration settings can be performed from the PC. Figure

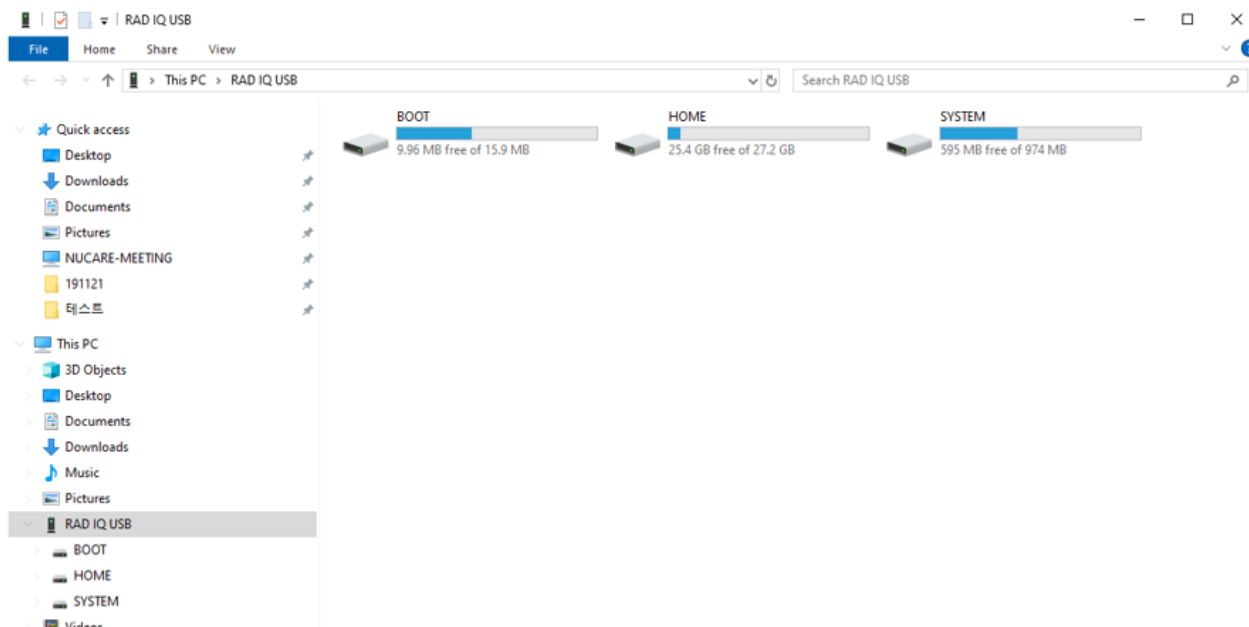


Figure 7.3.4: USB drive folder on PC

The event and configuration database on the SAM 940+ device is stored in the **Home\Data** folder, as shown in Figure 7.3.5.

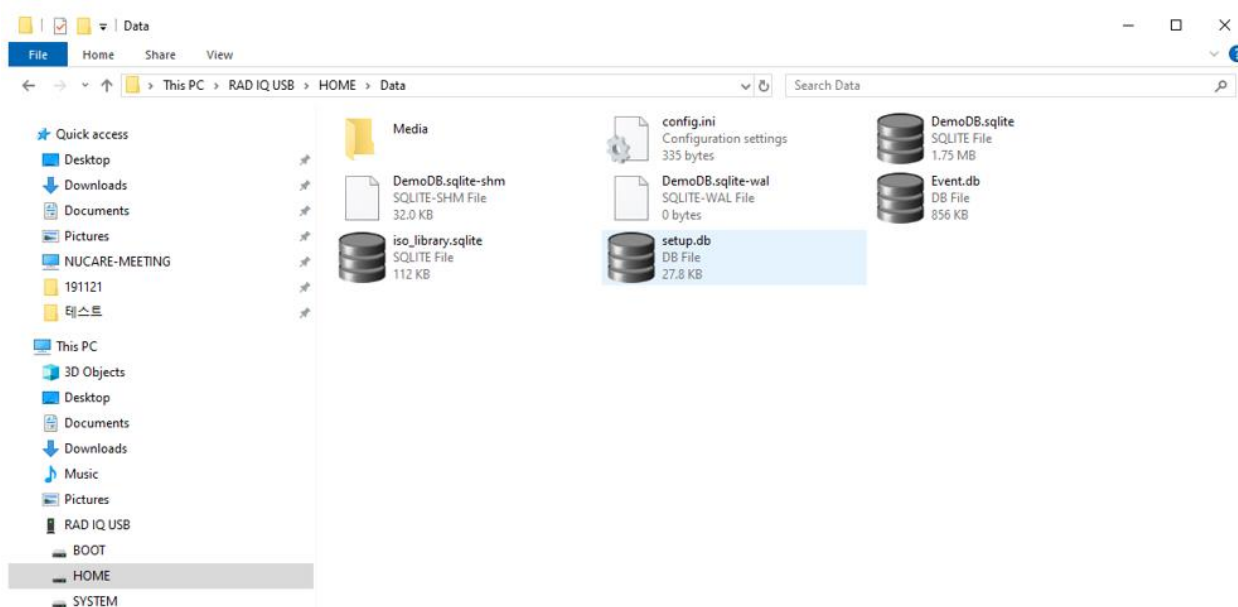


Figure 7.3.5: Event and configuration files are located in the device folder



In case the computer does not recognize the device when it is connected, you should turn off the device, turn it back on, and then reconnect it to the computer by unplugging and plugging in the USB cable again.

7.4. NAVIGATING THE PEAKID IV GUI & STRUCTURE

The home GUI (graphic user interface) of PeakID IV is shown in figure 7.4.1. It consists of tab menu bar on the left side of screen ① and several partitioned windows for spectrum display, command functions and other information. The partitioned windows and its contents are changed depend on selected menu.

PeakID IV's main menu is divided into Data analysis, Event Log and Configuration. The details of each menu is described in following sections.

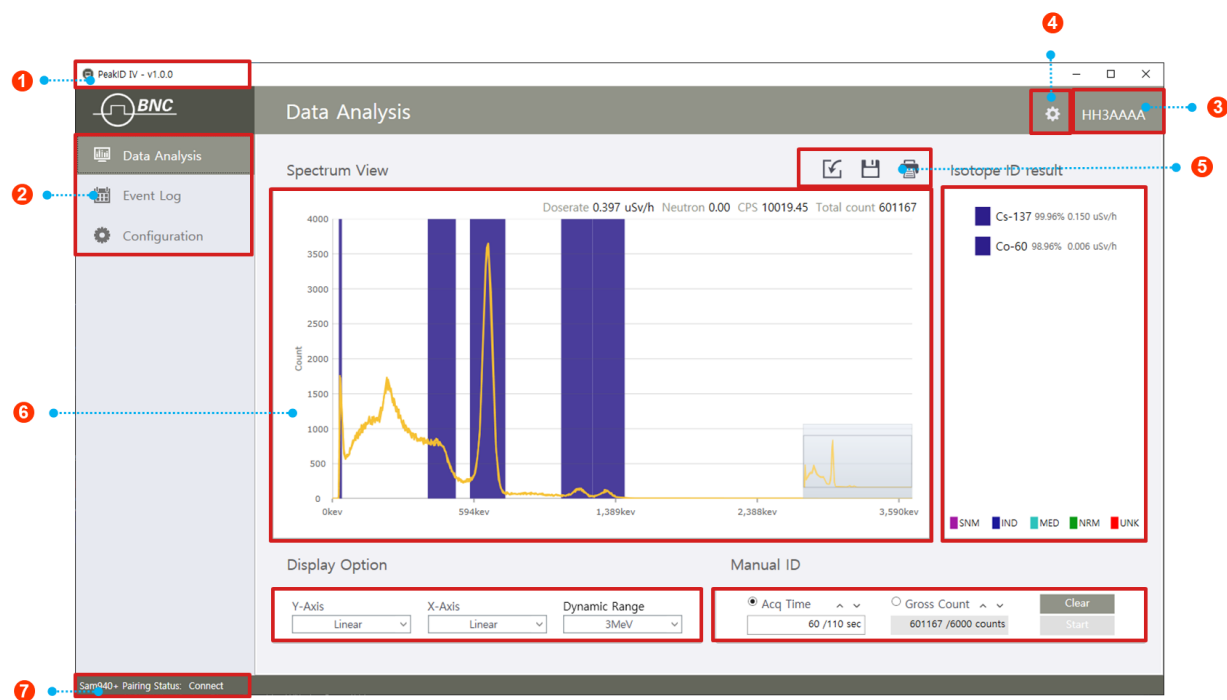


Figure 7.4.1. PeakID GUI

1	PeakID SW version	Information of PeakID IV SW version
2	Main Menu	Main menu's of PeakID IV: Data Analysis, Event Log and Configuration
3	Device Name	Paired device name: serial number of SAM940+
4	Configuration	General configuration for PeakID IV
5	Function icons	Function icons
6	Partitioned Windows	Partitioned windows
7	Connection Status	Connection Status of paired device

7.5. SPECTRUM MANIPULATION

Spectrum View window is an active screen that interacts with a mouse function. PeakID IV provides easy ways of manipulating spectrum using the combination of active window and mouse actions. In addition, “**index cursor**” supports an operator in analyzing the details of spectrum data.

7.5.1. Zoon in/out

Zoon in and out manipulation can be done using a mouse and its wheel. (figure 7.5.1.)

1. Place mouse curse at the ROI (region of interest) ①
2. Zoom in: scroll the mouse wheel up ②
3. Zoom out: scroll the mouse wheel down.



Spectrum View window includes a mini window at the bottom right corner called **“Chart Navigator”**. It is a replicate of Spectrum View in full dynamic range so that zoom in region is marked with a ROI box (figure 7.5.1.).



The **Chart Navigator** is also an active window. The Zoom in region can be moved by synchronizing the movement of ROI box on the chart navigator window.

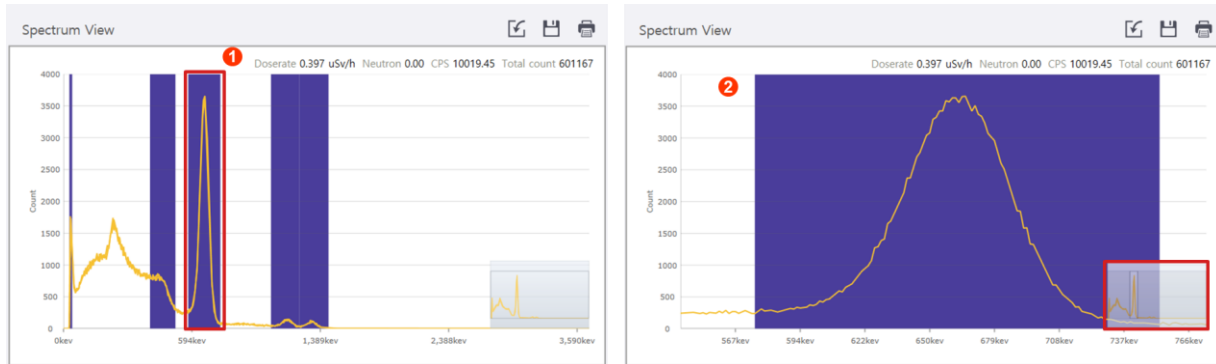


Figure 7.5.1 Spectrum zoom in example



Figure 7.5.2. Chart Navigator example

7.5.2. Index Cursor

Two thin cross lines, called **“index cursor”**, are showed up automatically along with information box as shown in figure 7.5.3. when the mouse cursor is on the **Spectrum view** area. The information box contains count, energy and channel value at the cross point of the spectrum.

The **Index Cursor** is moving along the spectrum contour when the mouse cursor is moving left/right.

1. Place mouse cursor at the ROI region.
2. Move the cross point at the desired point.

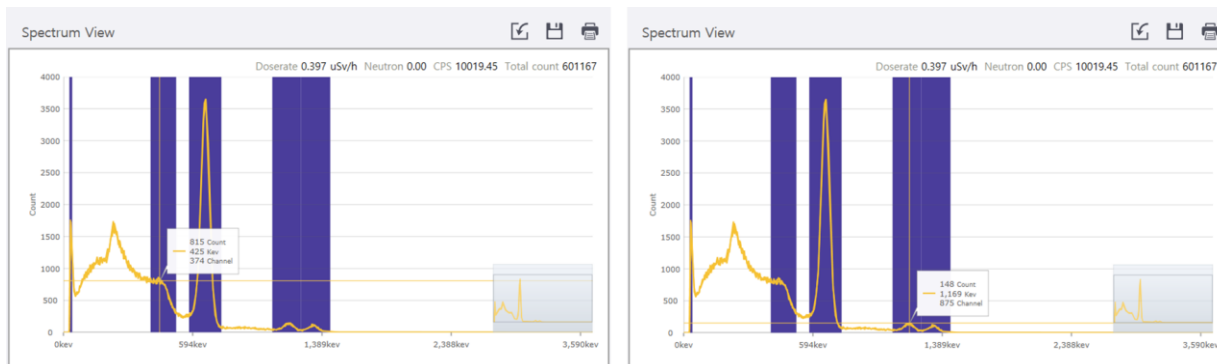


Figure 7.5.3. Index Cursor example

7.6. PEAKID CONFIGURATION

7.6.1. Open Configuration Screen

Configurable parameters for PeakID application software can be accessed by selecting configuration icon on the right top of the PeakID GUI.

1. Select configuration icon ①
2. Type in Administrator's password ②
3. PeakID IV Configuration window ③ will be popped up as shown in figure xx.
4. Set configuration parameters.



PeakID IV has two configuration menu. One is for PeakID IV itself and the other is for PeakAbout IV configuration. Refer to **7.6. Administrator Setup** section for PeakAbout IV setting.

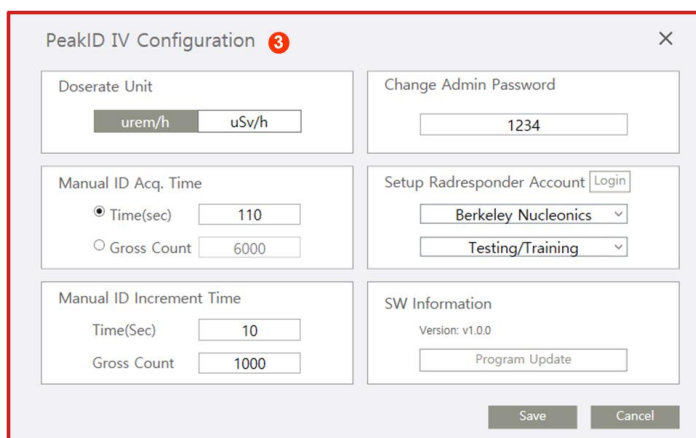
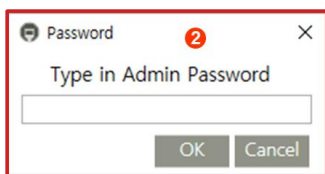


Figure 7.6.1. Configuration menu for PeakID

7.6.1. PeakID IV Configuration

PeakID IV configuration includes 6 setup parameters. Function of each parameter and usage is described in following sections.

Figure 7.6.2. PeakID IV Configuration category

1	Doserate Unit	Define Doserate unit
2	Manual ID Acq. Time	Manual ID acquisition time or count setup
3	Manual ID Increment Time	Define increment time for manual ID.
4	Change Admin. Password	Change new password for admin.
5	RadResponder	Radresponder accounting setup
6	SW Information	SW information and manual update

7.6.1.1. Dose Rate Unit

Rem/h or Sv/h unit can be used for dose rate unit.

Note: Default dose rate unit is rem/h.

7.6.1.2. Manual ID Acq. Time

Manual ID Acq. Time determines measurement time for manual ID. Either elapse time or total measured count can be used for measurement time.

Note: Default measurement time is 60 second.

7.6.1.3. Manual ID Increment Time

Define increment time for manual ID.

Note: Default increment time is 10 seconds.

7.6.1.4. Change Admin. Password

Change administrator's password.

Note: Factory setup is 1234. It is recommended to change it to more secured combination.

7.6.1.5. RadResponder Account



RadResponder is a free network for the rapid collection and management of radiological data during an emergency. When the SAM 940+ is operating within range of Wi-Fi, spectrum and event data can be uploaded to the RadResponder cloud in the appropriate format. RadResponder can be accessed on PDAs, smartphones, tablets, or any computer connected to the web, allowing it to be seamlessly employed at all levels of government during a radiological or nuclear emergency.



Before you can upload spectrum and event data to RadResponder, you must already have established a RadResponder account at www.RadResponder.net, and you must set up RadResponder Mode in PeakAbout IV.

The RadResponder set up is defined in the PeakID IV Settings menu. (figure 7.6.3)

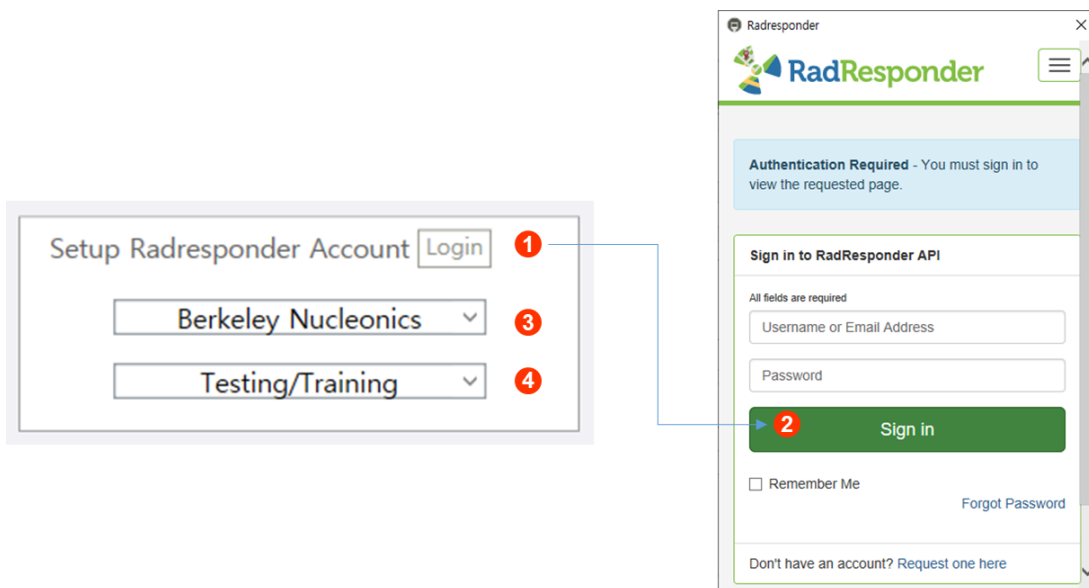


Figure 7.6.3. RadResponder settings example

1. Click Login button ①
RadResponder Sign In API popped up as shown in figure 7.18
2. Sign in with user name and password ② (It should be RadResponder Account)
3. Select Sponsor name from the pop up list ③
4. Select type of event from the pop up list ④



RadResponder requires to define **“Sponsor”** and **“Type of Event”** so that the information is attached together with event data when it is uploaded to the RadResponder cloud. The Sponsor list and type of event are differ from each RadResponder accounts.



Once you signed in RadResponder, PeakID IV automatically extract **“Sponsor list”** and **“types of event”** information assigned to the signed in user so that the Sponsor and type of event can be selected after login is done.



Event type should be determined in configuration (**refer to 7.9.4.12. RadResponder Account**) so that transmitting spectrum and event data to RadResponder as **defined Event type**. An Event that is transmitted as an Emergency Response will trigger an actual emergency response by RadResponder. This is not to be taken lightly. This setting should only be selected when the user will be monitoring actual emergency response or environmental conditions. An Event that is transmitted as Testing Training will be acknowledged and recorded in your RadResponder without triggering an emergency response. This setting is recommended when the user is performing equipment testing or participating in training exercises.

7.6.1.6. *SW information*

To upgrade to the latest PeakID software, follow the instructions in **Sections 7.1** and **7.2** to download and install the file **PeakID IV (vx.x.x).exe** to your PC.

7.7. DATA ANALYSIS MENU GUI

PeakID IV can remotely control the SAM 940+ when the unit is paired with a PC. The **Data Analysis** menu GUI is the gate way to perform remote manual ID, spectrum display and event analysis in real time.

Figure 7.7.1. shows major components and function buttons for **Data Analysis** menu GUI.

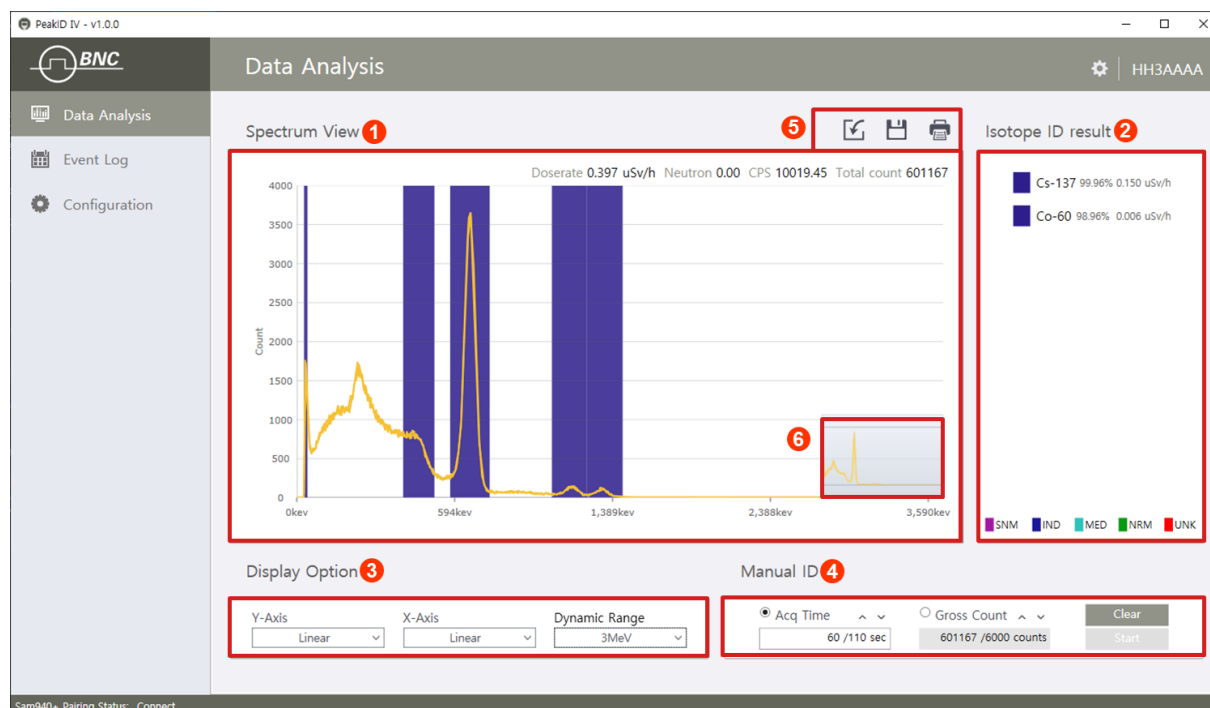


Figure 7.7.1. Data Analysis GUI

1	Spectrum View	Displays spectrum and count information
2	Isotope ID	Isotope ID result with classification color
3	Display Option	Setup X and Y-axis unit and dynamic range
4	Manual ID	Manual ID stop/start and Acq. Time or count setting
5	Function icons	Function icons for importing/exporting XML and print
6	Char Navigator	Duplicate of spectrum view in miniature scale

7.7.1. Spectrum View Window

Spectrum View window (1) consists of spectrum view window, count rate information (2) and chart navigator (3). It displays spectrum from a real time event measured from the SAM 940+. Count rate information is displayed on the top-right corner (2) and chart navigator provides duplicate of spectrum view in miniature scale (3).

Note: refer to the 7.5.1. section for manipulation of spectrum and usage of chart navigator.

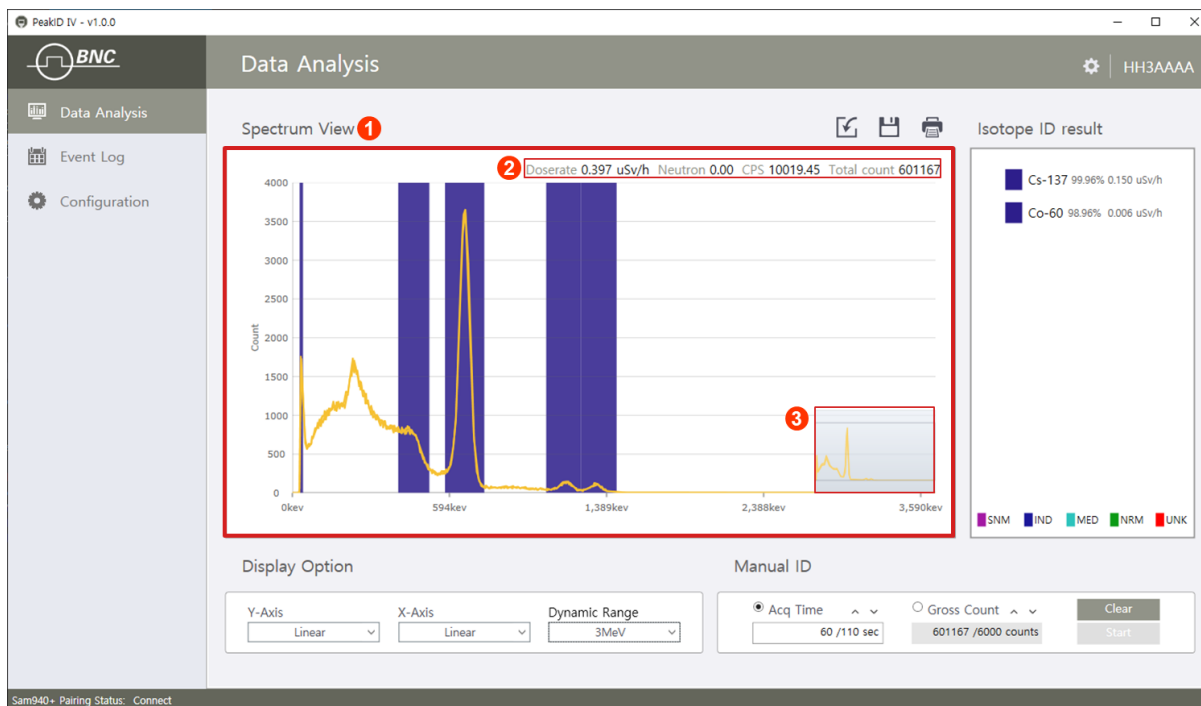


Figure 7.7.2. Spectrum View window example

7.7.2. Isotope ID Result Window

Isotope ID window ① shows the ID result along with classification information, confidence level and dose rate ②. Bottom of the ID window displays index color scheme of classification ③.

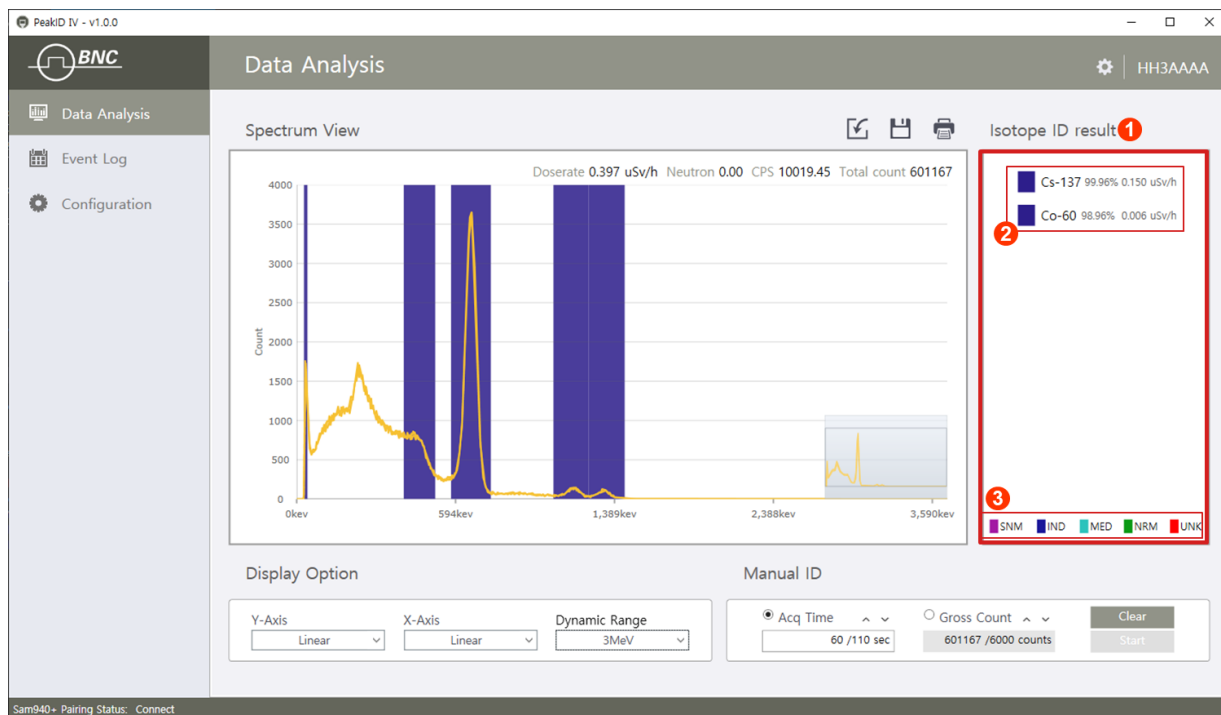


Figure 7.7.3. Isotope ID result window example

7.7.3. Display Option Window

Display Option manages display option of **Spectrum view**. X and Y axis have two display options per each and total of 4 combination of display option can be offered. Details of each display option is described in following sections.

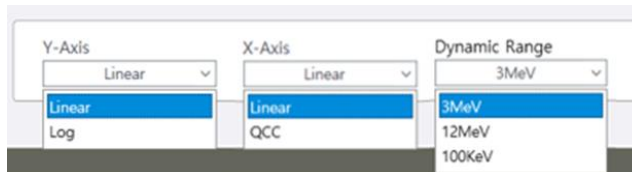


Figure 7.7.4. Options for X and Y-axis unit and Dynamic range

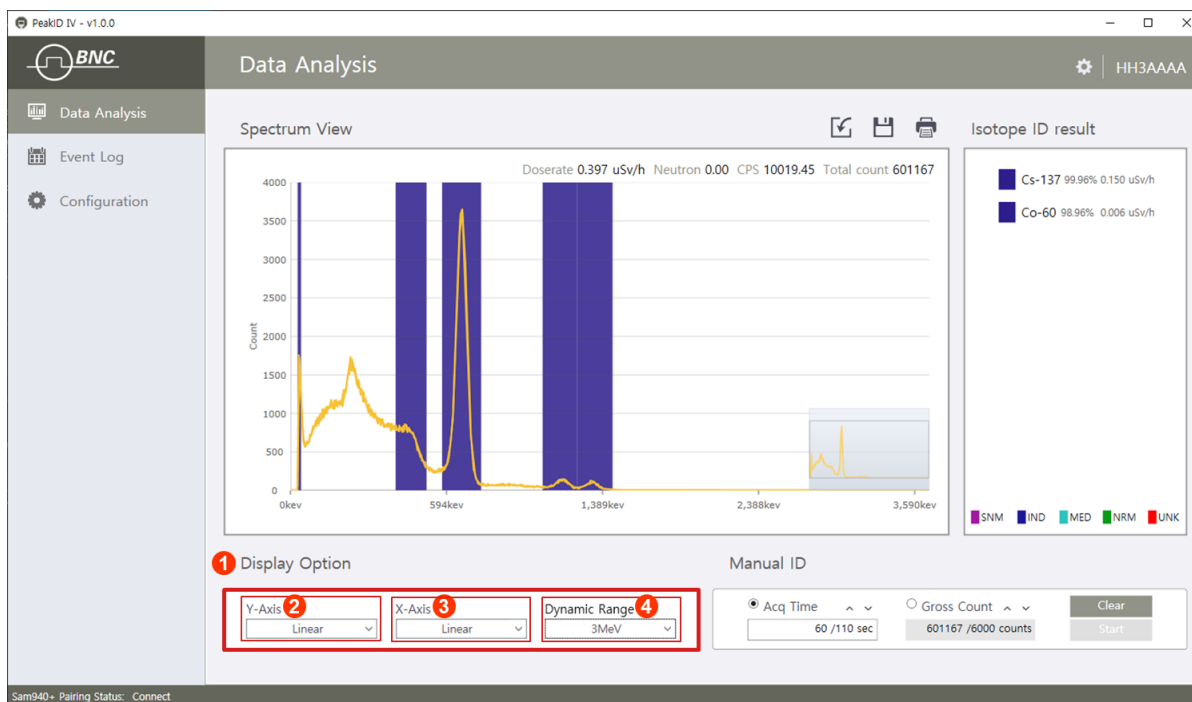


Figure 7.7.5. Display Option example

2	Y-Axis	Display option for Y-axis
3	X-Axis	Display option for X-axis
4	Dynamic Range	Dynamic Range selection

7.7.3.1. Y-axis display option

Spectral data can be displayed on either a linear or a logarithmic scale (Figure 7.7.6). To change between the two displays, click Y-Axis option and select desired display option.

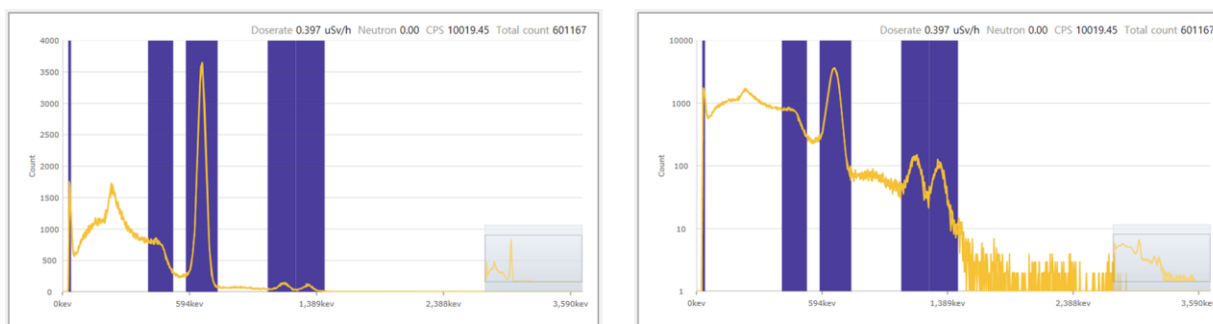


Figure 7.7.6. Linear vs. Logarithmic scale display example

7.7.3.2. Y-axis display option

X-axis unit can be selected either Linear or QCC (Quadratic Compress Conversion) (figure 7.7.7). Refer to the details of QCC spectrum display in Chapter 6. To change between the two displays, click X-Axis option and select desired display option.

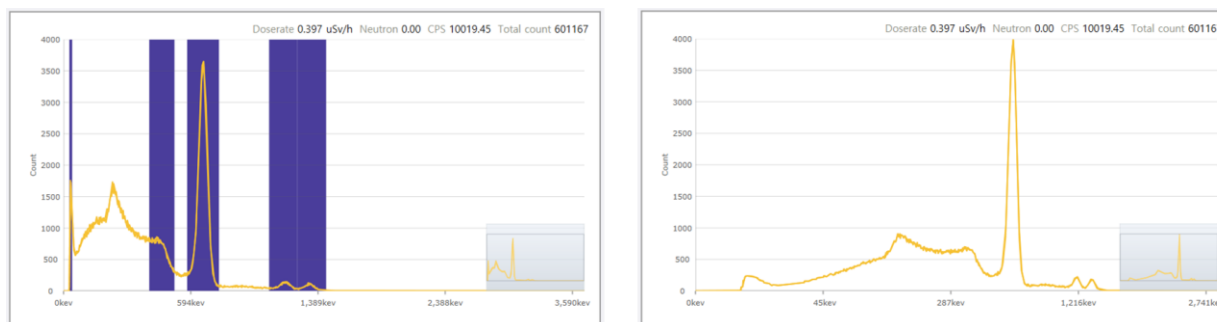


Figure 7.7.7. Linear vs. QCC display example

7.7.3.3. Y-axis display option

Specially designed high voltage divider scheme enables the SAM 940+ handle gamma energy up to 10 MeV. Dynamic range option offers three DR selections. Up to 150 KeV, 3 MeV and 10 MeV.

Figure 7.26 shows example of each different dynamic selection cases.



Note: 3MeV DR is the typical dynamic range setup for normal use. 100 KeV DR is for analyzing details of low energy region while 10 MeV DR is for high energy physics application cases.

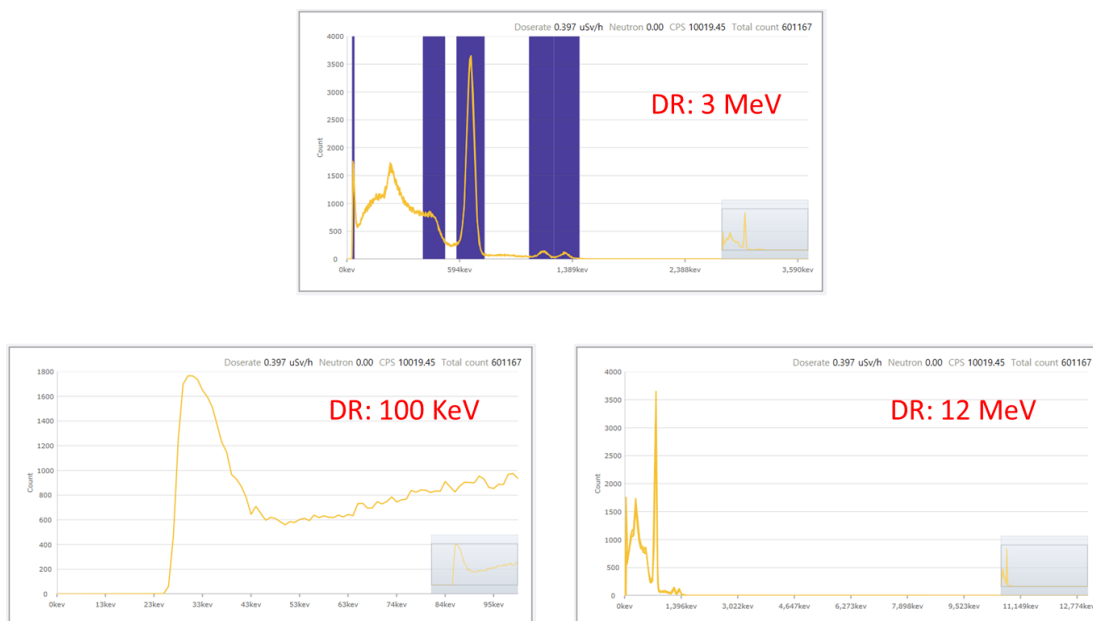


Figure 7.7.8. Dynamic Range example

7.7.4. Manual ID Window

Manual ID window consists of measurement time setup and command buttons for manual ID.

Acquisition time is either defined by total elapsed time ② or gross count ③. Manual ID start/stop or discard command can be executed by selecting a corresponding function button ④.

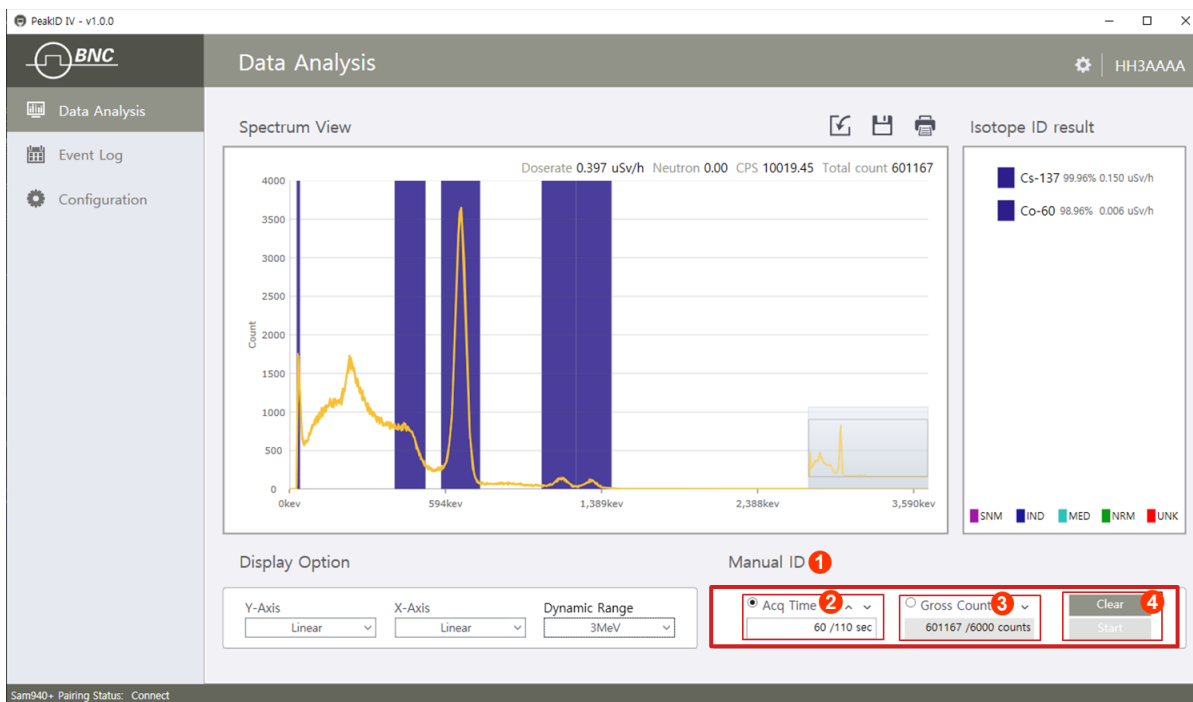


Figure 7.7.9. Manual ID command window



Increment/decrement time or gross count can be set by clicking the up/down arrow as shown in figure 7.7.10.

The 'Manual ID' panel contains two radio buttons: 'Acq Time' (selected) and 'Gross Count'. Each has an up/down arrow icon next to it. Below 'Acq Time' is a text box showing '60 /110 sec'. Below 'Gross Count' is a text box showing '601167 /6000 counts'. To the right are 'Clear' and 'Start' buttons.

Figure 7.7.10. Increment/decrement time or gross count

7.7.5. Function Icons for Data analysis menu GUI

Function icons ① of Data Analysis menu GUI provides importing/exporting XML file and print features. Details of each feature is described in following section.

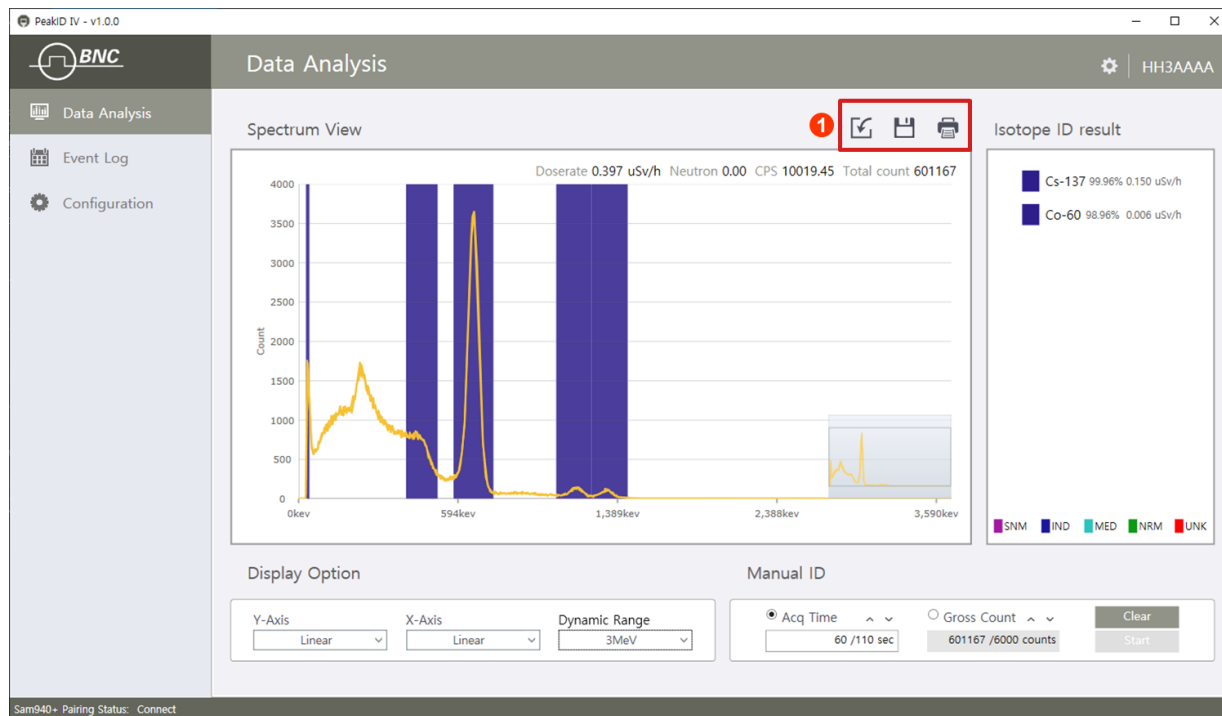


Figure 7.7.11. Data Analysis menu function buttons

7.7.5.1. Import XML file format Event

PeakID IV allows user to import XML file format specified in the ANSI 42.42 standard.

1. Click the  icon opens Explore window will appear as shown in figure 7.30.
2. Select the XML event from the list
3. Click Open button.

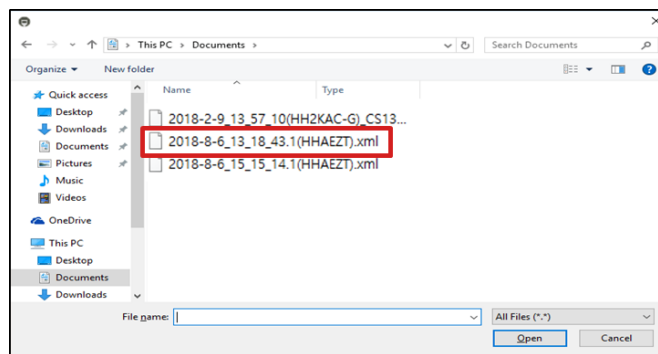


Figure 7.7.12. Import XML event example

7.7.5.2. Saving Event Data in XML Format (ANSI 42.42)

The spectrum and relevant data from an Event can be saved in the XML file format specified in the ANSI 42.42 standard.

1. Click the disk icon  (Figure 7.7.11.). The directory screen will appear.
2. Browse to the desired destination directory, select it, and click **Save**.

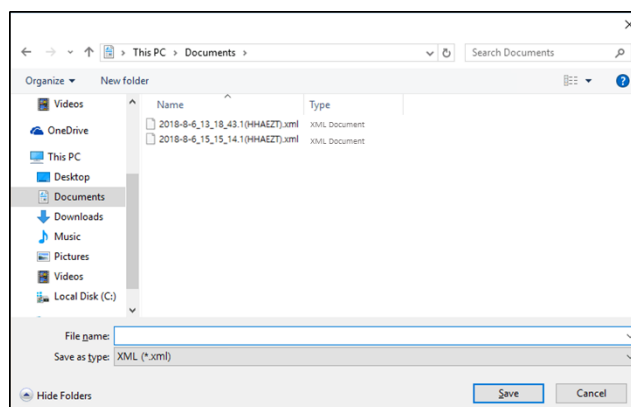


Figure 7.7.13. Save XML event example



Manual ID result by PeakID is not automatically saved in the Event Log folder generated by PeakAbout IV. You have to save it as an individual event at the end of manual ID.

7.7.5.3. Printing Event data

The spectrum and relevant data from an Event can be printed.

2. Click the printer  icon (Figure 7.7.11.).
3. The Measurement Report (Figure 7.7.13.) will appear in a separate window.

Measurement Report

Instrument Information

Instrument Type: Radionuclide Identifier
 Manufacturer: Berkeley Nucleonics Corp.
 Instrument Model: BNC
 Detector: NaI_2x2
 Coordinates: 0 0

Spectrum Information

Energy Units: KeV Acq Time: 120 sec
 Date: 2020-02-27 Time: 16:23:33
 Covariance Matrix: $\text{Term0} + (\text{Term1} * \text{Ch}) + (\text{Term2} * (\text{Ch}^2))$
 Calibration Coefficients: -0.000111008
 1.719696954
 6.229522519



Photon Data

AvgCPS: 831.0666667 Total Count: 99728
 Avg Dose Rate: 53.83 urem/h Avg Neutron: 0 cps
 Max DoseRate: 165.17 urem/h Max Neutron: 0 cps

Analysis Result

Name	Type	Doserate	Confidence
Cs-137	IND	19.12 urem/h	99.95

Figure 7.7.13. Printed measurement report example

7.8. EVENT LOG GUI

PeakID allows to access event data that has been downloaded to PeakID from the SAM 940+ (in both cases: pairing and USB cable) or retrieved from a database previously saved on the PC.

7.8.1. Navigating the Event Log menu GUI

Event Log menu GUI provides Event list, spectrum windows and several configuration and analytical windows. Details of each windows and functions are described in following sections.

In addition, PeakID IV can replay the event under various display conditions and calibration configuration.

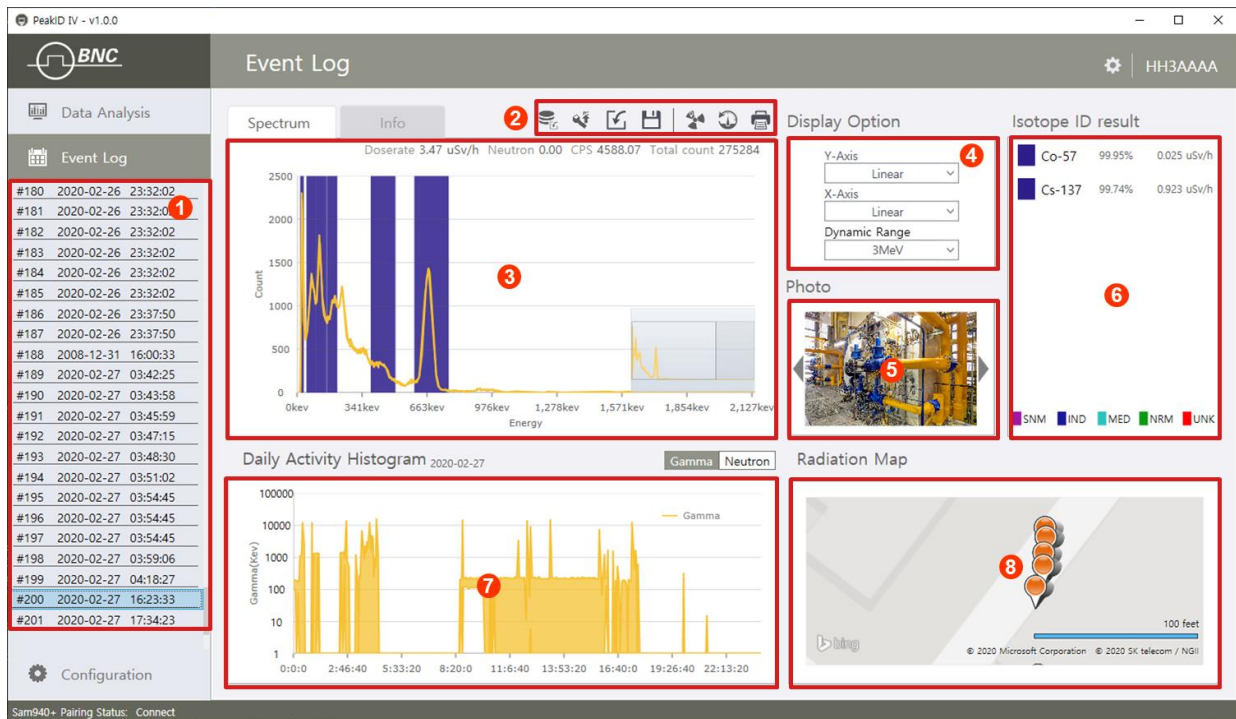


Figure 7.8.1. Event Log menu GUI example



Spectrum view and Event information windows can be toggled by selecting the menu tap. Figure 7.8.2. shows spectrum and Info window.

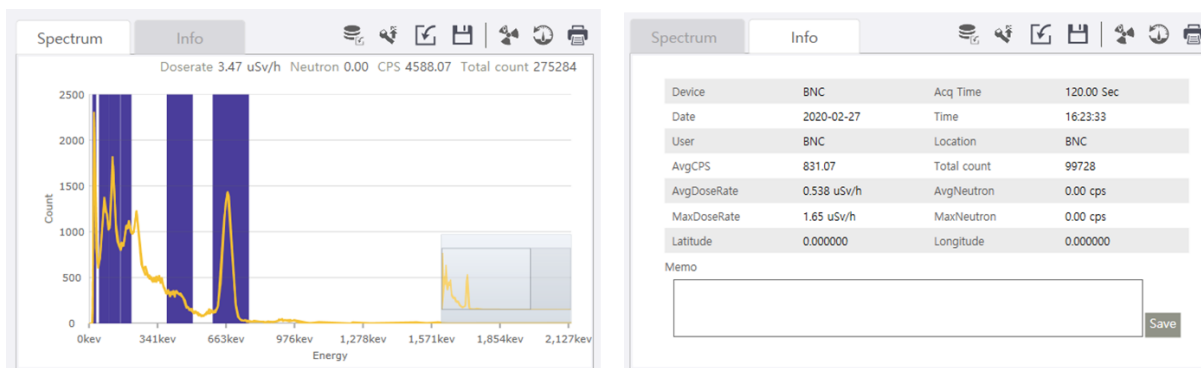


Figure 7.8.2. Spectrum vs. Info window example

1	Event List	Displays Events in numerical order. Includes date and user.
2	Download buttons	Allows the user to load event data to PeakID by clicking either Load from PDA or Load from DB (a database previously saved on PC)
3	Event spectrum/Info	Displays spectrum and general information for the selected Event
4	Display option	X and Y-axis display option and Dynamic range setup
5	Photos	Displays photos and video if any were attached
6	Identification	Displays nuclides identified in the selected Event
7	Real-time activity	Displays activity as a function of time for the selected date
8	Map	Displays GPS location and activity information

7.8.2. Managing Event Log

PeakID IV manages Event log in various ways. The Event Log DB can be directly downloaded from the SAM 940+, a PC database or XLM file format. Following sections describe methods downloading event log to the PeakID IV.

7.8.2.1. Download Event Data from the SAM 940+ to a PC via WiFi

1. Ensure that the SAM 940+ is properly connected to the PC (refer to **Section 5.4: Connecting the SAM 940+ to a PC**).
2. On the PC, open PeakID IV.
3. Click **Load from SAM 940+ ic** . Download confirm message window is popped up as shown in figure 7.8.3. ①
4. Select an Event from the list to analyze.

Downloaded Events are displayed in the Event List area ②.

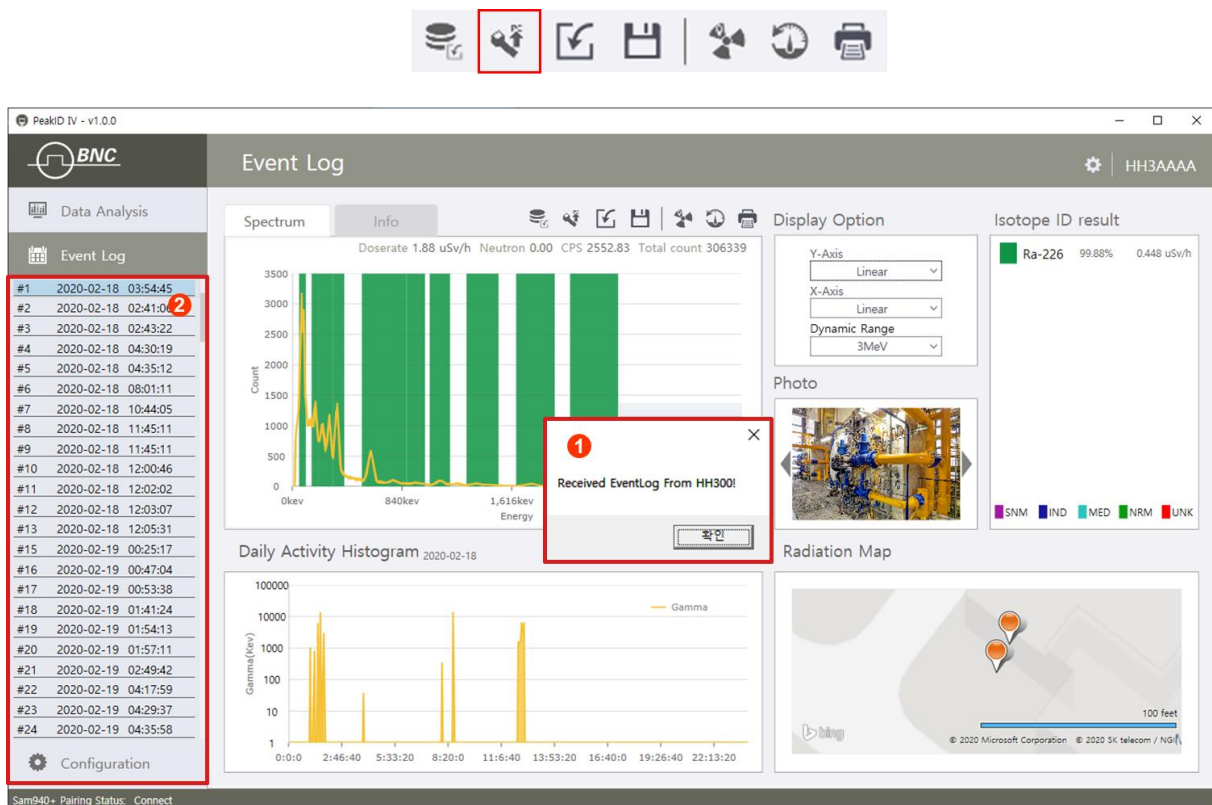


Figure 7.8.3. Download from the SAM940+ example

7.8.2.2. Download Event Data from the SAM 940+ to a PC via USB Cable

1. Ensure that the SAM 940+ is properly connected to the PC via USB Cable
2. On the PC, open PeakID IV.
3. Click **Load from SAM 940+** icon .
4. Select the desired folder to store the event database. The database from the device will be copied to this folder shown in Figure 7.8.4. Download confirm message window is popped up as shown in figure 7.8.5.
5. Select an Event from the list to analyze.

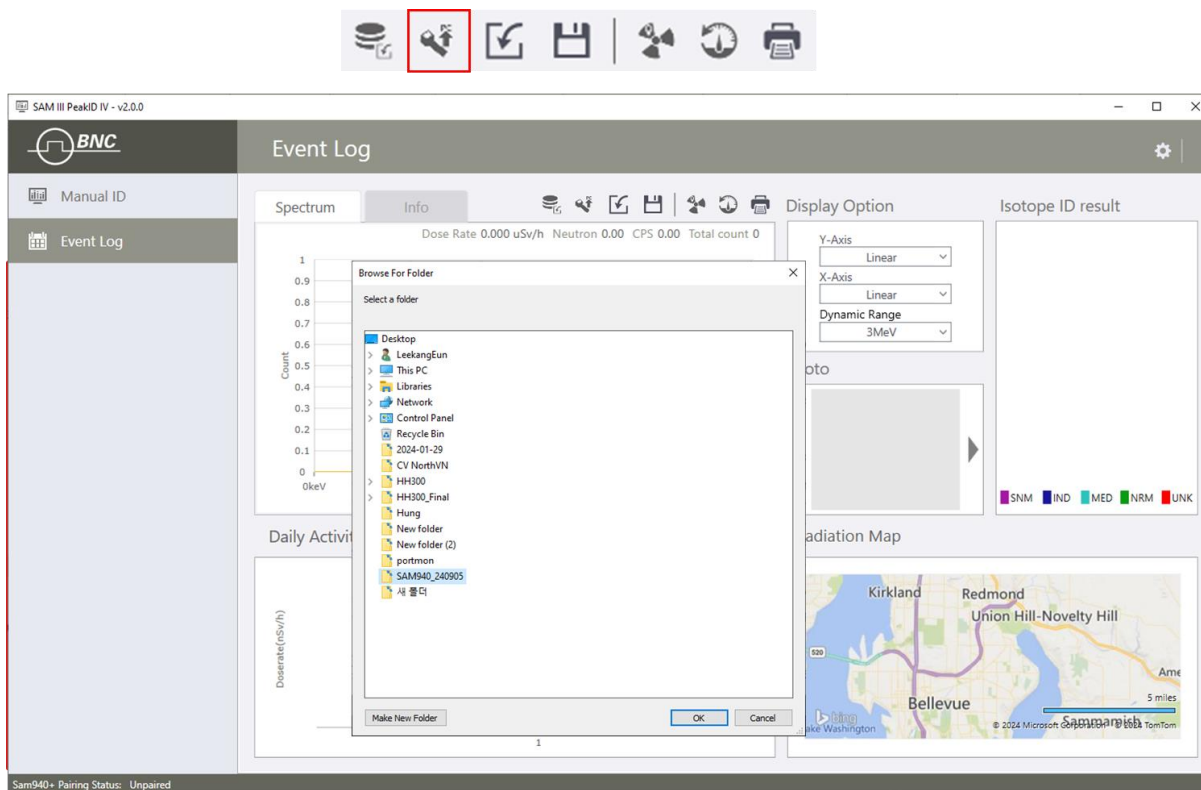


Figure 7.8.4. Select the desired folder to store the event database example

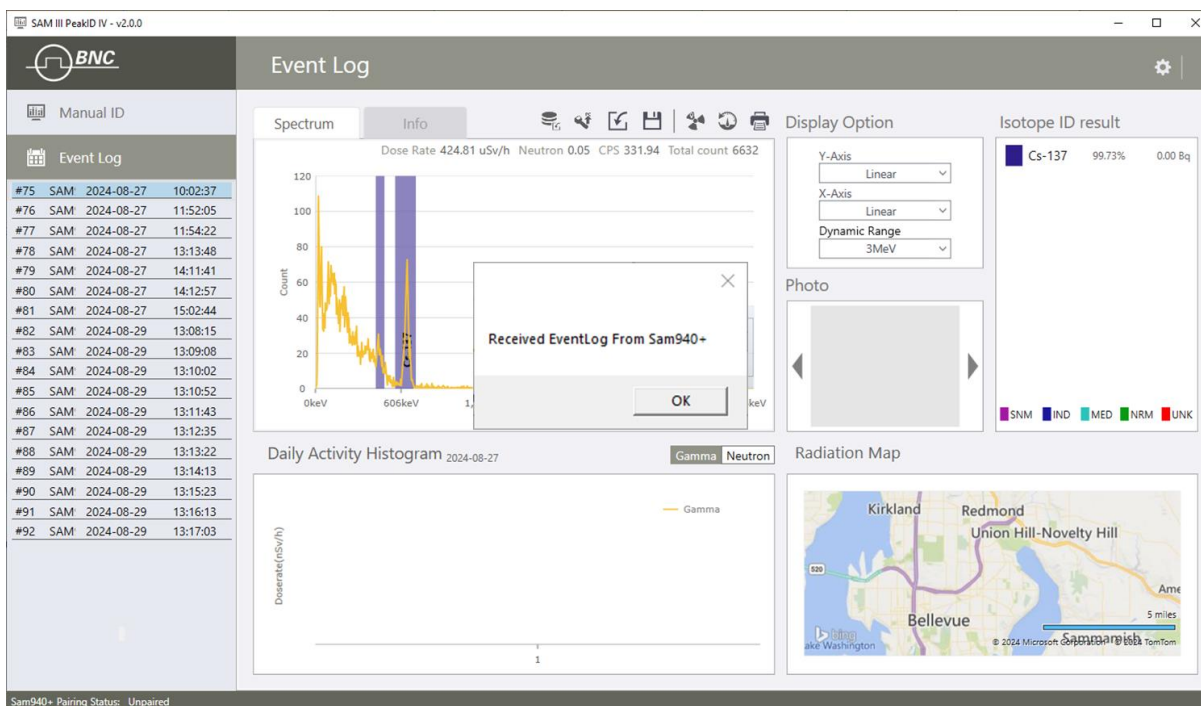


Figure 7.8.5. Download from the SAM 940+ example via USB cable

7.8.2.3. Load Event Data from a PC Database

All the Events collected from multiple SAM 940+ units may be stored in a single database and managed using PeakID IV. It is not necessary to connect the unit to a PC in order to manage or analyze Events loaded from a PC database.

1. Click **Load from DB**  (Figure 7.8.6). The database browsing window pops up. ①
2. Click on the desired folder to select it. Folders are labeled with the date of collection of the event data.
3. When a folder is selected, the Event(s) from that folder will appear in the Event List area ②. Select an Event from the Event List to display the spectrum and other information.

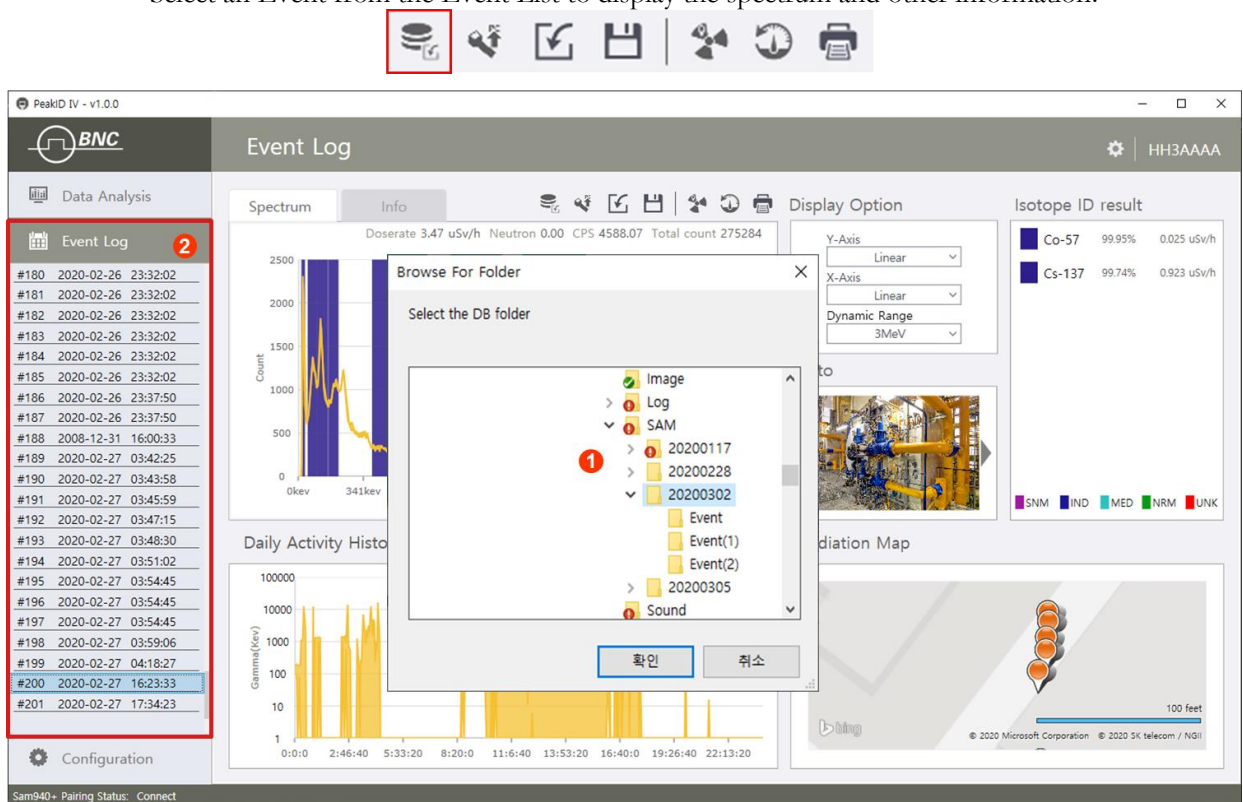


Figure 7.8.6. Load data from a PC database example

7.8.2.3. Importing XML file format

PeakID IV allows user to import XML file format specified in the ANSI 42.42 standard.

1. Click the disk icon . The directory screen will appear (figure 7.8.5).
2. Browse to the desired destination directory, select it, and click **Open**.

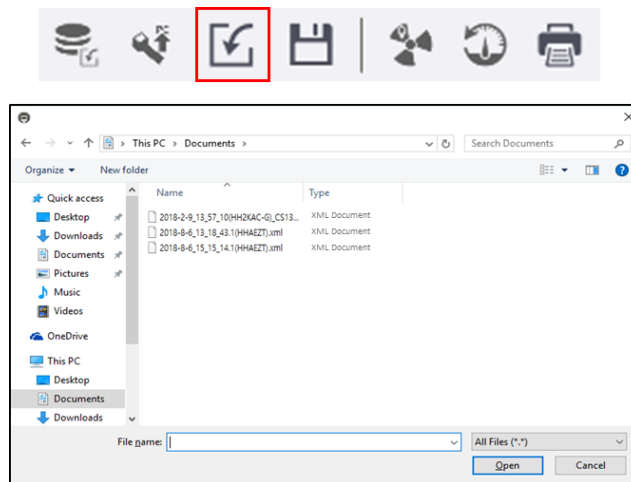


Figure 7.8.7. Importing XML file format

7.8.2.4. Save as XML file

The spectrum and relevant data from an Event can be saved in the XML file format specified in the ANSI 42.42 standard.

1. Select the Event from the Event List
2. Click the disk icon  in the lower left corner of the screen. The directory screen will appear. (figure 7.8.8)
3. Browse to the desired destination directory, select it, and click **Save**.

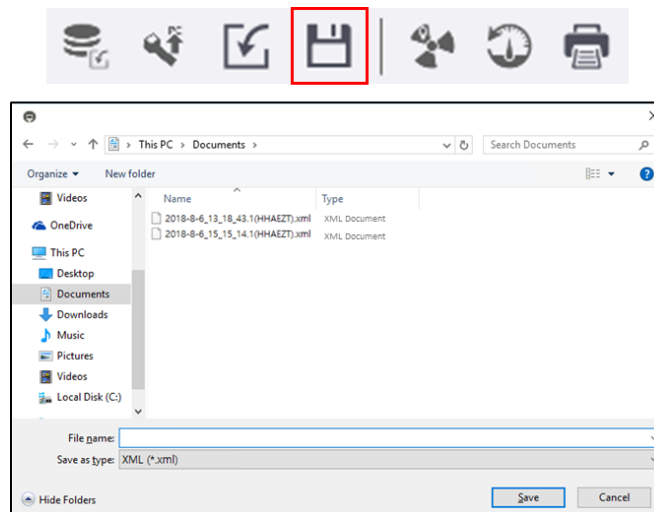


Figure 7.8.8. Save as XML file example

7.8.2.5. Send event to RadResponder

The spectrum and relevant data from an Event can be send to RadResponder cloud. (Refer to RadResponder section).

1. Select the Event from the Event List
2. Click the disk icon  in the lower left corner of the screen (Figure 7.8.9.) The RadResponder screen will appear.
3. Select either single event or multiple event.
4. Select event to be sent.
5. Click Grant button to send the selected event to RadResponder cloud.



The default setting for Event ID is Testing Training (refer to section 7.9.4.12. RadResponder account setup).

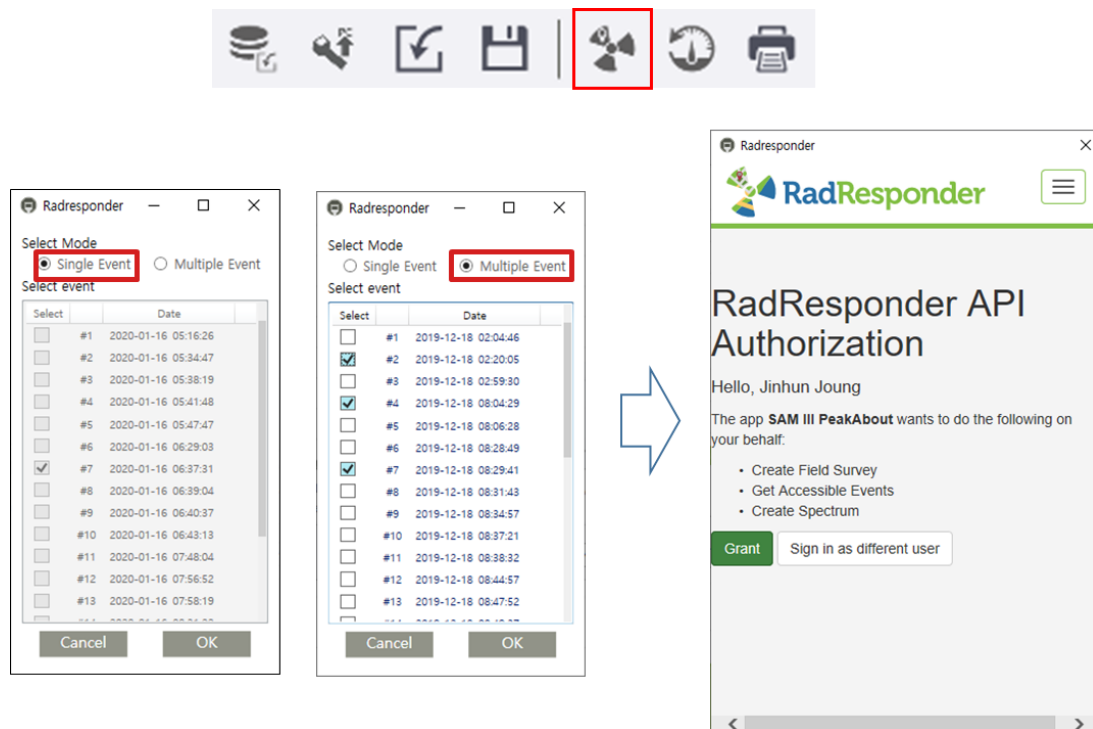


Figure 7.8.9. Send event to RadResponder example

7.8.2.6. Recalibration

PeakAbout and Peak ID use the 2nd order polynomial fit to convert measured channel value to corresponding energy. To achieve 2nd order polynomial fit minimum of 3 data points are required and they are 32 keV and 662 keV from Cs-137 and 1,461 keV from K-40.

However, recalibration allows user to replay spectrum analysis with customized energy calibration coefficients and library setting.

1. Select  icon
2. Recalibration window pop up as shown in figure 7.8.10.
3. Check select box to assign channel to energy line
4. Select OK

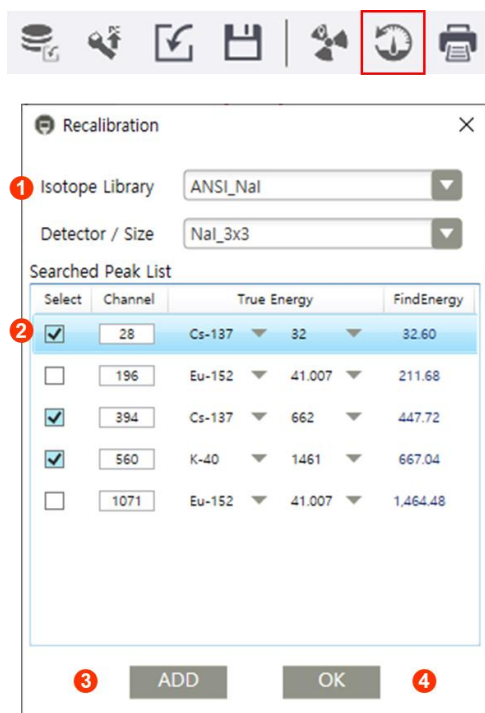


Figure 7.8.10. Recalibration setup window example

7.8.2.7. Print

The spectrum and relevant data from an Event can be printed.

1. Select the Event from the Event List
2. Click the printer  icon. Refer to the section xx for printing format and instructions.

7.9. CONFIGURATION MENU GUI

7.9.1. Navigating the Event Log GUI

PeakID IV Configuration menu allows user to configure the SAM 940+ settings. Some parameters that need frequent changes can be set by using PeakAbout IV configuration menu. The other hand, PeakID Configuration menu provides full access and management of all configurable parameters in operating the SAM 940+.

The configuration GUI consists of User setting, Administrator Setting and Library setting. Administrator login is required to use advanced features.

The screenshot displays the SAM940+ Configuration GUI. The interface includes a sidebar on the left with navigation options: Manual ID, Event Log, Configuration (highlighted with a red box and number 1), User (highlighted with a red box and number 5), Admin (highlighted with a red box and number 6), and Library. The main area is divided into two sections: User and Admin. The User section contains fields for Name (BNC), Location (BNC), Detector (NaI_2x2), S/W version (1.0.5), Alarm Ringtone (Siren), and Alarm Volume (80% and ON). The Admin section contains various settings including Isotope Library (ANSI_NaI), Sleep Mode (300 Sec), Manual ID Acq. Time (60 Sec), Manual ID Increment Time (10 Sec), Background Acq. Time (60 Sec), Calibration counts required (200000 count), Dose Rate Unit (uSv/h), Display Option (Linear), Gamma Alarm Threshold (Mode: FIX, Sigma: 3, Health: 10000), Neutron Alarm Threshold (Neutron(cps): 0.8), Reach Back Account Info (Receiver: mark.slattery@berke, Sender: bncapplications@gm, Password: bncsupport, Server: smtp.gmail.com, Port: 465), Setup Radresponder Account (Login ID: mark.slattery@berkeley, Password: msBNC1234i@#\$), Device WiFi Setup (Name: BNC WAN, Password: 1a2a3a4a5a), Admin Password (****), Time Zone (America/Los_Angeles), Setup Sequential Mode (Sequential On/Off: Off, Measurement Time: 60, Pause Time: 5, Repeat Count: 5), Event Erase (Erase button), and Factory Default Reset (Restore Default button). The bottom status bar indicates 'Sam940+ Pairing Status: Disconnected'.

Figure 7.9.1. Configuration Menu GUI example.

1	Configuration sub menus	Configuration has User, Administrator and Library sub menus
2	Copy to an USB	Copy configuration to an USB device
3	Upload to SAM940+	Upload configuration to SAM940+ via WiFi network

4	Download from 940+	Download configuration from 940+ via WiFi network
5	User setup	User name and location, Language and alarm ringtone setup
6	Administrator setup	Password protected configuration items

7.9.2. Copy configuration to an USB device

Configuration parameters can be copied to an external USB device for backup and/or downloading the parameters to the SAM 940+ device.

1. Insert an USB device to a PC
2. Select USB device icon ①
3. Confirmation message will be appeared as shown in figure 7.9.2. Click OK.
4. Select the USB device
5. Select folder to save the configuration parameters

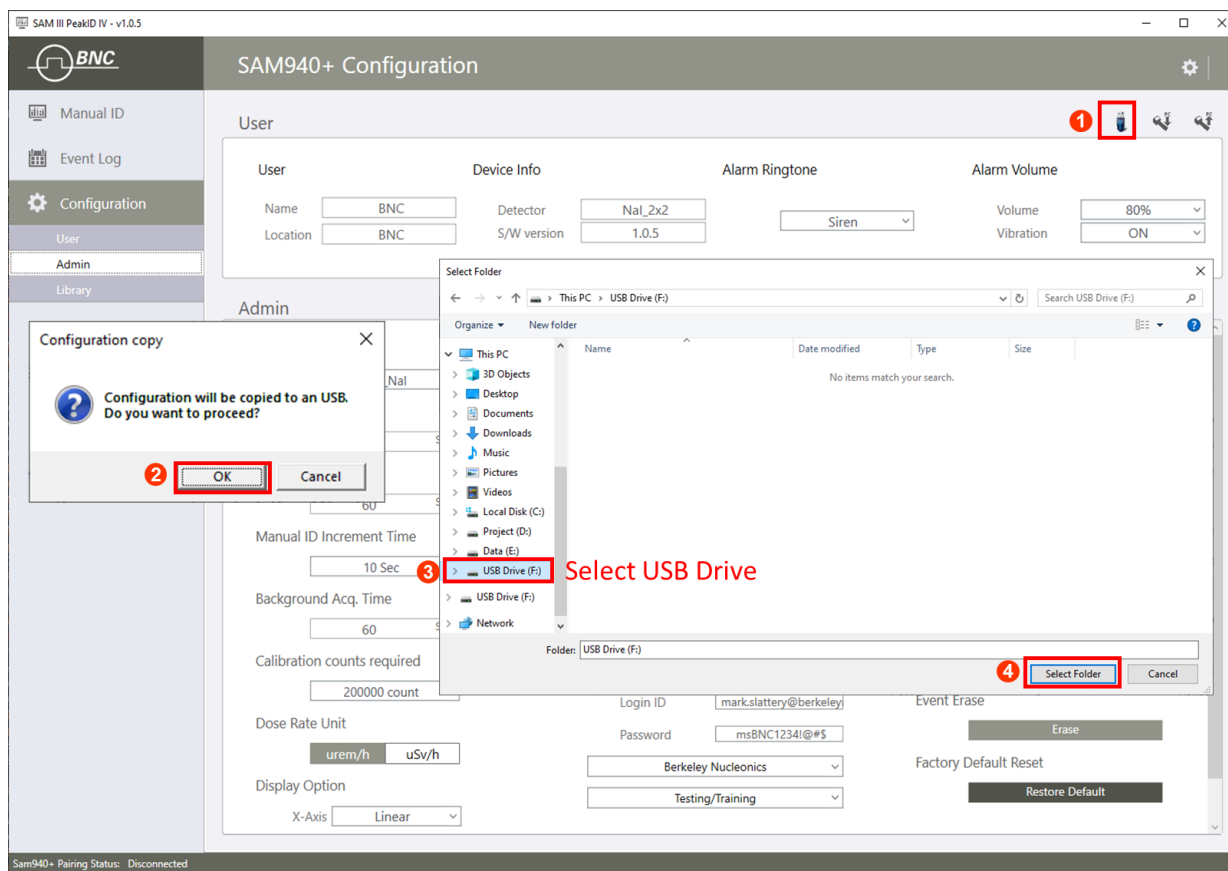


Figure 7.9.2. Copy configuration to an USB device.



Refer to the section xx for download configuration from an USB to the SAM 940+.

7.9.3. Upload/Download configuration via WiFi network

All setup parameters configured and saved by PeakID IV can be transferred to SAM 940+ or vice versa using WiFi network.

7.9.3.1 Download configuration parameters from the SAM 940+ Click

1. icon 
2. The saved parameter of SAM 940+ are downloaded/overwritten to PeakID IV.

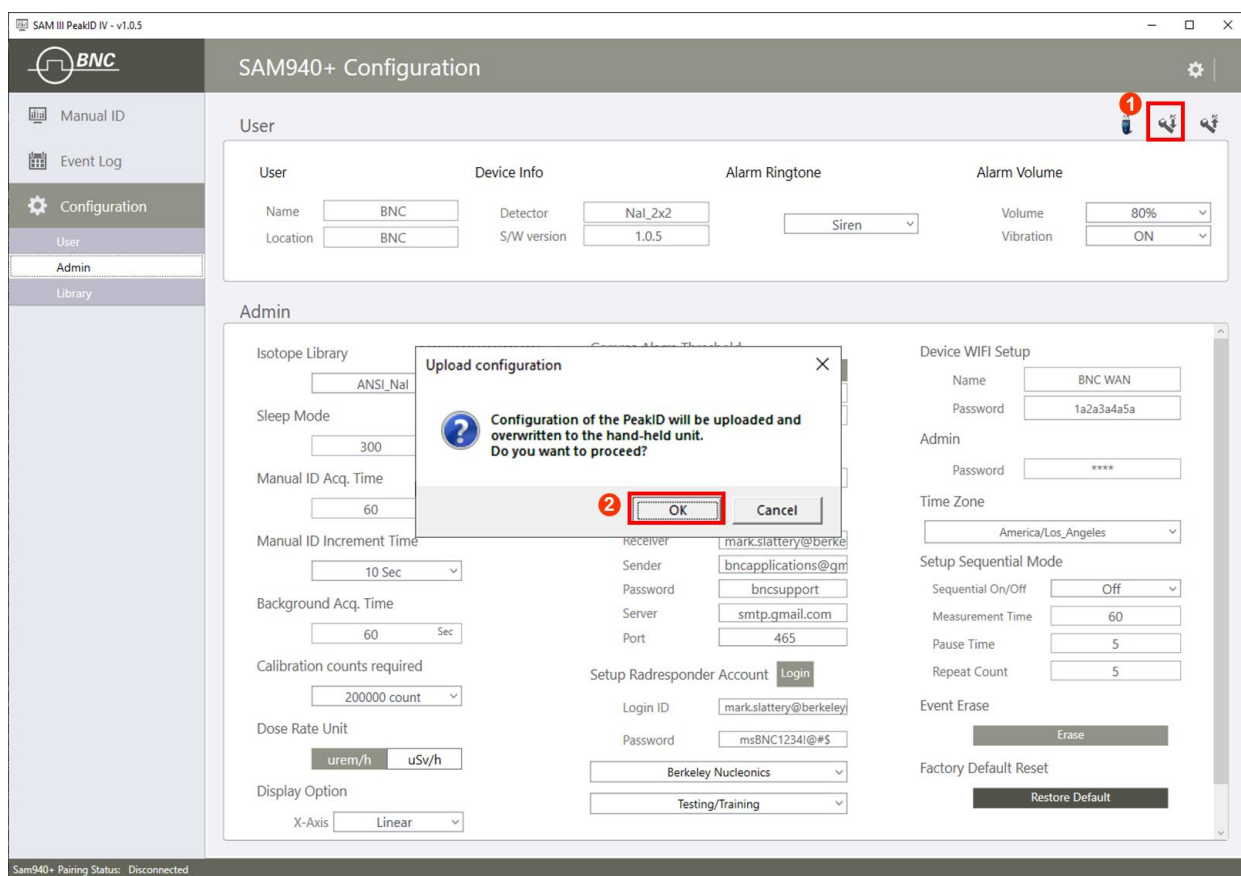


Figure 7.9.3. Download configuration from the SAM 940+.

7.9.3.2 Upload configuration parameters to the SAM 940+ Click

1. icon 
2. The edited/saved parameters in PeakID IV are uploaded/overwritten to the SAM 940+

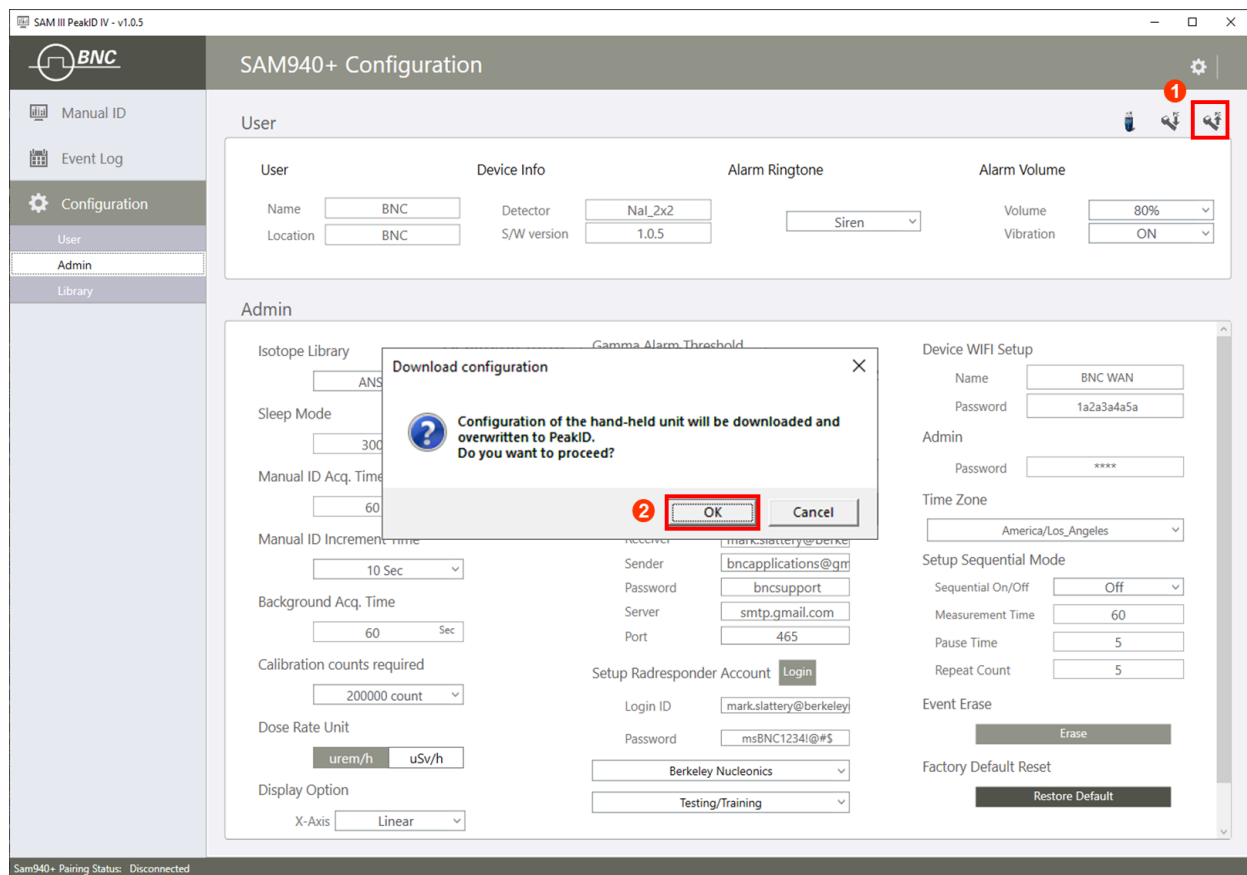


Figure 7.9.4. Upload configuration to the SAM940+.



The SAM940+ and a PC should be paired to use this feature. Refer to the section xx for pairing SAM 940

+ . 7.9.4. Upload/Download configuration via USB Cable

All setup parameters configured and saved by PeakID IV can be transferred to SAM 940+ or vice versa using USB cable. Connect device to PC by USB cable, and ensure that the SAM 940+ is properly connected to the PC via USB Cable

7.9.4.1 Download configuration parameters from the SAM 940+

1. Click  icon
2. Select the desired folder to store the configuration file. The file from the device will be copied to this folder shown in Figure 7.9.5. Download confirm message window is popped up as shown in figure 7.9.6.

3. The saved parameter of SAM 940+ are downloaded/overwritten to PeakID IV.

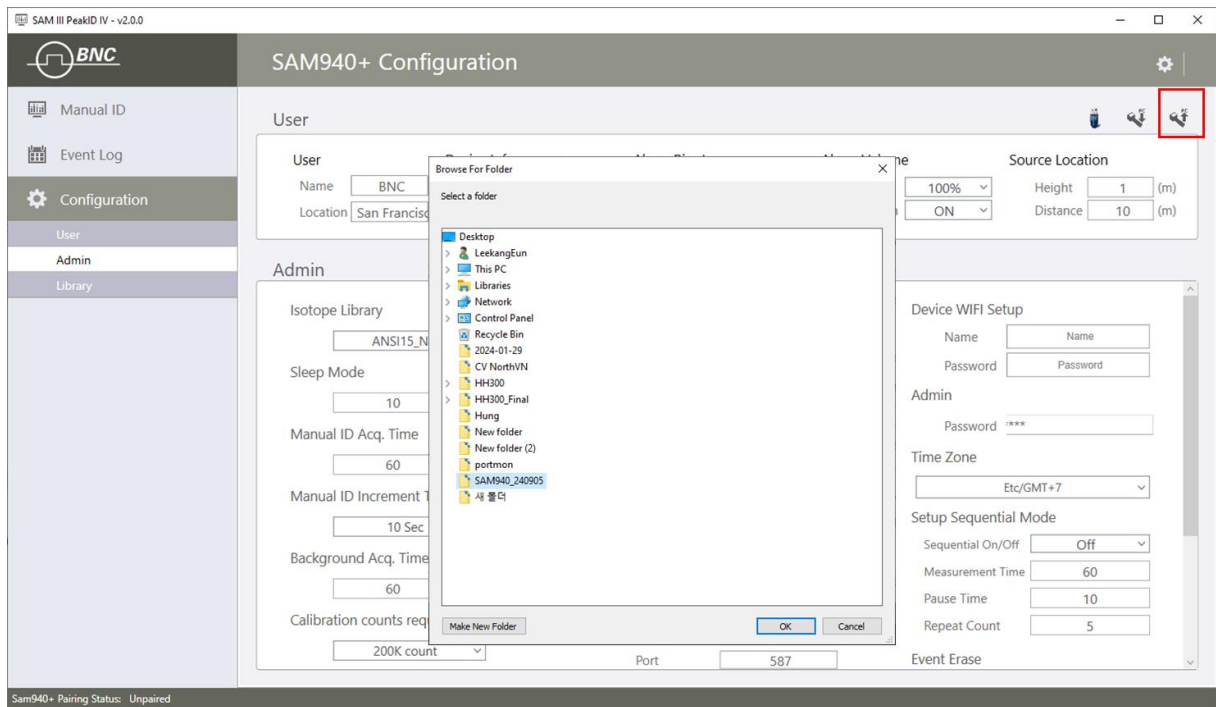


Figure 7.9.5. Select the desired folder to store the configuration file

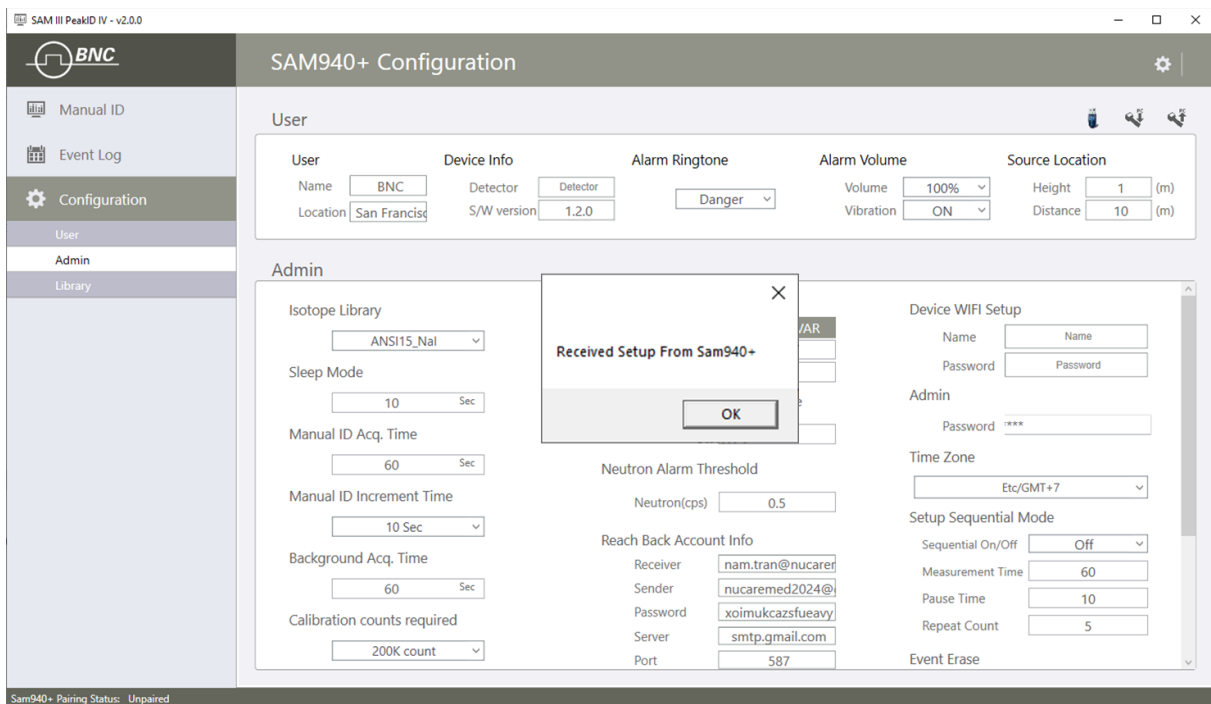


Figure 7.9.6. Download configuration from the SAM 940+.

7.9.4.2 Upload configuration parameters to the SAM 940+ 1.

Click 

2. The edited/saved parameters in PeakID IV are uploaded/overwritten to the SAM 940+

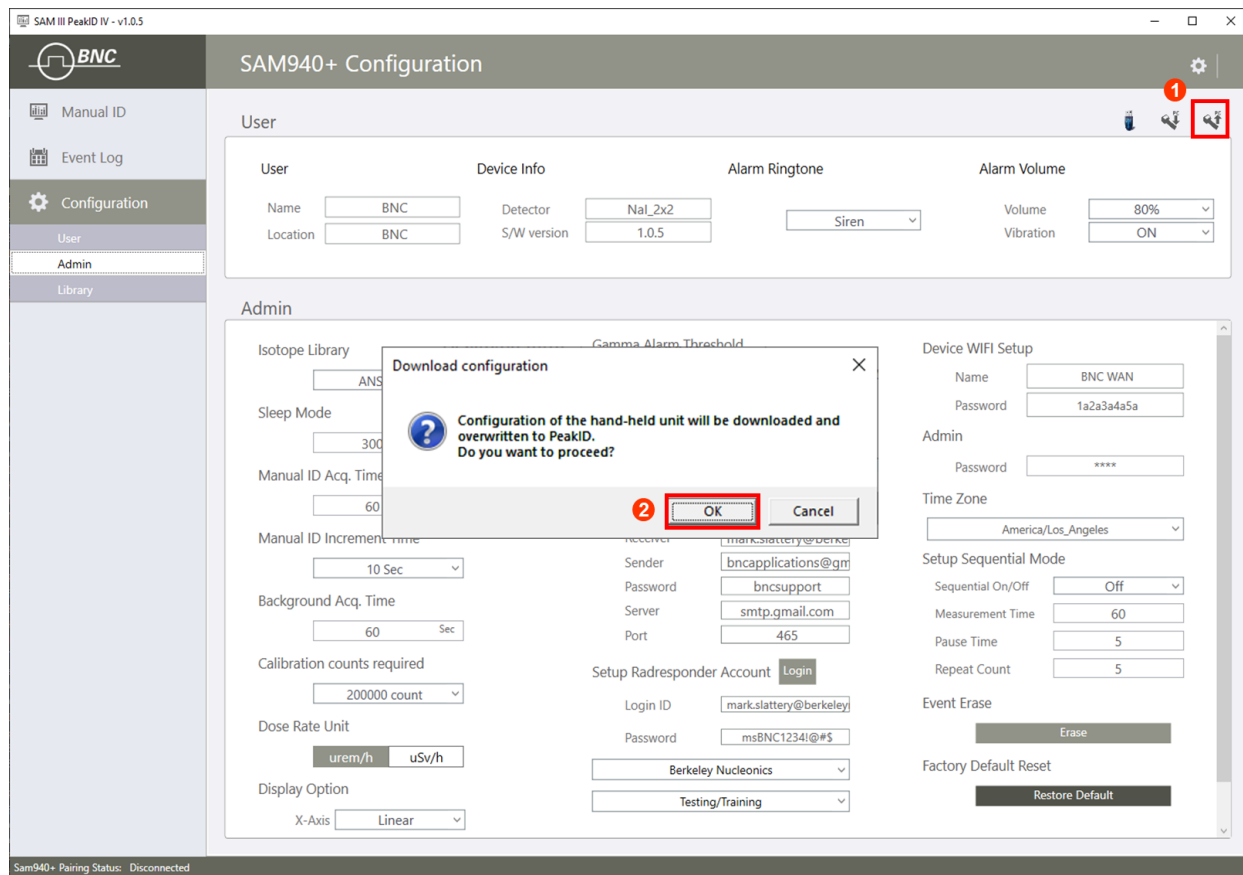


Figure 7.9.7. Upload configuration to the SAM 940+.



The device should be rebooted to apply the changed settings.

7.9.5. User setup

User setting allows user to set User name, Location, Language and Alarm Ringtone.

The screenshot shows the SAM940+ Configuration software interface. The left sidebar has a 'User' tab selected, indicated by a red circle with the number 1. The main area is titled 'SAM940+ Configuration'. Within this area, the 'User' section is highlighted with a red box, indicated by a red circle with the number 2. This section contains the following fields:

- User:** Name (BNC), Location (BNC)
- Device Info:** Detector (NaI_2x2), S/W version (1.0.5)
- Alarm Ringtone:** Siren (dropdown)
- Alarm Volume:** Volume (80% dropdown), Vibration (ON dropdown)

Below the 'User' section is the 'Admin' section, which contains various configuration options including Isotope Library, Sleep Mode, Manual ID Acq. Time, Background Acq. Time, Calibration counts required, Dose Rate Unit, Display Option, Gamma Alarm Threshold, Neutron Alarm Threshold, Reach Back Account Info, Setup Radresponder Account, Device WIFI Setup, Admin, Time Zone, Setup Sequential Mode, Event Erase, and Factory Default Reset.

Figure 7.9.8. User setup example

7.9.5.1. Enter User Name and Search Location

Entering these settings will save the operator's name and the search location with the Event data.

7.9.5.2. Device info.

Display device information and software version of PeakAbout IV.



Only English language is supported currently. Consult with Nucare, Inc. for other language support if needed.

7.9.5.3. Alarm Ringtone setup

There are six alarm ringtones to choose from: Siren, Morning call, Bell, Trumpet, Telephone call, and Beep.



The default alarm ringtone is Siren.

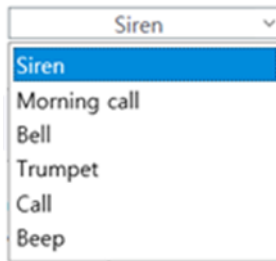


Figure 7.9.9. Ringtone types



The “Beep” ringtone is different from the other choices in that the rate of the “beep” increases as detected energy increases in count rate.

7.9.5.4 Alarm volume setup

User can define alarm volume. 0 volume represents mute mode.

7.9.5.5. Vibration setup

User can define vibration mode. Select On/Off vibration mode.

7.9.6. Administrator setup

You must enter the Administrator password before you can open the Administrator setup management tool.

1. Click the **Admin** menu (Figure 7.9.10). ①
2. Enter the Administrator password ②. The default password is **1234**. The Admin configurable List is shown in figure 7.9.10. ③

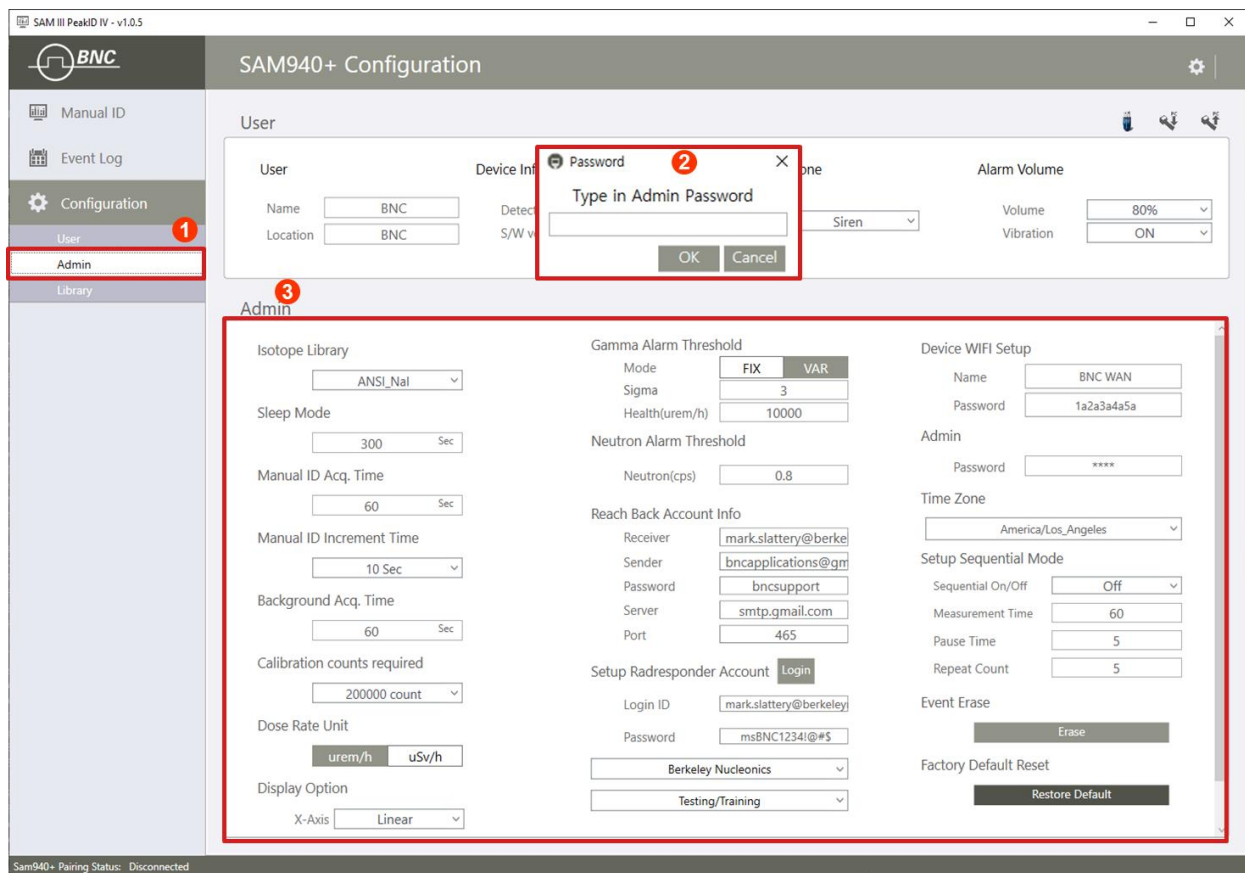


Figure 7.9.10. Admin menu GUI example

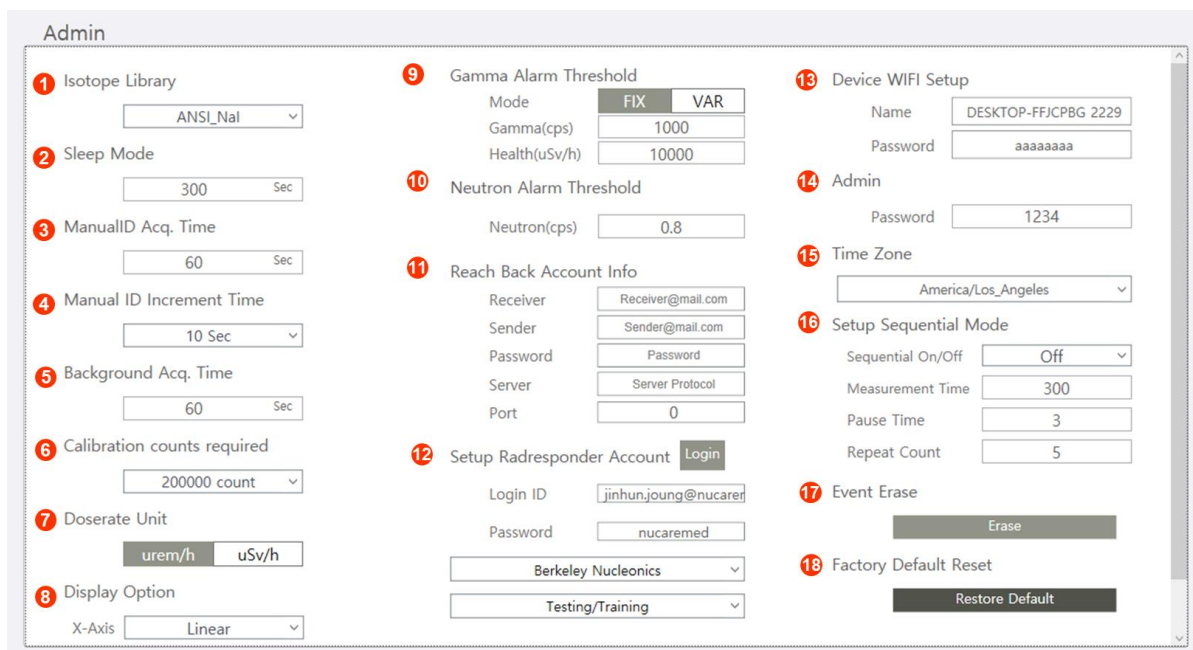


Figure 7.9.11. List of Configuration by Admin.

7.9.6.1. Isotope Library List.

The Library List in PeakAbout IV for SAM 940+ contains eight libraries. Four libraries are optimized for sodium iodide (NaI) detectors. Four libraries are optimized for high-resolution detectors like cerium bromide (CeBr).

1. Open and select the Library from the list.

The default library is ANSI_NaI (ANSI_CeBr if your SAM 940+ has a high-resolution detector).

✓ **ANSI_NaI and ANSI_CeBr are compliant with ANSI N42.34-2006**

²⁴¹Am, ¹³³Ba, ⁵⁷Co, ⁶⁰Co, ¹³⁷Cs, ¹⁵²Eu (not ANSI required), ¹⁸F, ⁶⁷Ga, ¹²³I, ¹²⁵I, ¹³¹I, ¹¹¹In, ¹⁹²Ir, ⁴⁰K, ²³⁷Np, ²³⁷Np HE, ²³⁹Pu, ²²⁶Ra, ^{99m}Tc, ²³²Th, ²⁰¹Tl, ²³³U, ²³⁵U HE, ²³⁵U, ²³⁵U HE, ²³⁸U

✓ **IND_NaI and IND_CeBr are specialized with additional Industrial classified isotopes**

²⁴¹Am, ¹⁰⁹Cd, ⁵⁷Co, ⁶⁰Co, ¹³⁴Cs, ¹³⁷Cs, ¹⁵²Eu, ¹¹¹In, ¹⁹²Ir, ⁴⁰K, ⁸⁵Kr, ⁵⁴Mn, ²²Na, ²⁰⁴Tl

✓ **MED_NaI and MED_CeBr are specialized with additional Medical classified isotopes**

⁷⁷Br, ⁵⁸Co, ⁶⁰Co, ⁶⁷Co, ⁵¹Cr, ¹³⁷Cs, ¹⁸F, ⁶⁷Ga, ¹²⁴I, ¹³¹I, ¹⁹²Ir, ⁴⁰K, ⁹⁹Mo, ²⁴Na, ⁸¹Rb, ⁸²Rb, ⁸³Rb, ¹²⁴Sb, ⁴⁶Sc, ⁷⁵Se, ¹⁵³Sm, ⁸⁵Sr, ⁸⁹Sr, ^{99m}Tc, ²⁰¹Tl, ¹³³Xe, ⁸⁸Y, ⁹⁰Y, ⁸⁸Zr, ⁸⁹Zr

✓ **ANSI15_NaI and ANSI15_CeBr are compliant with ANSI N42.34-2015**

²⁴¹Am, ¹³³Ba, ⁵⁷Co, ⁶⁰Co, ¹³⁷Cs, ⁶⁷Ga, ¹³¹I, ¹⁹²Ir, ⁴⁰K, ²³⁹Pu, ²²⁶Ra, ^{99m}Tc, ²³²Th, ²⁰¹Tl, ²³⁵U, ²³⁸U, DU, HEU, RGPu, WGPu

7.9.6.2. Sleep Mode.

To save power consumption and maximize operation time, the SAM 940+ falls in sleep mode when it is not used for a certain time.

The default sleep mode time is 300 seconds.

7.9.6.3. Manual Identification Acquisition Time.

To measure and record an event, Manual Isotope Identification must be performed. Manual ID can be performed whether the SAM 940+ is in an alarm condition or not.

1. Open the **MANUAL ID ACQ. Time** menu:

The default Manual ID Measurement Time is 60 seconds.

7.9.6.4. Manual Identification Adjustment Time.

When Manual ID is initiated, an event spectrum begins. While the Manual ID is running, you may decide that you will need to record a longer measurement. Without interrupting or restarting your Manual ID, you can extend or reduce the measurement time.

The Manual Identification Adjustment Time is defined in the Administrator Settings menu.

1. Open the Manual ID Adj. Time menu:

The default Manual ID Adjustment Time is 10 seconds.

7.9.6.5. Background measurement Time.

Reliable background information is critical to good nuclide identification and alarm performance. PeakAbout IV uses an advanced NORM rejection algorithm to minimize false alarms due to fluctuations of Naturally Occurring Radioactive Material in the background.

1. Open the **BACKGROUND MEASUREMENT** menu:

The default Background Measurement Time is 60 seconds.



It is good practice to select a Background Measurement Time value that is equal to your Manual Identification Measurement Time value. Refer to **Section 4.4.6.7: Manual Identification Measurement Time**.

7.9.6.6. Calibration measurement Count.

Although Auto Calibration and Auto Stabilization features adjust the calibration parameters of the SAM 940+ to correct for drift caused by environmental changes in ambient temperature, if drift of more than $\pm 20\%$ between the stabilized peak and the original peak (Cs-137 calibration peak) occurs, then you must perform manual calibration using a Cs-137 source. Refer to **Section 4.4.6.16: Perform Energy Calibration**.

1. Open the **CALIBRATION** menu:

The default Calibration Count value is 200,000 counts (200 x 1000).

7.9.6.7. Dose Rate measurement Unit.

The dose rate unit will apply automatically to dose rate values that are displayed in the Main screen, the Manual ID, and the Event files. The value of the Health Safety Threshold will also be automatically adjusted.

The default Dose Rate Unit is rem/h.

7.9.6.8. Display option.

The display option allows user to select x-axis option for QCC (Quadratic) spectrum analysis. Refer to the section xx for QCC display.

The default X-axis display option is Linear.

7.9.6.9. Select Alarm Mode and Alarm Threshold.

In the Administrator Settings menu, PeakAbout IV provides two modes for alarm triggering:

Fixed Alarm Mode is a basic method that assumes the background environment exhibits a very small amount of fluctuation. The alarm is triggered when the count rate exceeds the operator-determined

threshold value. This method can cause false alarms if there is substantial background fluctuation due to Naturally Occurring Radioactive Material (NORM), as is often the case when the system is moving across a large area while actively monitoring.

Variable Alarm Mode is a more advanced method that continuously samples the previous 10 seconds' worth of background fluctuation and calculates the standard deviation above background (σ , or sigma) of the fluctuation. Using this method prevents alarms due to changing background (and eliminates the need to perform new background measurements). Instead, alarms are triggered when real gamma peaks exceed the standard deviation of the dynamic threshold value. We recommend using Variable Alarm Mode when moving across a large area that may be subject to changes in background.

1. Select Alarm mode
2. Set Gamma alarm threshold
3. Set Health alarm threshold

The default threshold value for Fixed Alarm Mode is 1000 counts per second (cps), which is optimal for many applications. A typical background environment produces around 200 cps for a 3 x 3-inch NaI detector, or lower if a high-resolution detector option was chosen. A threshold value lower than 1000 is usually unnecessary and is more likely to generate false alarms. In situations where the background environment is known to be above normal, it may be necessary to select a higher value such 2000 or 3000 cps to minimize false alarms.

The default threshold value for Variable Alarm Mode is 4 (Sigma) which is optimal for many applications. The typical range is 1 to 5 (Sigma). Selection of a lower value creates slack in the threshold criteria and increases the likelihood of false alarms. Selecting a higher value can narrow the threshold criteria and cause significant gamma activity to be overlooked if it passes very quickly.

For the user's safety, PeakAbout III features a separate Health Safety Alarm with a different ringtone and visual signal. When the safety threshold is exceeded, the distinctive Health Safety ringtone sounds, and the Operating Message displays "DANGER" (Figure 4.46). The Health Safety Alarm takes precedence over all other alarm conditions.

The Health Safety Alarm threshold is defined in the Administrator Settings menu.

For your protection, PeakAbout III also features a separate and unique alarm which is triggered when the Health Safety Threshold is exceeded. Refer to **Section 4.4.6.3: Select Health Safety Threshold**.

The default value for Health Safety Threshold is 10,000 $\mu\text{rem/h}$ (100 $\mu\text{Sv/h}$).

7.9.6.10. Select Neutron Threshold.

If your SAM940+ includes the neutron detection option, then it contains either solid-state sensor arrays on a ^6Li substrate (Domino) or a tube filled with ^3He gas, integrated with moderator packaging inside the detector unit. PeakAbout III will display neutron activity on the Main screen in counts per second (cps) (Figure 4.44).

The default value for Neutron Threshold is 0.8 CPS which is optimal for most applications.

7.9.6.11. Reach Back Email Account.

When operating within range of Wi-Fi, the SAM 940+ allows you to transmit spectrum and event data directly to a prearranged location such as a U.S. DOE reach-back facility, group leader, command center, etc. Data from an Event that you select is emailed in N42-compatible XML format to the specified recipient. The system is optimized for Gmail (the email service developed by Google). Before you can transmit spectrum and event data, you must setup email information.

7.9.6.12. RadResponder Account.

In the United States, we have partnered with FEMA to implement RadResponder, a free network for the rapid collection and management of radiological data during an emergency. When the SAM 940+ is operating within range of Wi-Fi, spectrum and event data can be uploaded to the RadResponder cloud in the appropriate format. RadResponder can be accessed on PDAs, smartphones, tablets, or any computer connected to the web, allowing it to be seamlessly employed at all levels of government during a radiological or nuclear emergency.

You must already have established a RadResponder account at **www.RadResponder.net** before you can use this feature.

1. Click Login button
2. Sign in with RadResponder ID/Password
3. Click Grant button
4. Select Sponsor and type or event.



Once the login information (user ID/password, Sponsor and event type) is saved, you don't need to repeat this procedure when sending event to RadResponder server unless Sponsor and/or event type need to be changed.

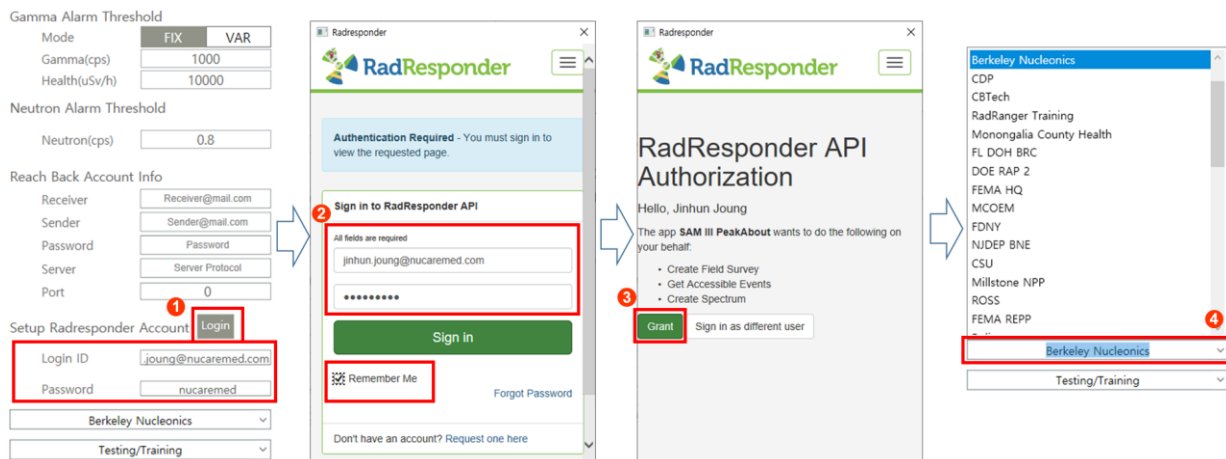


Figure 7.9.12. List of Configuration by Admin.

7.9.6.13. WiFi setup

Select available WiFi network and login. The SAM 940+ and a PC should be the same WiFi network to exchange data between.

7.9.6.14. Change Admin. Password

Change administrator's password.

Note: Factory setup is 1234. It is recommended to change it to more secured combination.

7.9.6.15. Time Zone

Event data has a date/time stamp and proper date/time zone has to be setup.

7.9.6.16. Setup Sequential mode

The purpose of Sequential Mode is to generate a series of measurements separated by short pauses, one right after the other. The sequence of measurements is generated by a single keypress, instead of requiring the operator to manually begin each individual Event. All Events in the sequence have the same Measurement Time, and the Pause Time between Events is the same. The sequence continues for a specified Number of Measurements or until the user interrupts and cancels the sequence. Sequential Mode is useful for plotting a series of events while the SAM940+ moves along a continuous path.

For each Event in the sequence, the SAM 940+ collects the date, time, GPS position, overall count rate, overall dose rate, count rate and dose rate for each identified isotope, and other typical event data (including the identified neutron rate if the neutron detection option is installed). After the survey, all the spectral data from the entire sequence may be batch processed (to a spreadsheet, for example) with a single command.

7.9.6.17. Erase Event

Erase the event data saved in the SAM 940+.

7.9.6.18. Restore Factory Hardware Configuration

The Restore Factory HW Configuration setting is used to reset parameters like calibration and high voltage bias for the installed detector among other settings. Situations in which this operation is useful are extremely rare, and the procedure should only be performed by a factory technician. Contact Nucare, Inc. before performing this operation.

7.9.7. Library setup

You must enter the Administrator password before you can open the Isotope Library management tool and upload an Isotope Library to the SAM 940+.

7.9.7.1. Navigating Library Menu GUI

Library Menu GUI consists of Library Name (2), Isotope List (3), Isotope Info (4), Photopeak and BR info (5) and icons (6) for transferring Library to the SAM 940+. Refer to the following sections for details of each.

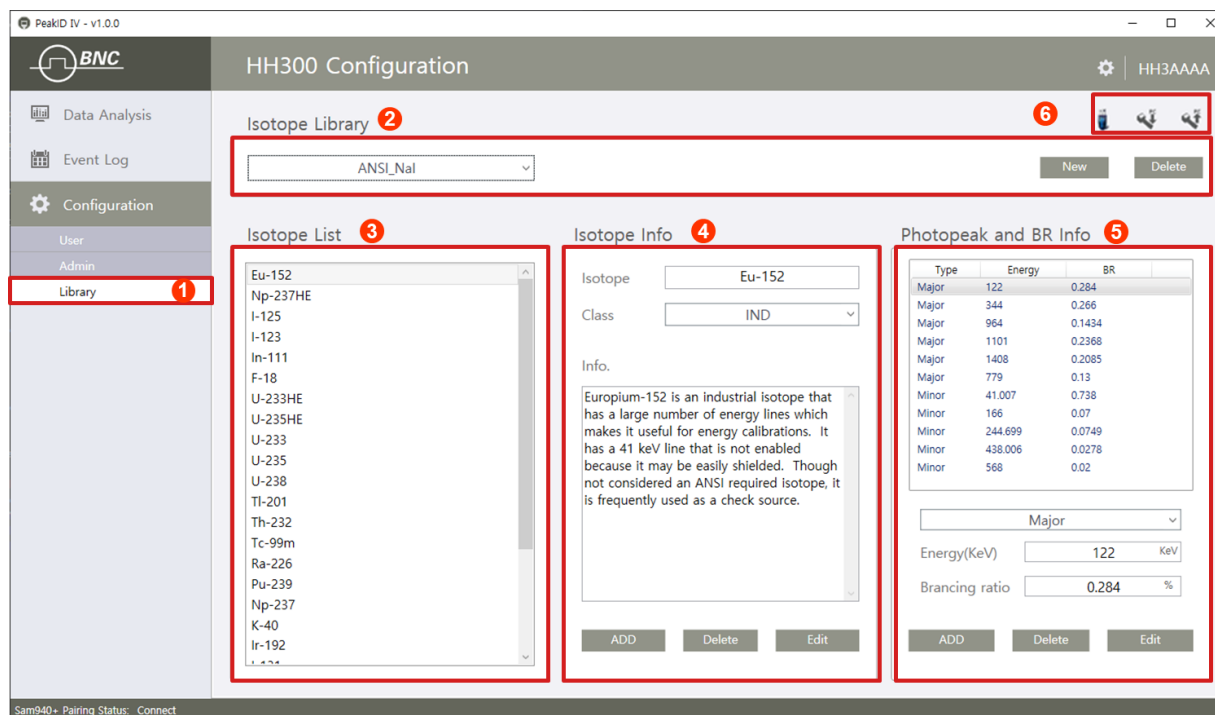


Figure 7.9.13. Library menu GUI.

7.9.7.2. Isotope Name

The SAM 940+ provides total of 6 libraries. (refer to the section xx). Edit existing library, Add a new library and delete library from the list are managed here.

7.9.7.3 Isotope List

Isotope List displays the member of isotopes in the selected library.

7.9.7.4. Isotope Info

Isotope Info displays the information of selected isotope in the isotope list.

Admin can Add, Delete or Edit the isotope.

7.9.7.5. Photopeak and BR info

Photopeak and BR info displays the photopeak(s) and its BR (Branching Ratio) of the selected isotope in the selected library.

Admin can Add, Delete or Edit the isotope

information 7.9.7.6. Upload to SAM 940+

Upload the library information to the SAM 940+ (or vice versa) can be done by clicking the icon.

CHAPTER 8. REMINDER AND USUFUL TIPS

- ✓ Make sure there are no radiation sources in the vicinity of the SAM 940+ before turning the power on.
- ✓ Make sure there are no radiation sources in the vicinity of the SAM 940+ during Auto Calibration.
- ✓ Make sure there are no radiation sources in the vicinity of the SAM 940+ before performing a Background Measurement.
- ✓ Wearing polarized lenses such as sunglasses can make the LCD display appear blank. We recommend that you do not wear polarized lenses while using SAM 940+.
- ✓ Do not open the unit. Opening the unit may void your warranty. Any repairs or alterations should only be performed by technicians who are authorized by NuCare, Inc.

APPENDIX: SPECIFICATIONS

Detectors			
Gamma	2x2 inch NaI(Tl), CeBr ₃ . And PIN detector (for high dose rate)		
Neutron	1.5x1.5 or 1.5x2 inch CLYC or CLLBC		
External Probe	Custom size gamma or neutron probe. Pancake (alpha, beta and gamma)		
Performance			
Energy Range	20 keV – 3 MeV		
Linearization	Real-time linearization by firmware		
Dose rate range (ID mode)	0 – 100 μSv/h		
Dose rate range (Count mode)	0-10 mR/h (NaI), 10 mR/h-10 R/h (PIN)		
Sensitivify	> 1500 cps / μSv/h @Cs-137		
Stabilization	Sourceless automatic real-time stabilization		
Identification	ANSI N42.34 compatible		
MCA channel	16 bit 2048 ch.		
Library categories	SNM, IND, MED, NORM		
Typical resolution	NaI(Tl) < 7%, CeBr3 < 5%, LaBr3 < 4%, CLLBC < 4% @ 662 keV		
BATTERY			
Type	replaceable Li-ion		
Operation time	15x2h in sleep mode, Max run time 8x2h		
Backup battery adapter	4 AA alkaline batteries (> 2h. backup)		
PHYSICAL			
Dimensions (W x D x H)	102 x 243 x 97 (mm)	Weight	1.3 kg
ENVIRONMENTAL			
Operating Temperature	-20°C(-4°F)–50°C(122°F)	Protection Rating	IP65
Relative Humidity	10 to 80%, non condensing	Testing Condition	IEC 62706, EN 61326
DISPLAY		INPUT/OUTPUT	
Type	LCD	USB	micro USB 3.0
Size	3.5inch	Bluetooth	Class 4.0
Resolution	480 x 320, configurable up to 1920x1080	WLAN	802.11n Wireless LAN
MISCELLANEOUS		SOFTWARE	
GPS	22tracking, 66channels, 1-10Hz	Reach-back	ANSI N42.42 or CSV event data
CCD Camera	CMOS 8MP	Application SW	PeakAbout (application SW)
GYRO sensor	Linearity 0.1 ± % FS		PeakID (PC version SW)
ACCESSORIES			
Case	Pelican style carrying case	Charger	USB charger / Vehicle Battery charger
Holster	Belt holster	Connection cable	mini USB cable



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